

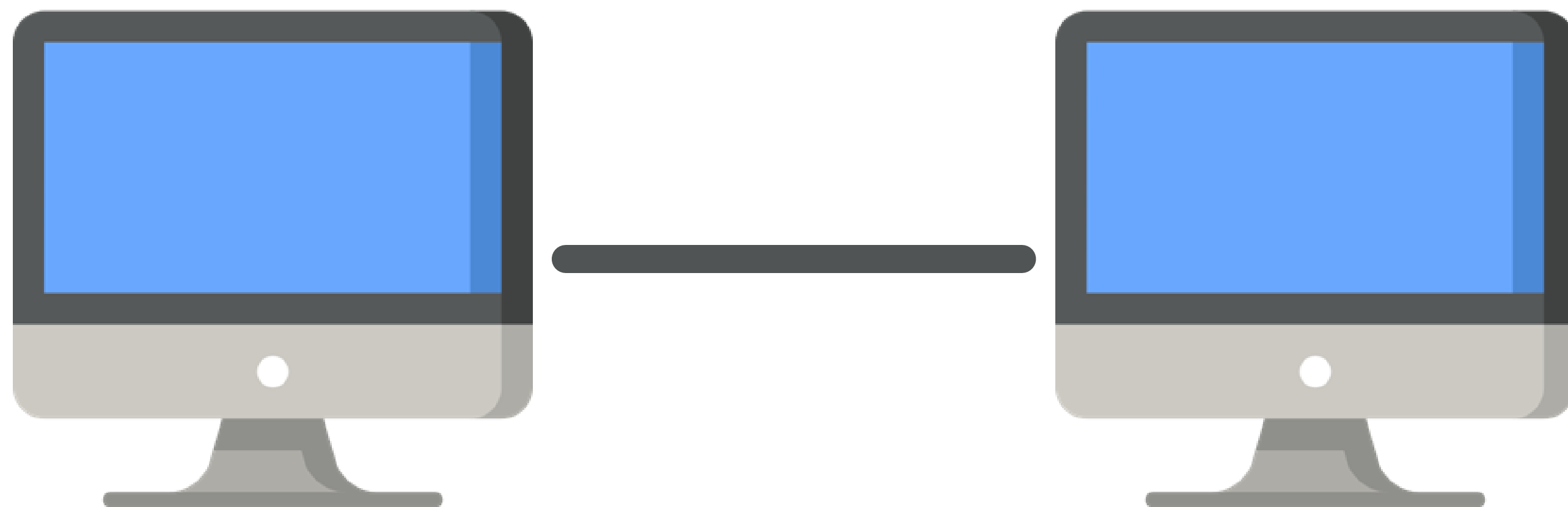
Computer Networking

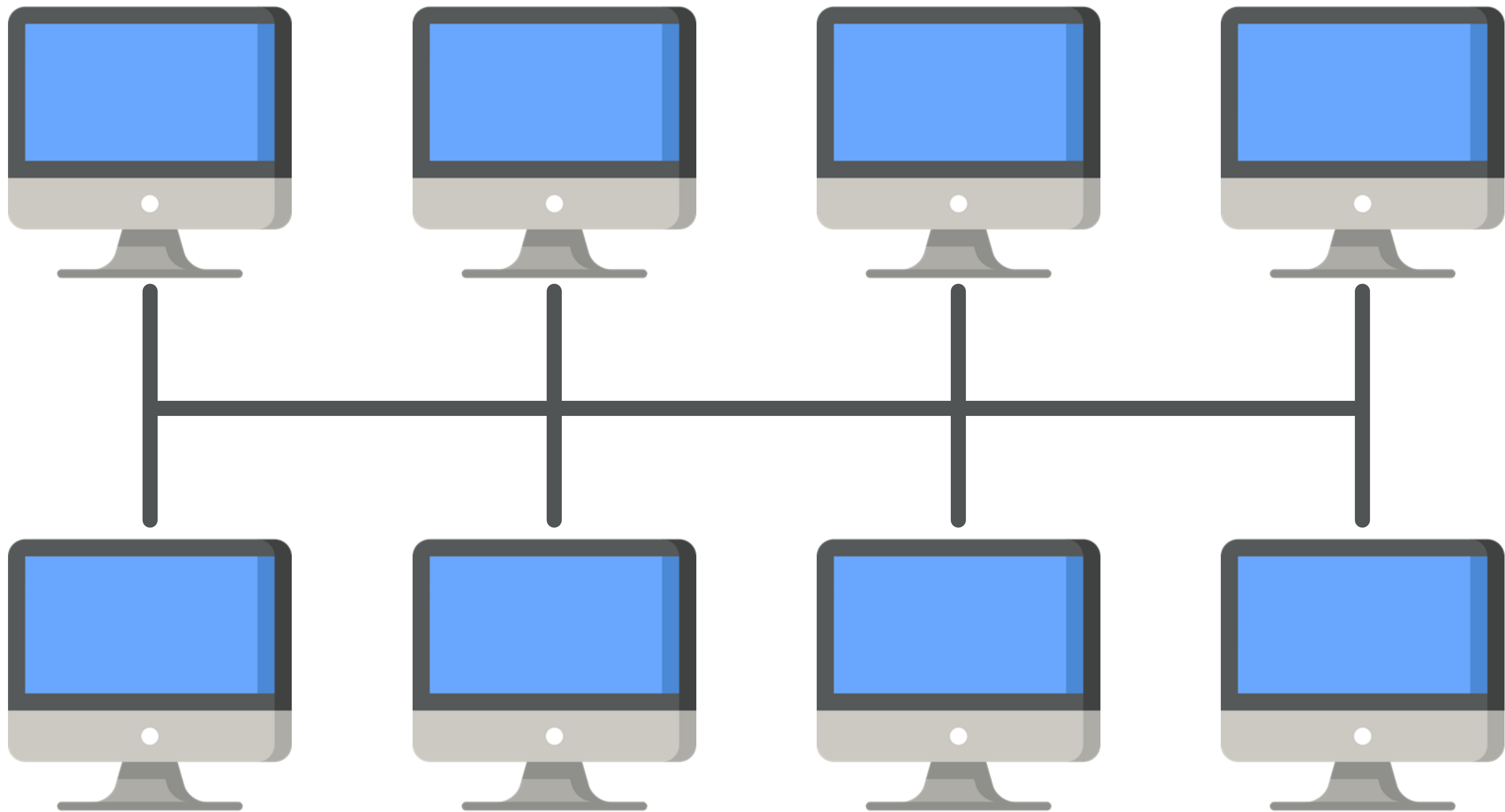
An Overview

Agenda

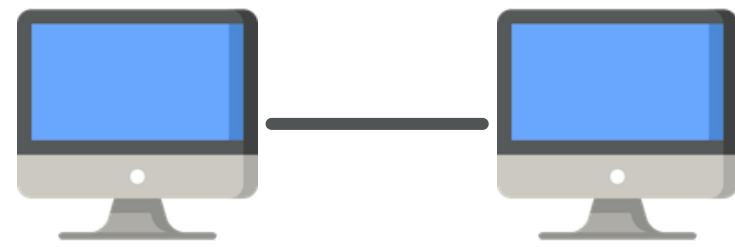
- Making a **Simple Network** prototype
- Some **History**
- Standardization: **OSI & TCP/IP** Models
- Some **Network Devices**
- Some **Protocols**
- **A Packet's Life Cycle**

Making a **simple network**
prototype

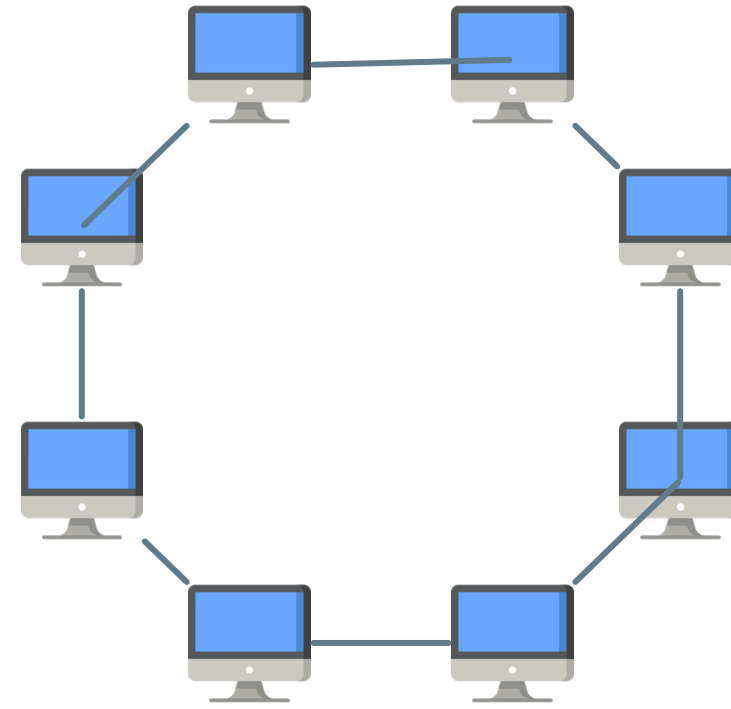




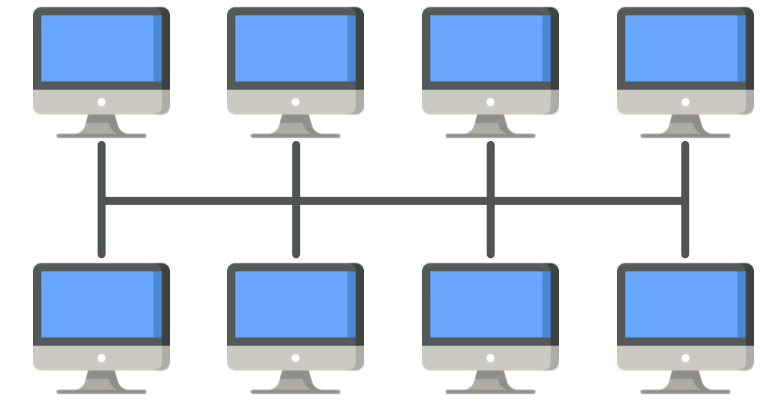
We may also have other
Network Topologies



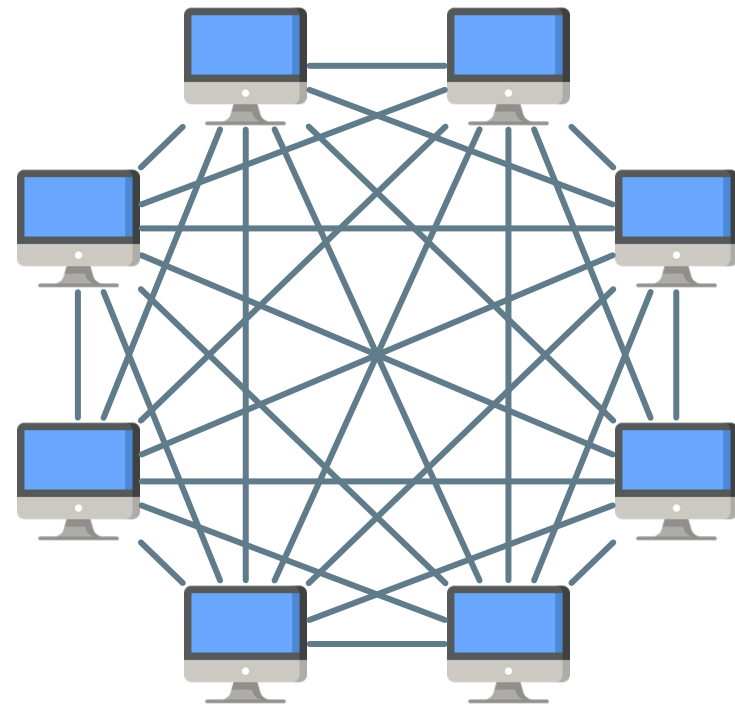
Point-to-point Topology



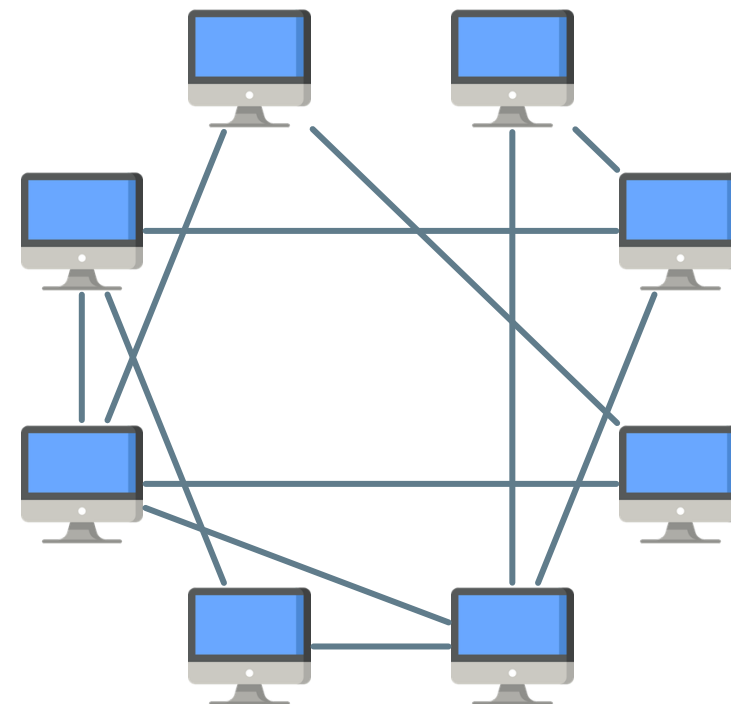
Ring Topology



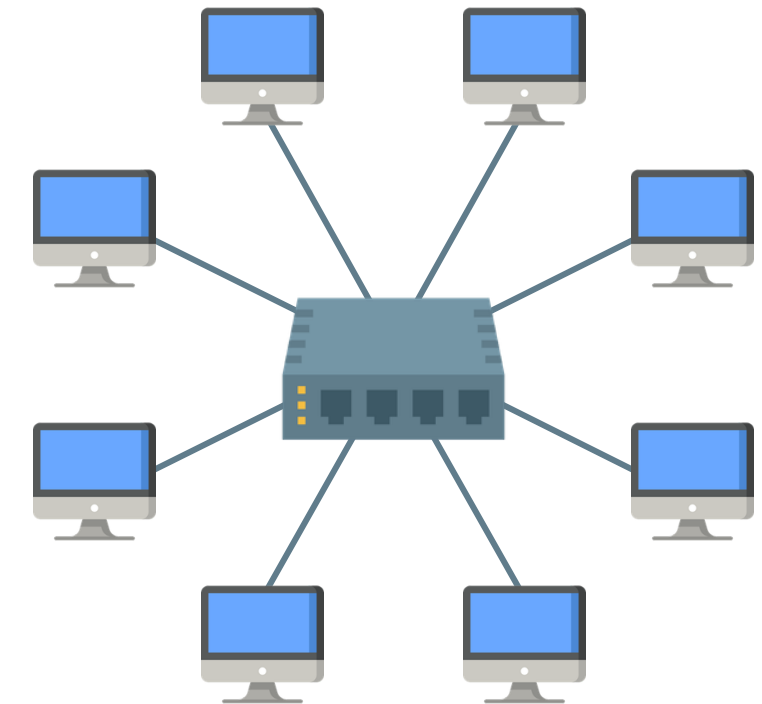
Bus Topology



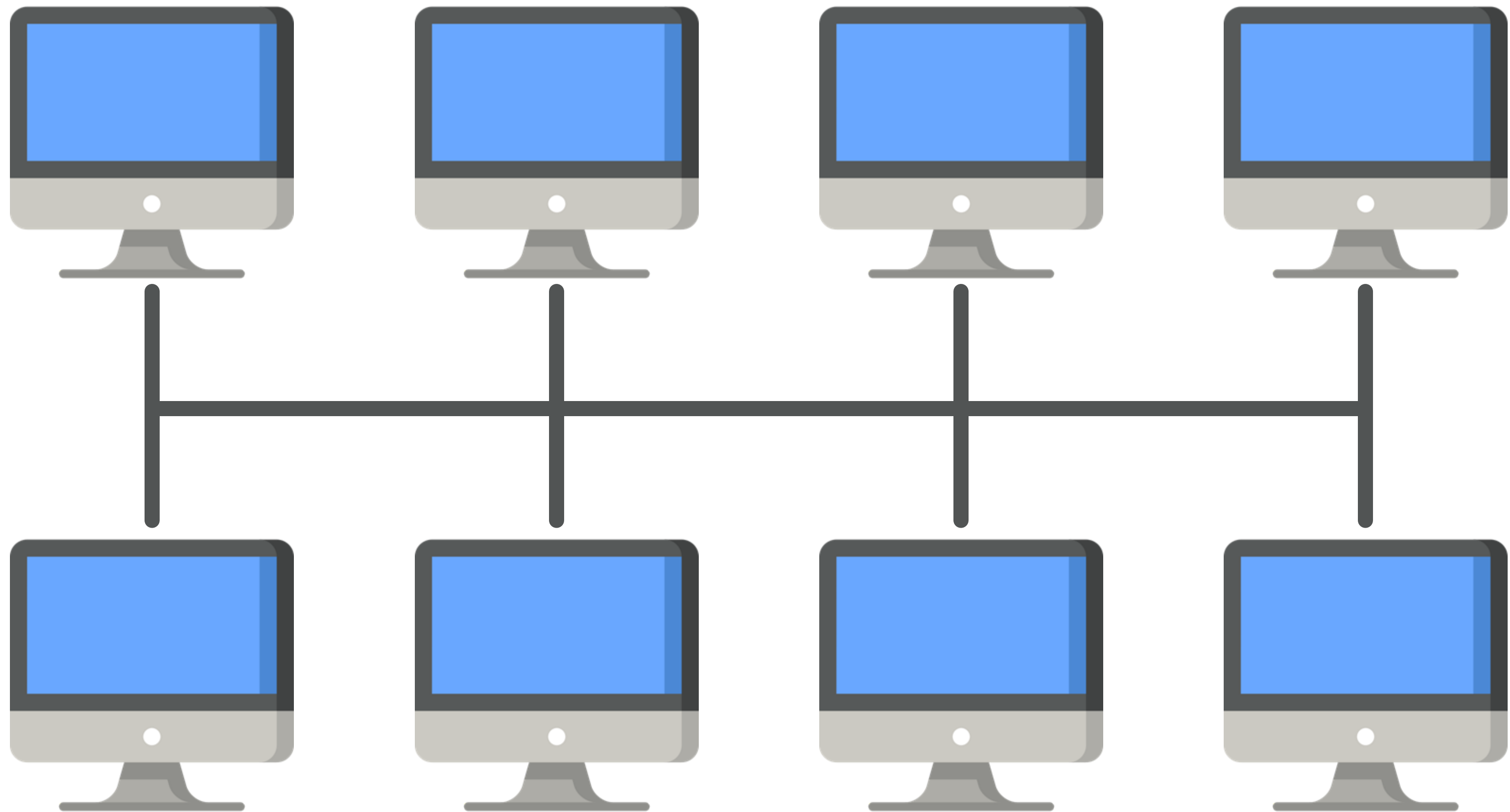
Fully Connected Topology



Mesh Topology



Star Topology

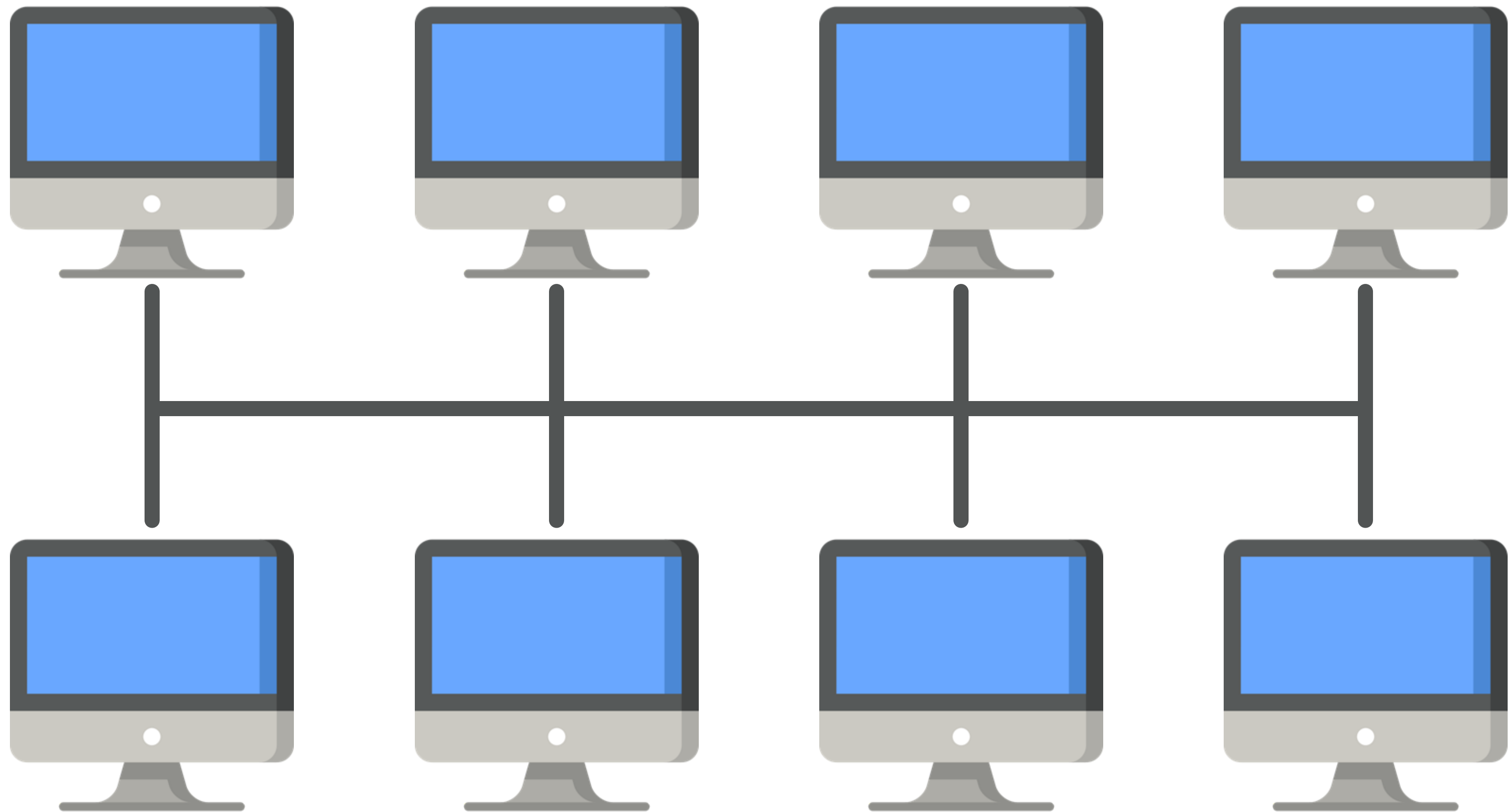


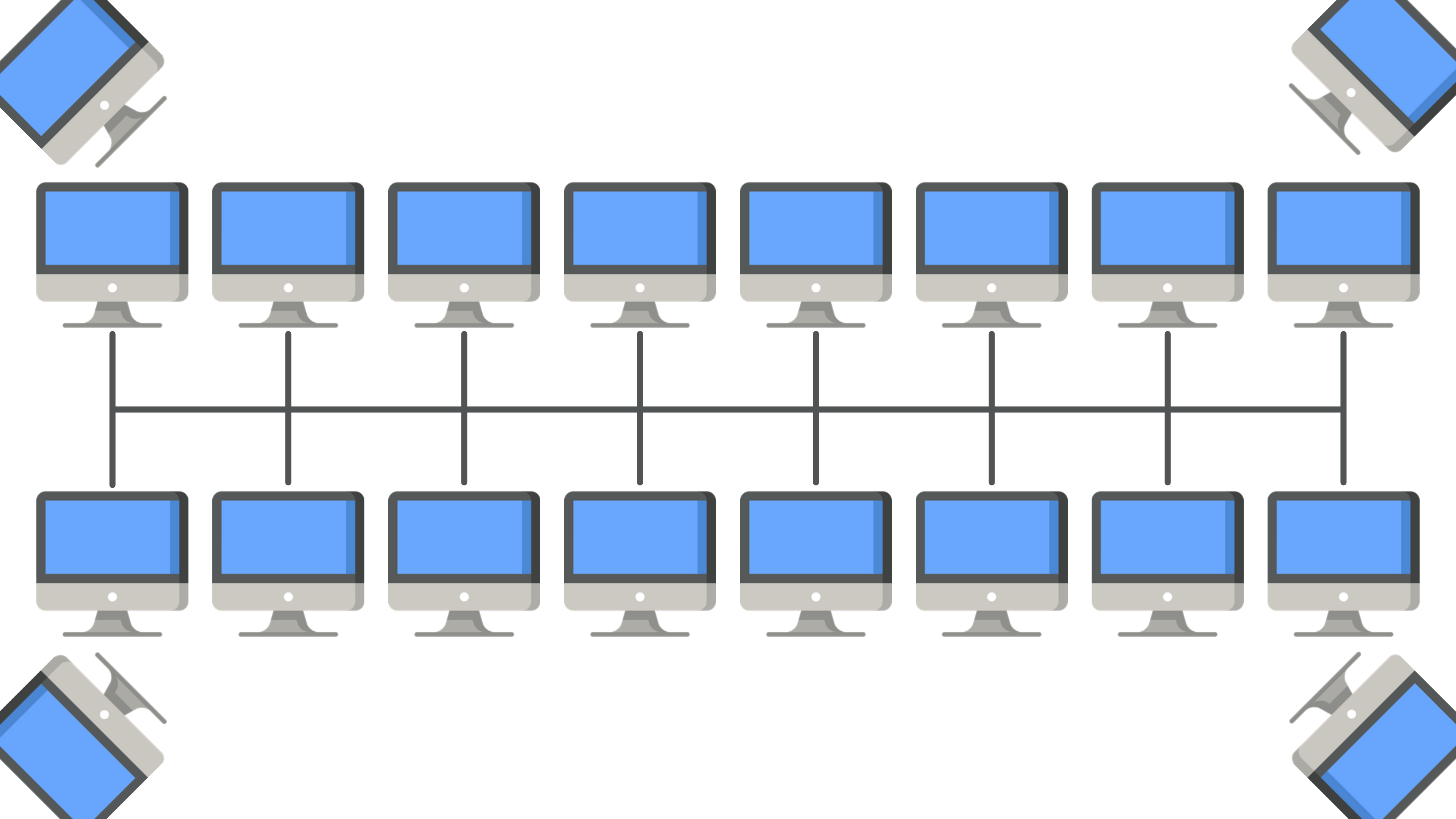
Problems:

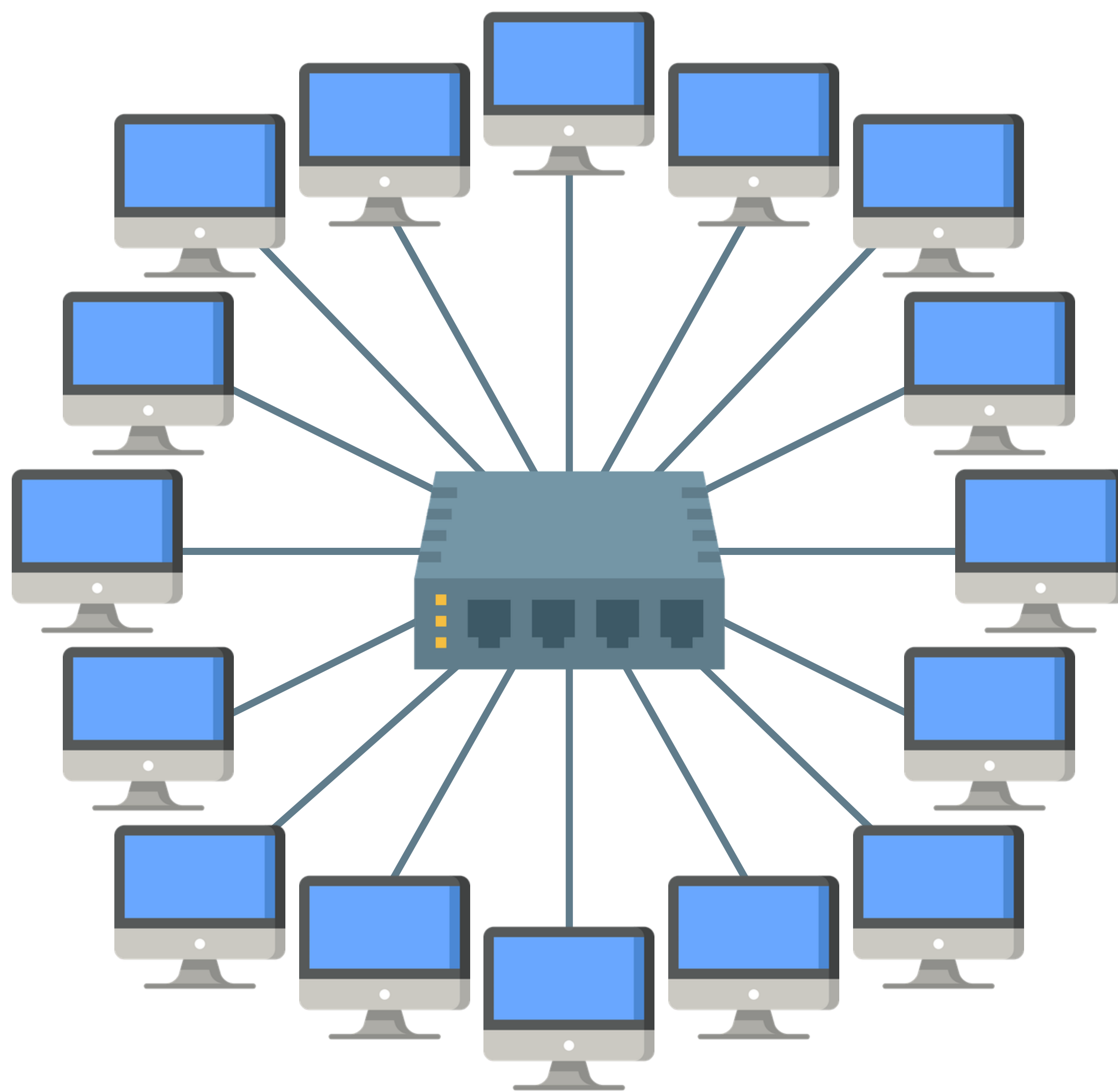
- **Collision** may occur..
- How would a device know **who is sending**, and how could it know if it is the intended **receiver**?

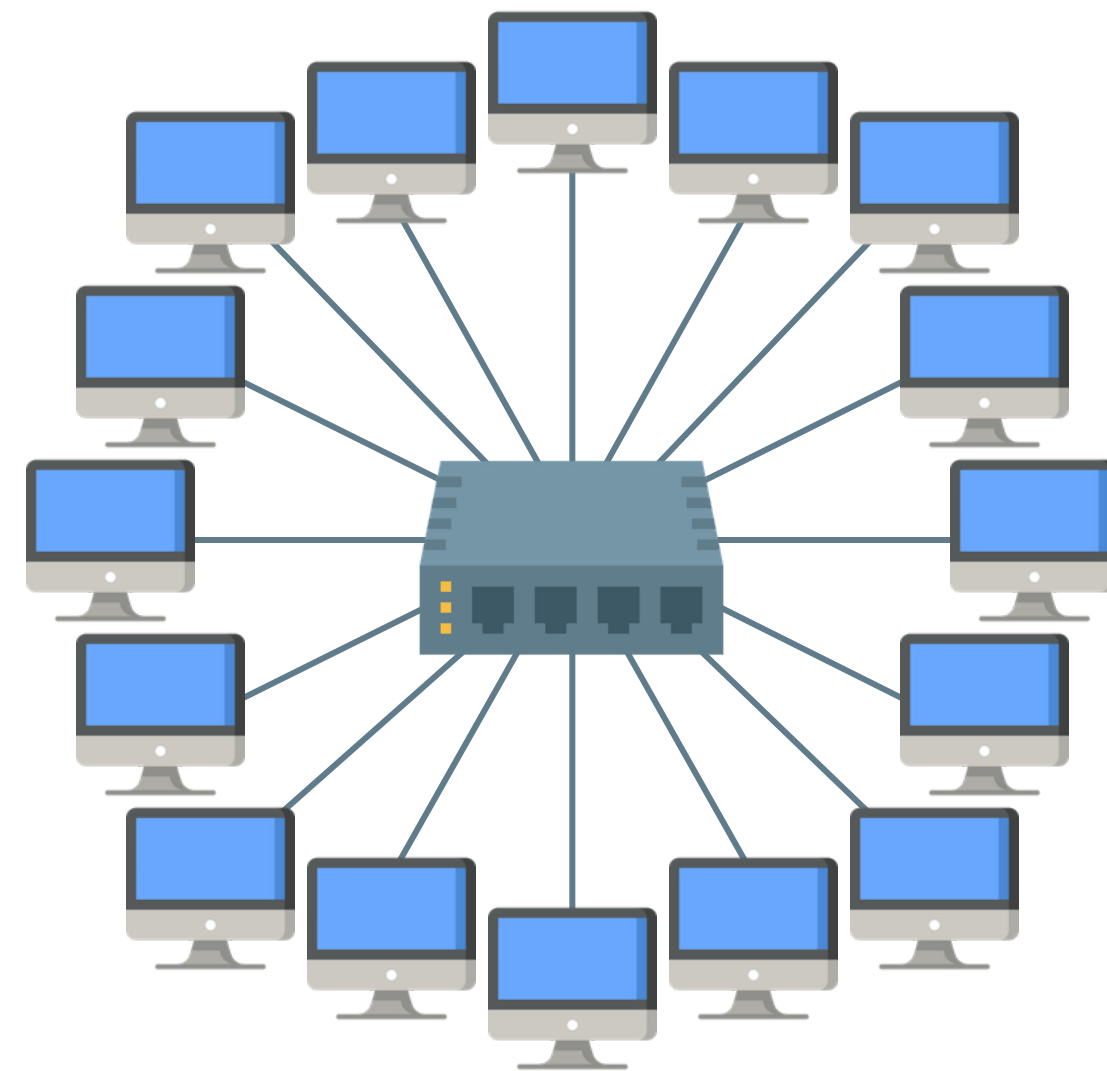
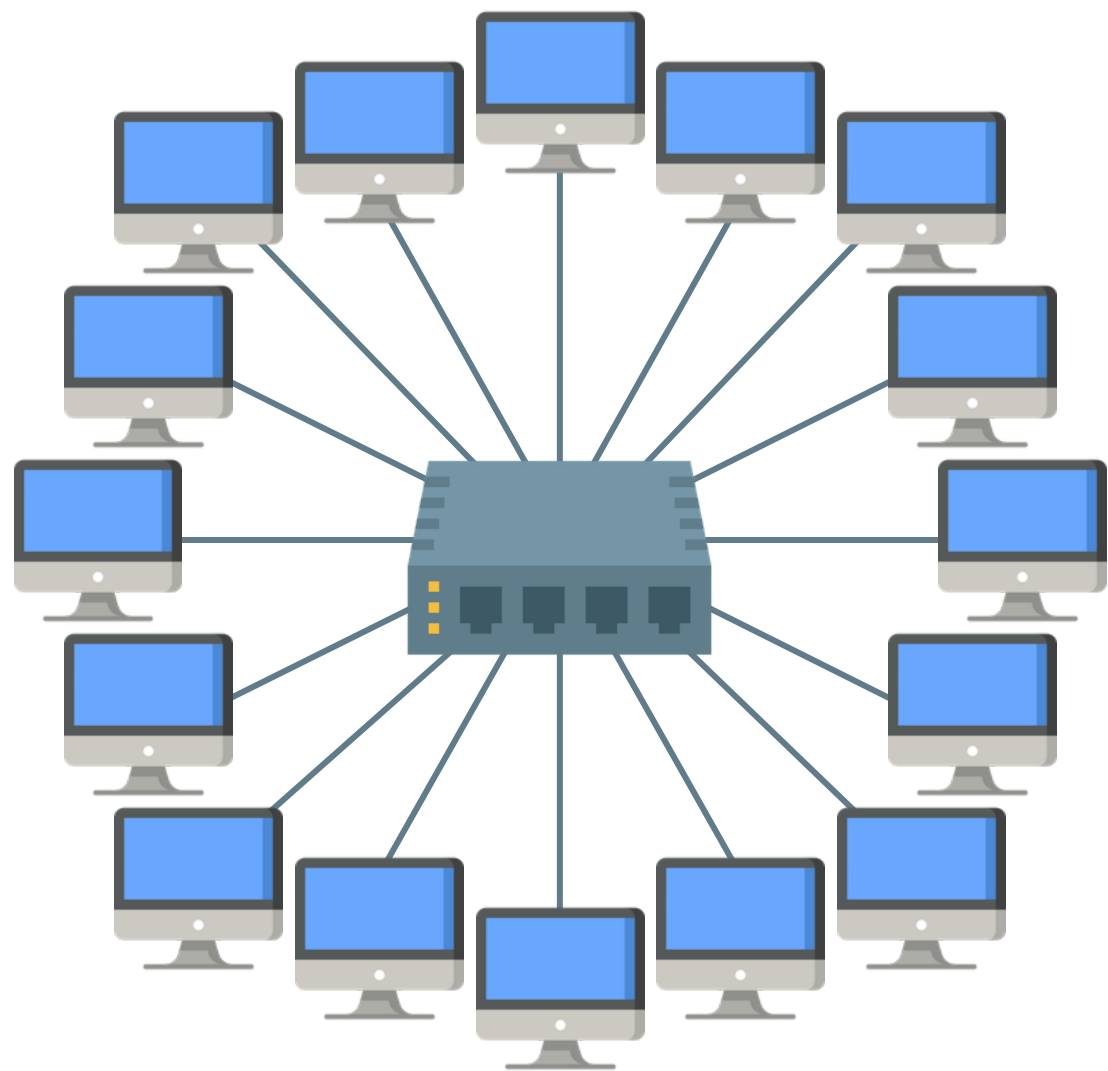
Solutions may be:

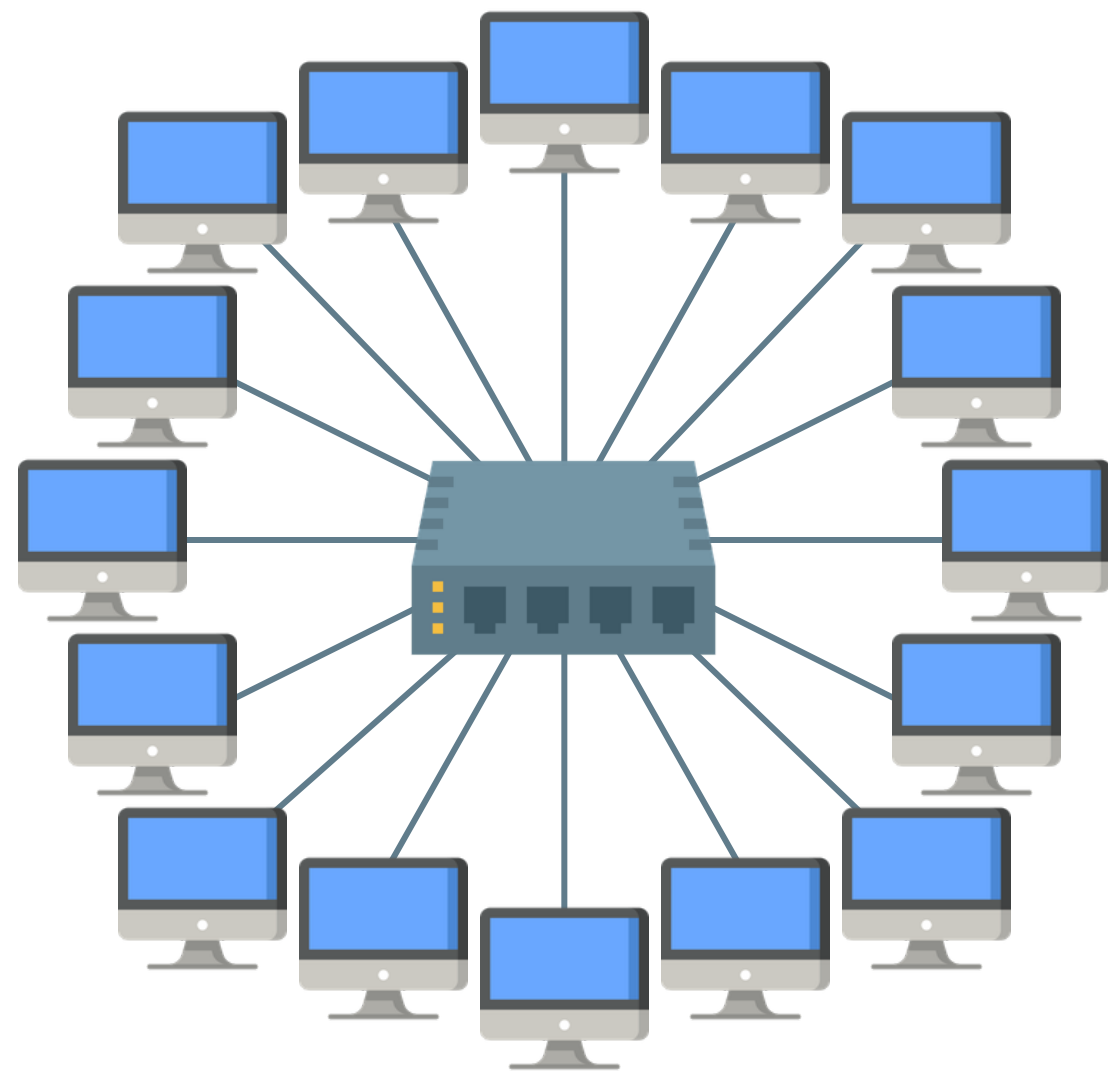
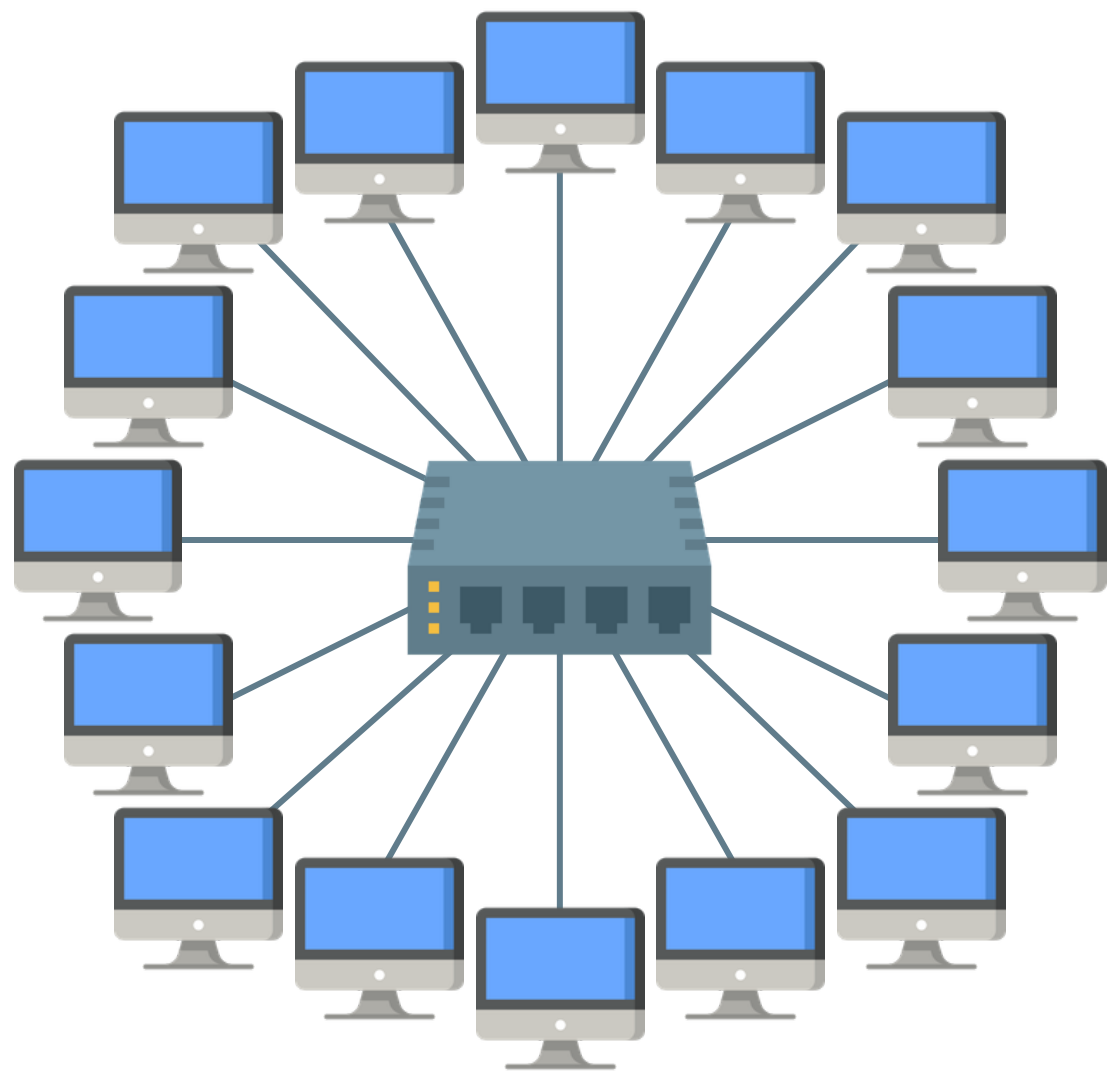
- Some sort of **synchronization** (or collision detection mechanism).
- Some sort of **addressing** (which would be added to the message).

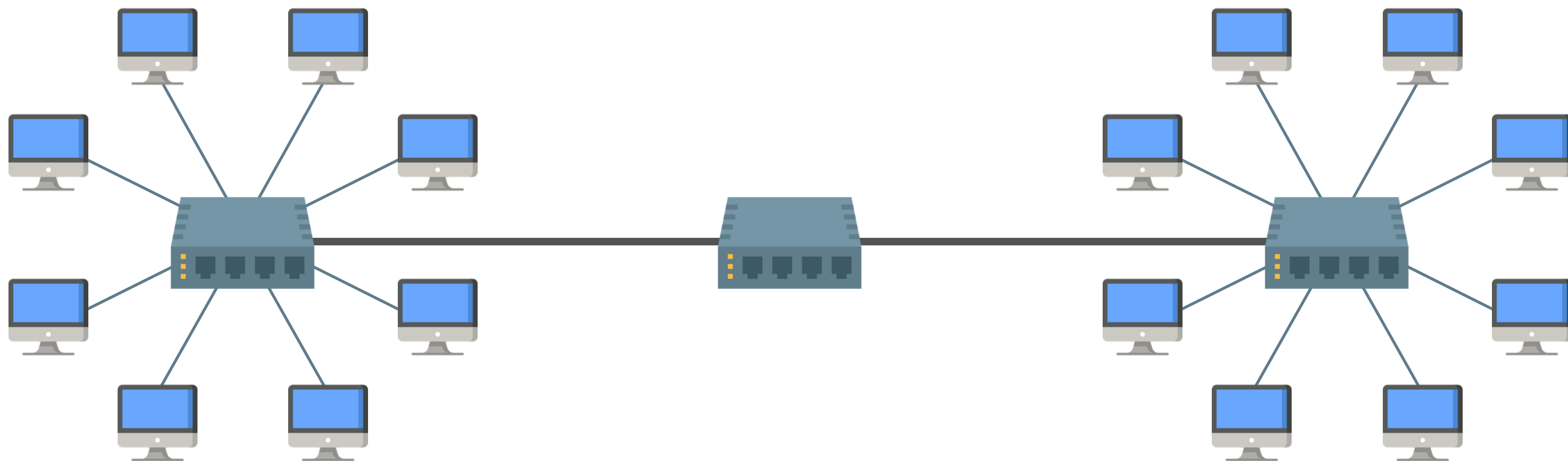






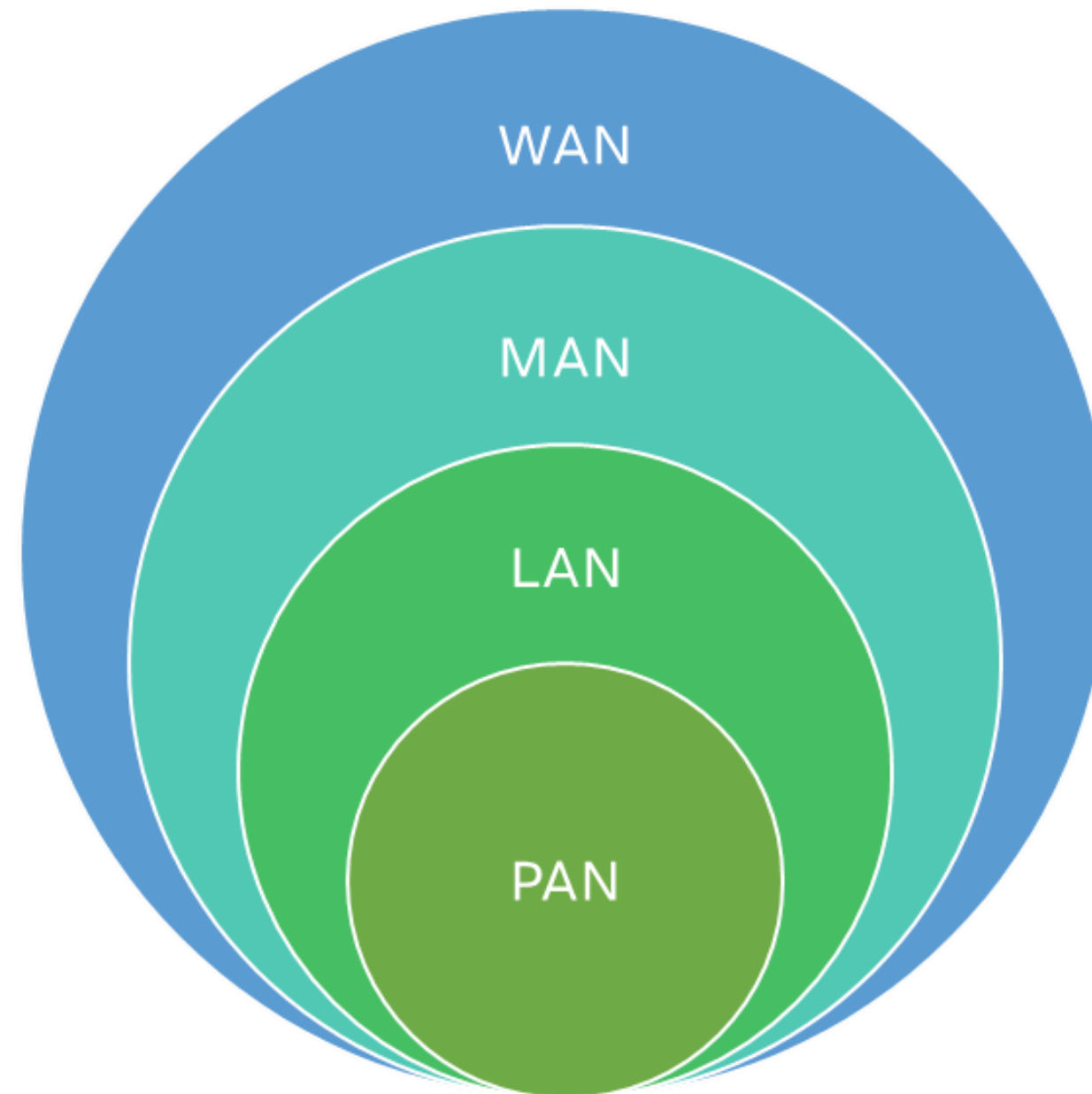


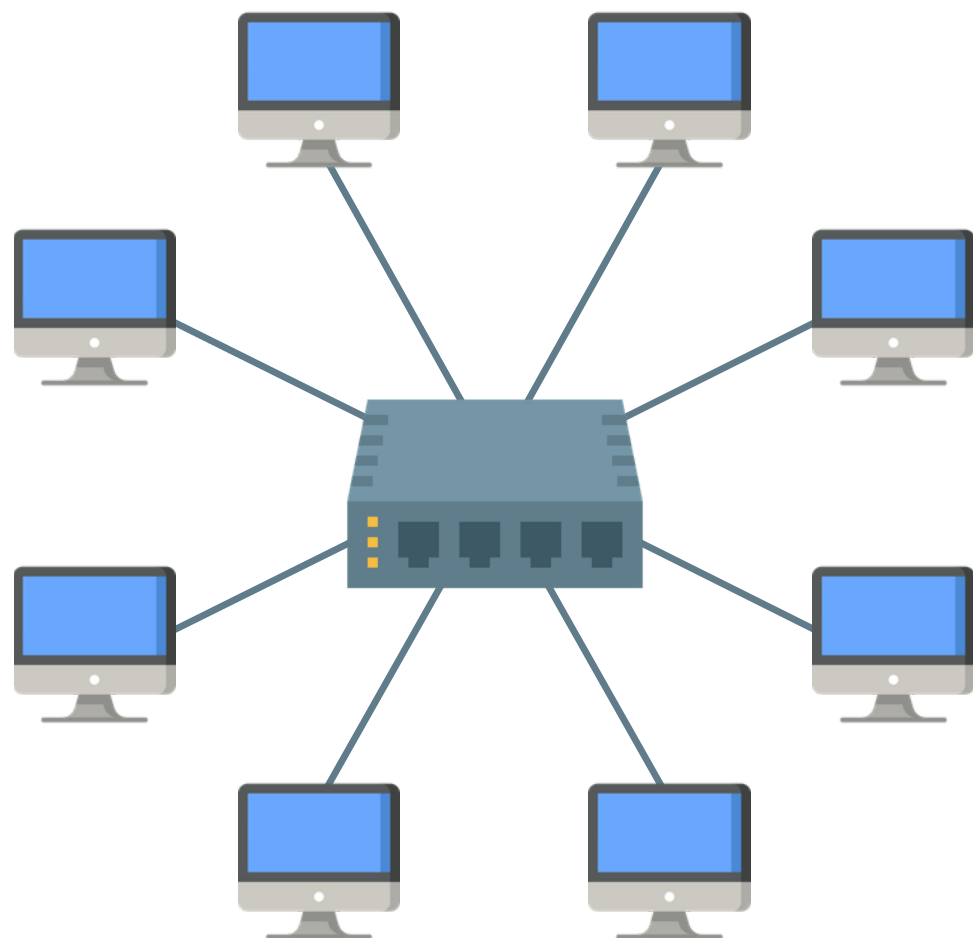




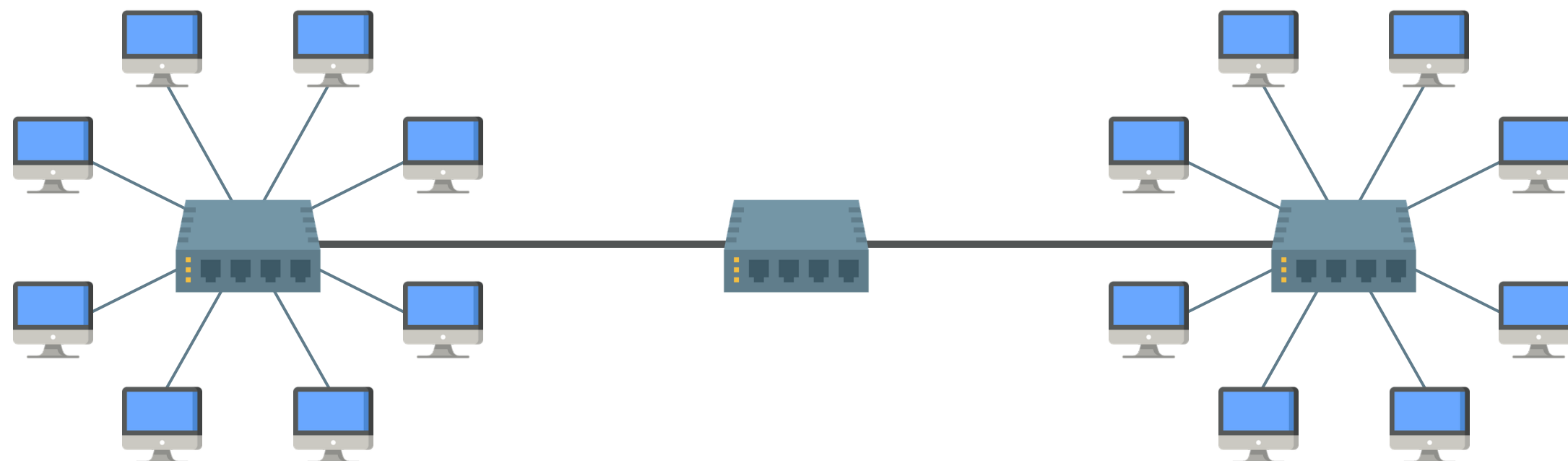
After connecting **different networks**, networks can come in **different sizes**

Network Types

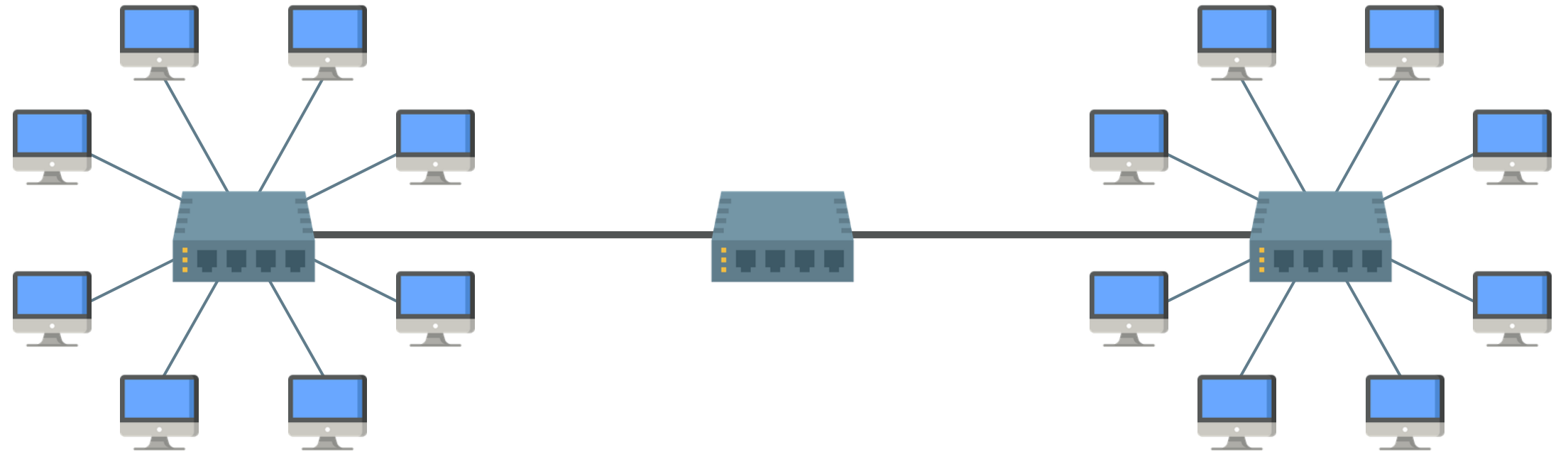




LAN



WAN



Our **Problem** now is that
scaling seems **hard**, and
maybe **inefficient**

History time

Some early networks

many independent networks started in different parts of the world with different drivers, from Europe to the US to Russia..

OGAS – ~1962, USSR

Proposed nationwide network for economic planning (never fully implemented).

NPL Network – 1967, UK

First practical packet-switching network (Donald Davies).

SAGE – 1958, USA

Real-time military radar/computer network.

CYCLADES – 1970, France (concept started late 1960s)

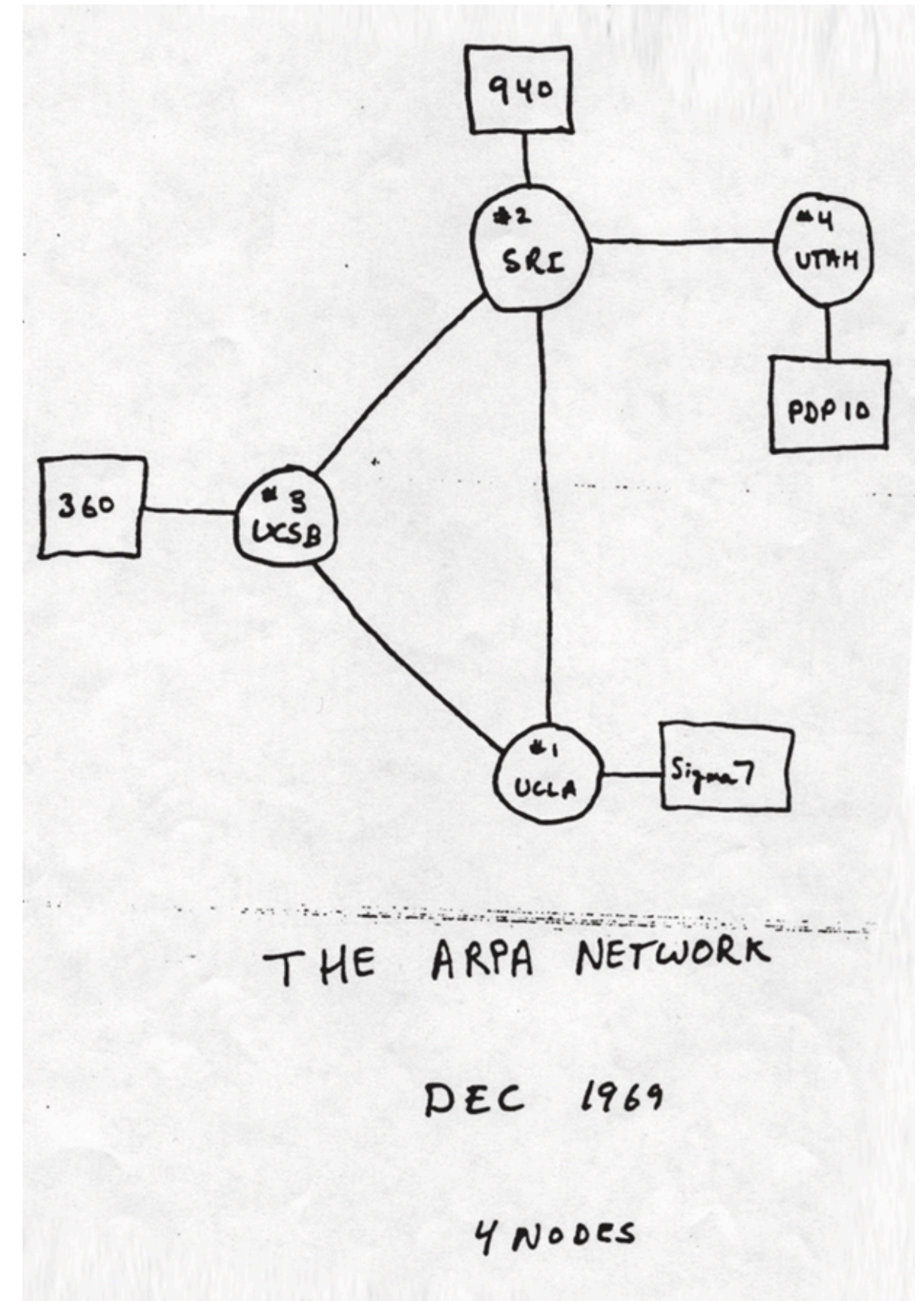
Introduced end-to-end principle; influenced TCP/IP design.

ARPANET

The precursor to the internet

The initial ARPANET configuration linked four main participants:

- UCLA
- SRI
- UCSB
- University of Utah School of Computing.



The **First message** ever sent via the **ARPANET**

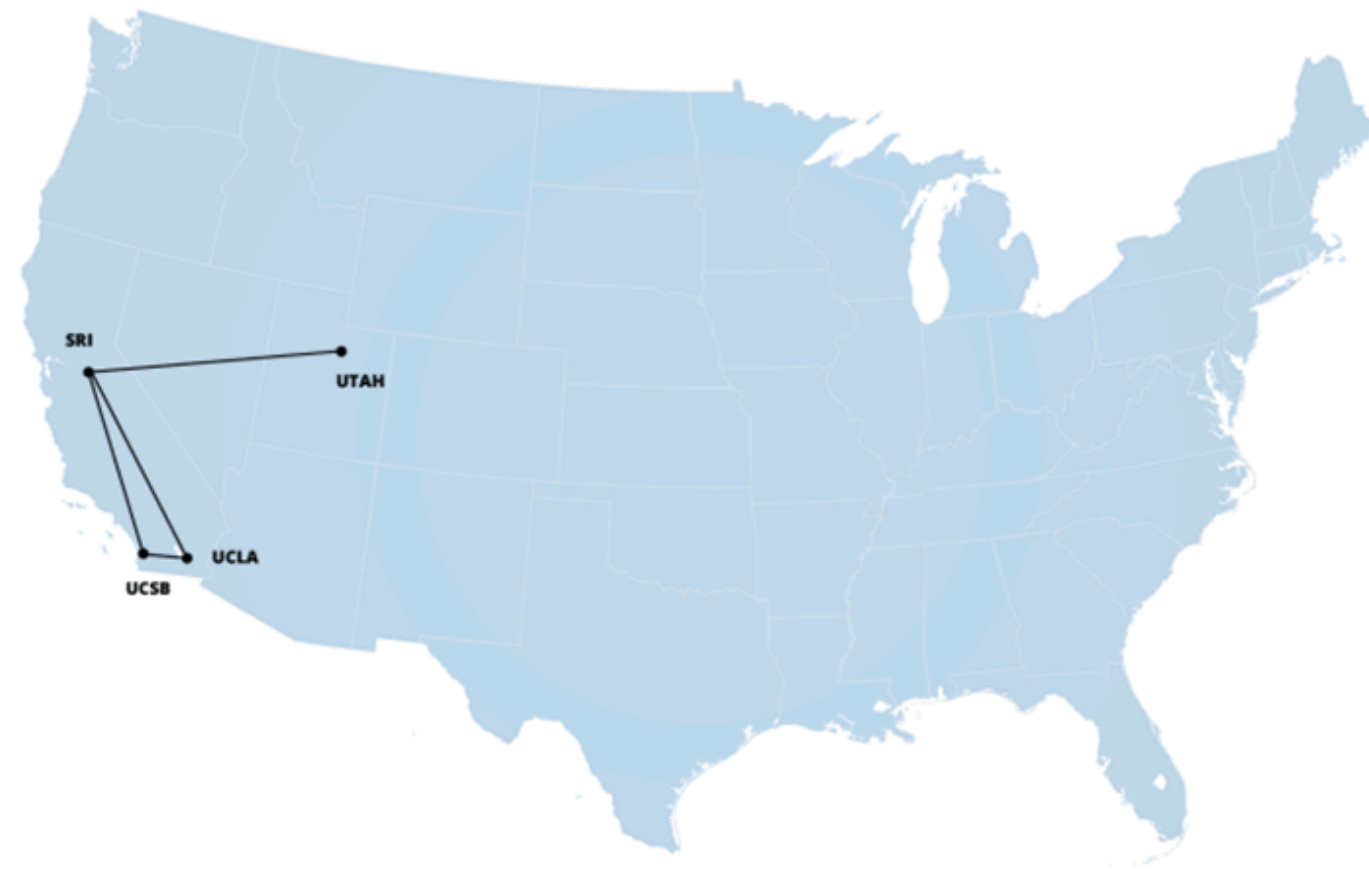
10:30 pm PST on
29 October 1969

29 OCT 69	2100	LOADED OP. PROGRAM	SK
		EOIC BEN BARKER	
		BBV	
	22:30	Talked to SRI	CSL
		Host to Host	
		Left op. program	CSL
		running after sending	
		a host dead message	
		to imp.	

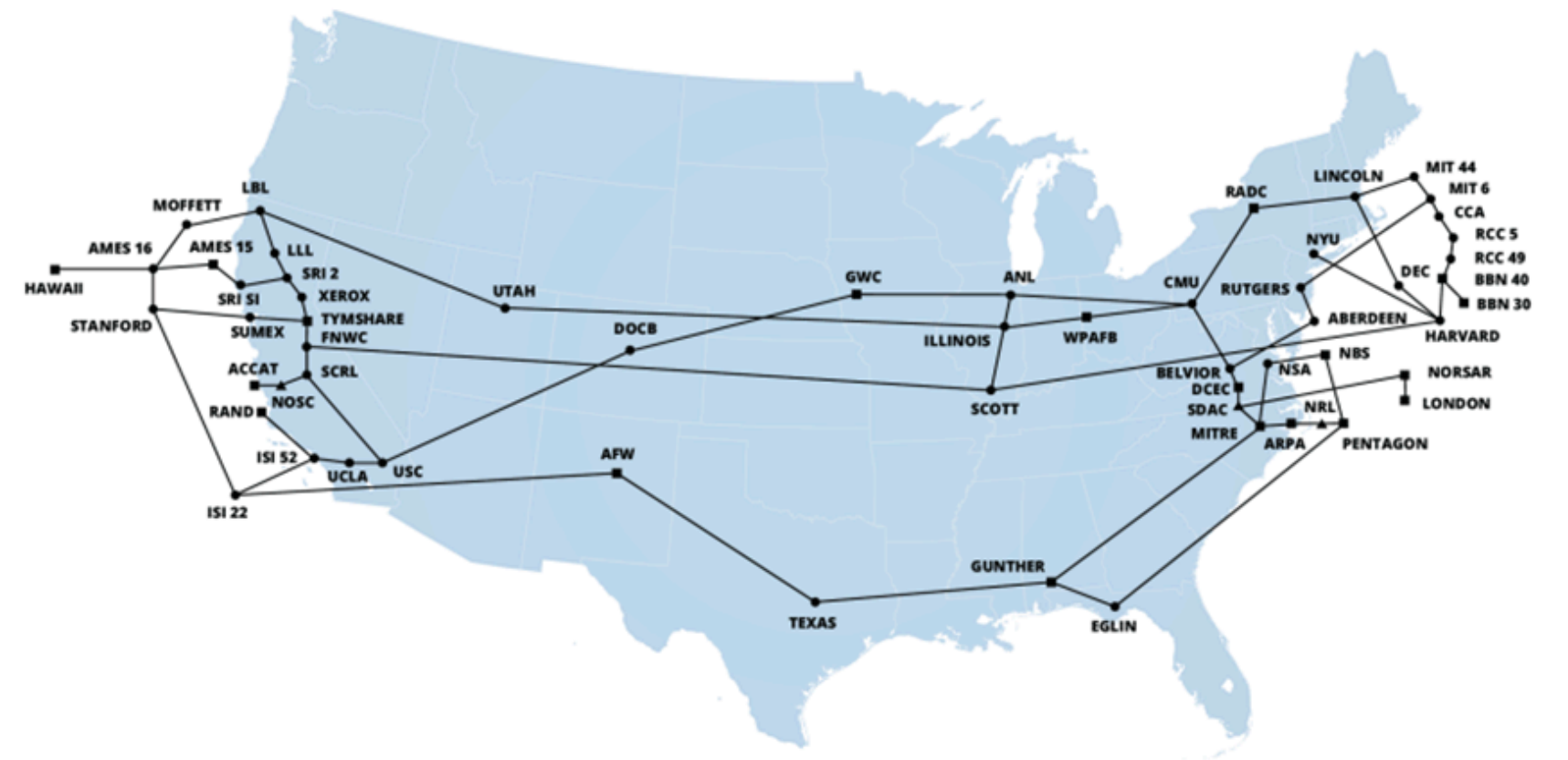
First ARPANET IMP log

This IMP Log excerpt, kept at UCLA, describes setting up a message transmission from the UCLA SDS Sigma 7 Host computer to the SRI SDS 940 Host computer

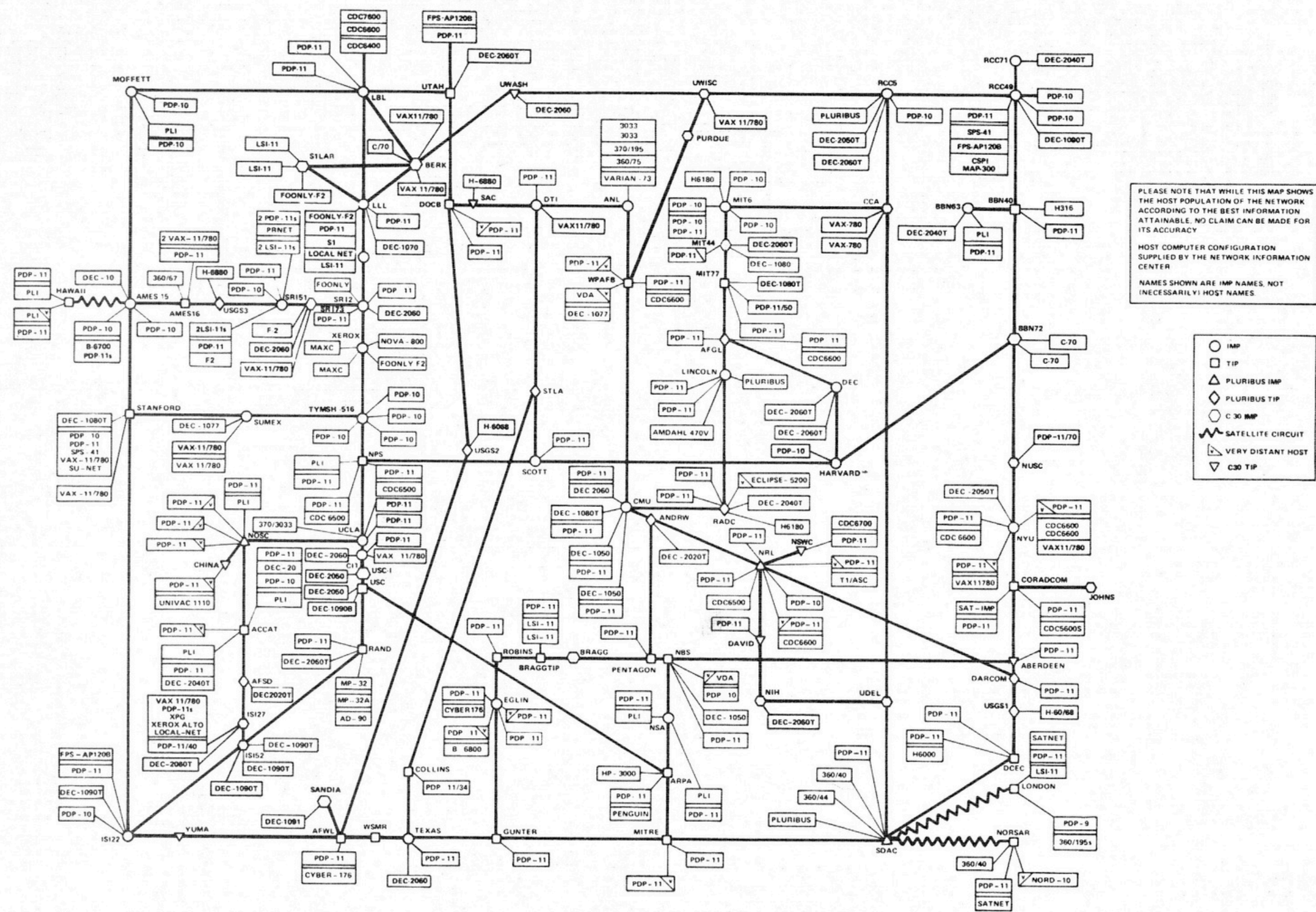
1969 ARPANET



1977 ARPANET



ARPANET LOGICAL MAP, DECEMBER 1981



vendor-developed with proprietary standards networks:

- IBM's **SNA**
- DEC's **DECnet**
- Xerox's **XNS**

government-sponsored networks:

- **ARPANET** in **US** (the DoD Darpa project)
- **CYCLADES** in **France**
- **EuroNet** in **EU**
- **NSFnet** in the **US**
- **NSL** network in **UK**

Too many Networks with
different Standards

To **connect** them, they, the
relevant authorities, needed to
agree on some standard

That produced
the **Protocol Wars.**

Protocol?

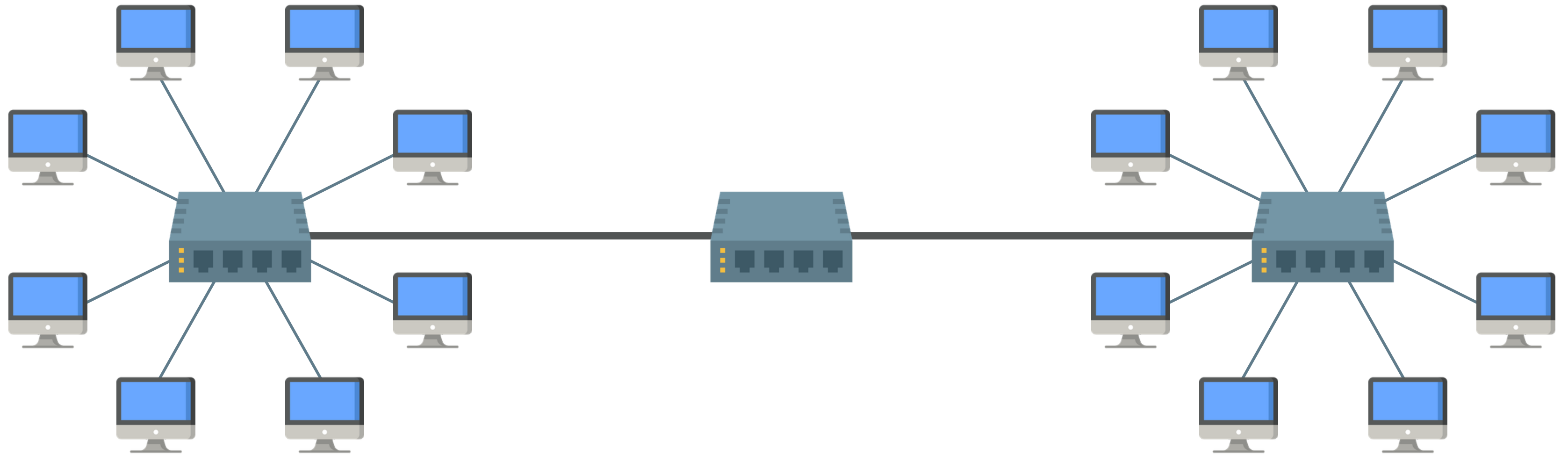
But, What is a **Protocol**?

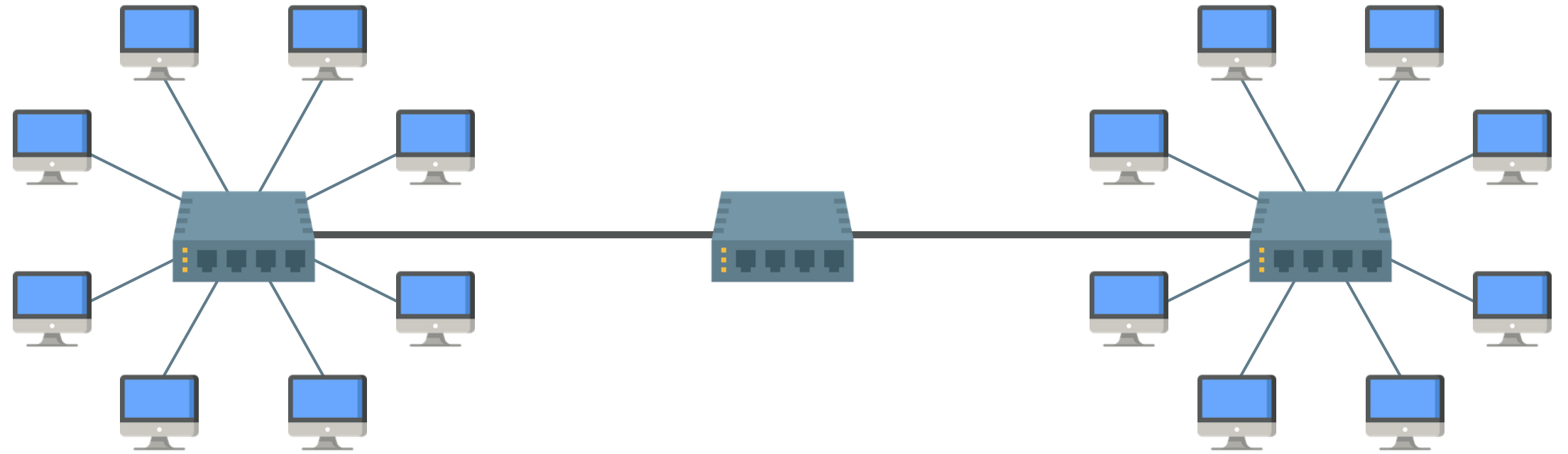
**A Protocol is an agreement on
how to communicate.**

Before knowing what the
Protocol Wars ended with, we
need to ask a **question**

What do we need now?

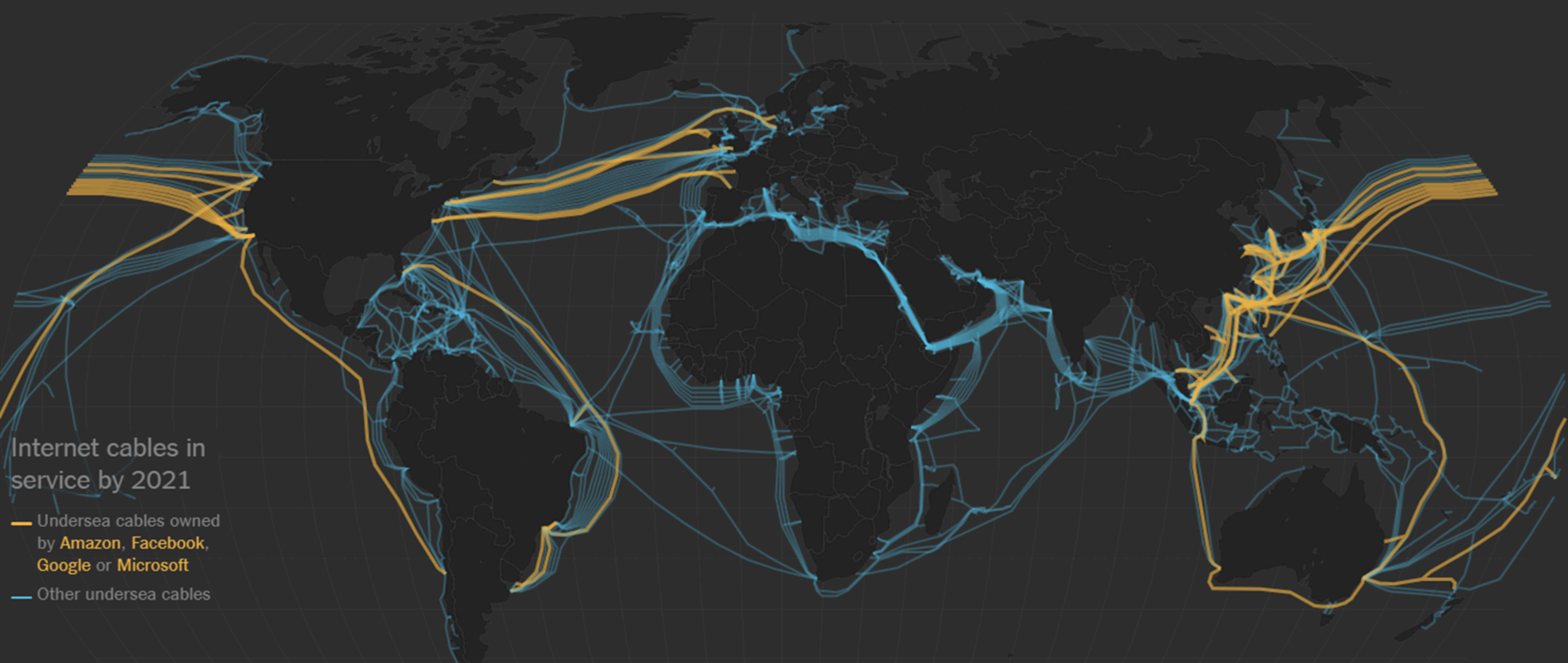
What do we need now?





We need to know **how to scale**
our network and **how to get**
from **one network** to **another**?

The Protocol Wars ended with
the following..



Internet cables in
service by 2021

- Undersea cables owned
by Amazon, Facebook,
Google or Microsoft
- Other undersea cables

OSI & Internet Protocol Suite

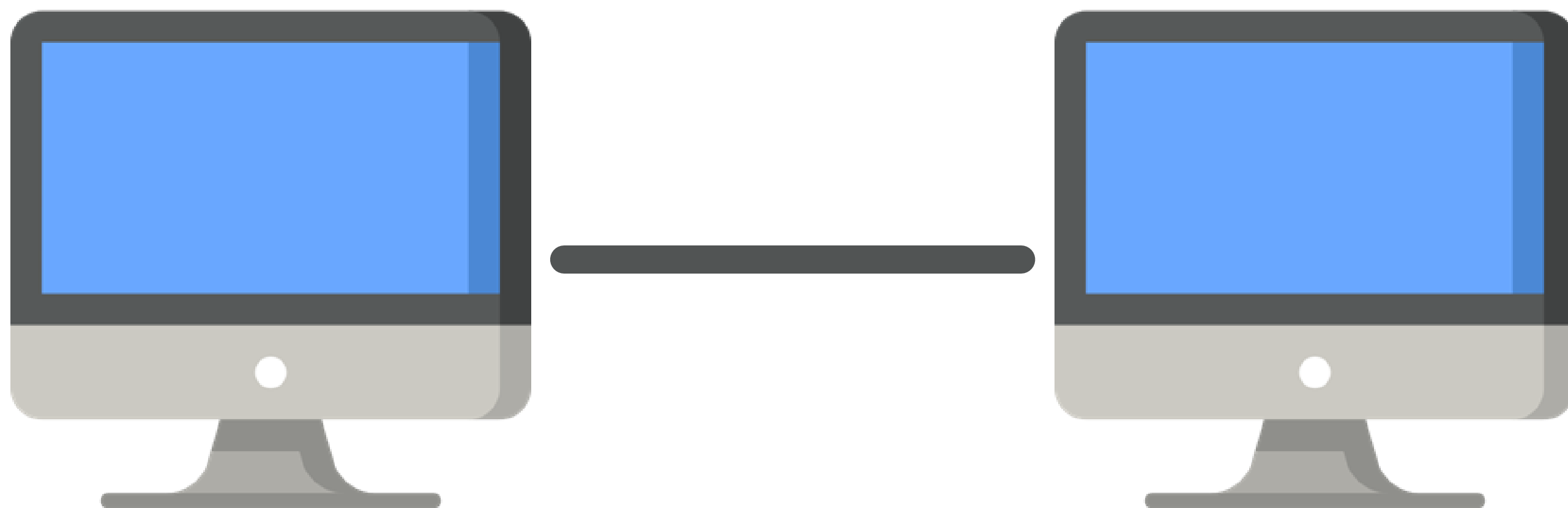
OSI Model	TCP/IP Model
Application Layer	Application layer
Presentation Layer	
Session Layer	
Transport Layer	Transport Layer
Network Layer	Internet Layer
Data link layer	Network Access
Physical layer	

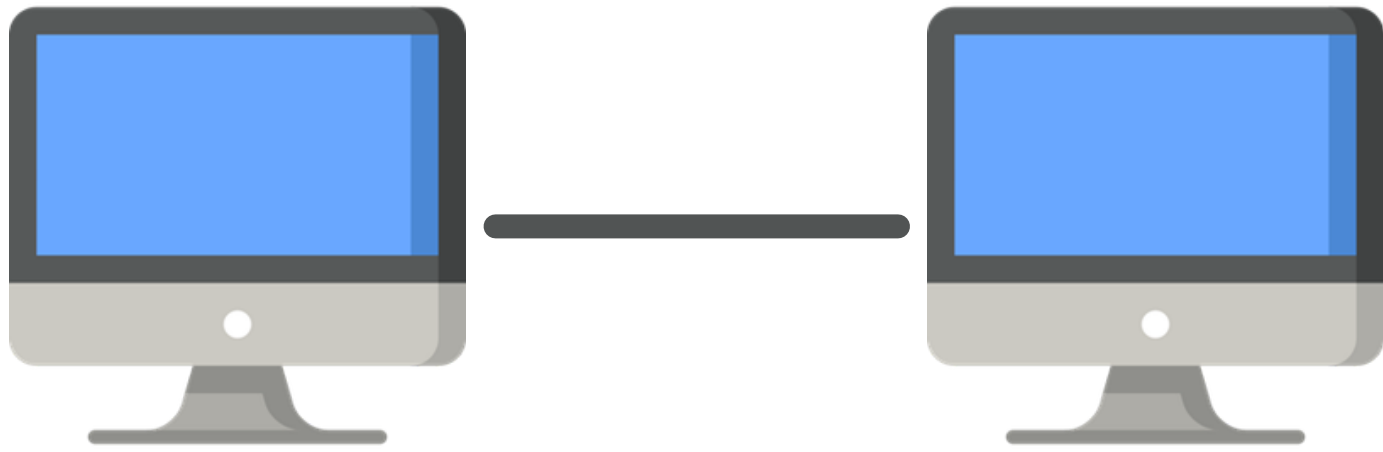
Why two models?

**The OSI Reference Model offers a
great layering functions scheme,
and its layering numbering is the
one used in industry,
while the TCP/IP Model is the one
implemented.**

We'll walk through the **layers**, but, we'll
need to explain some **critical concepts**
first.

Remember the **network** we made?







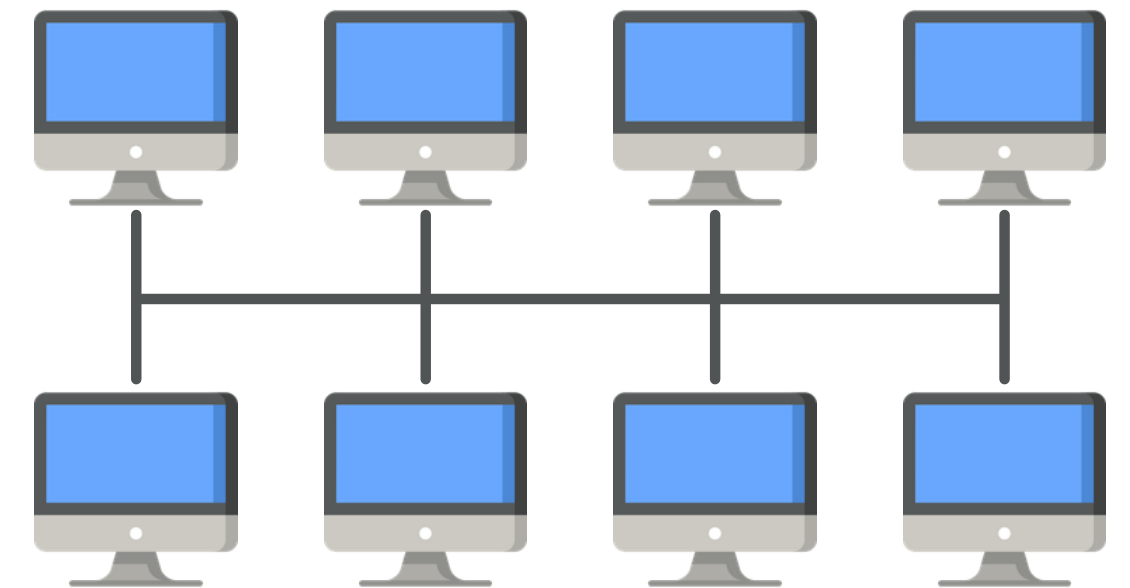
Problems:

- **Collision** may occur..
- How would a device know **who is sending**, and how could he know if he is the **intended receiver**?

Solutions may be:

- Some Sort of **synchronization** (or collision detection mechanism).
- We need some sort of **addressing** (which would be added later to the message).

Remember that

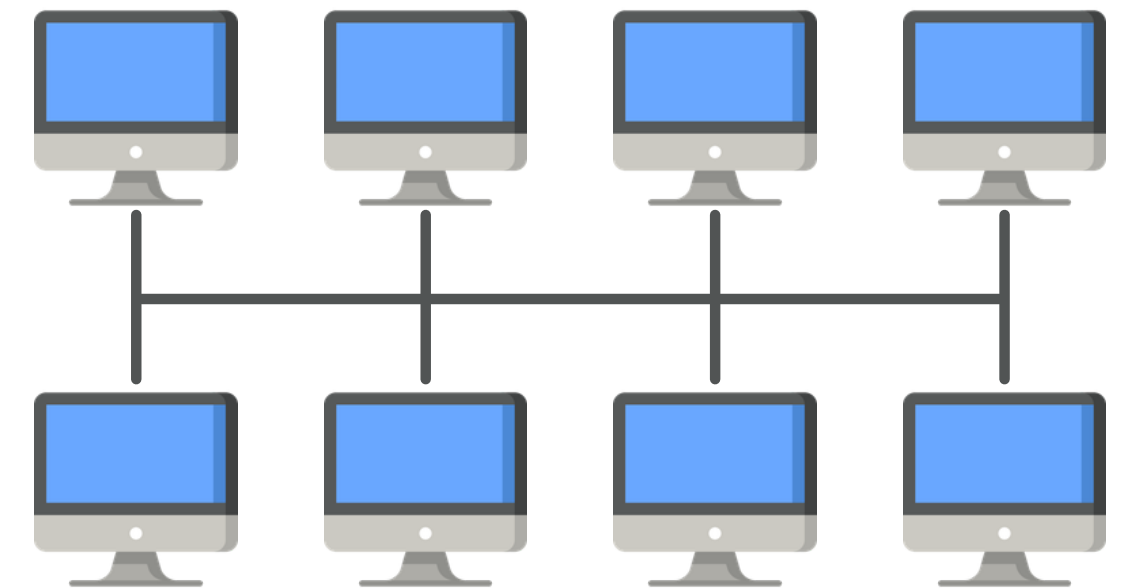


Problems:

Remember that

- **Collision** may occur..
- How would a device know **who is sending**, and how could he know if he is the **intended receiver**?

Real Solutions actually are:



- using **CSMA/CD, CSMA/CA, ...** for collision detection and avoidance.
- using, what is now called, **MAC Address** and sending it with our data.

MAC Address

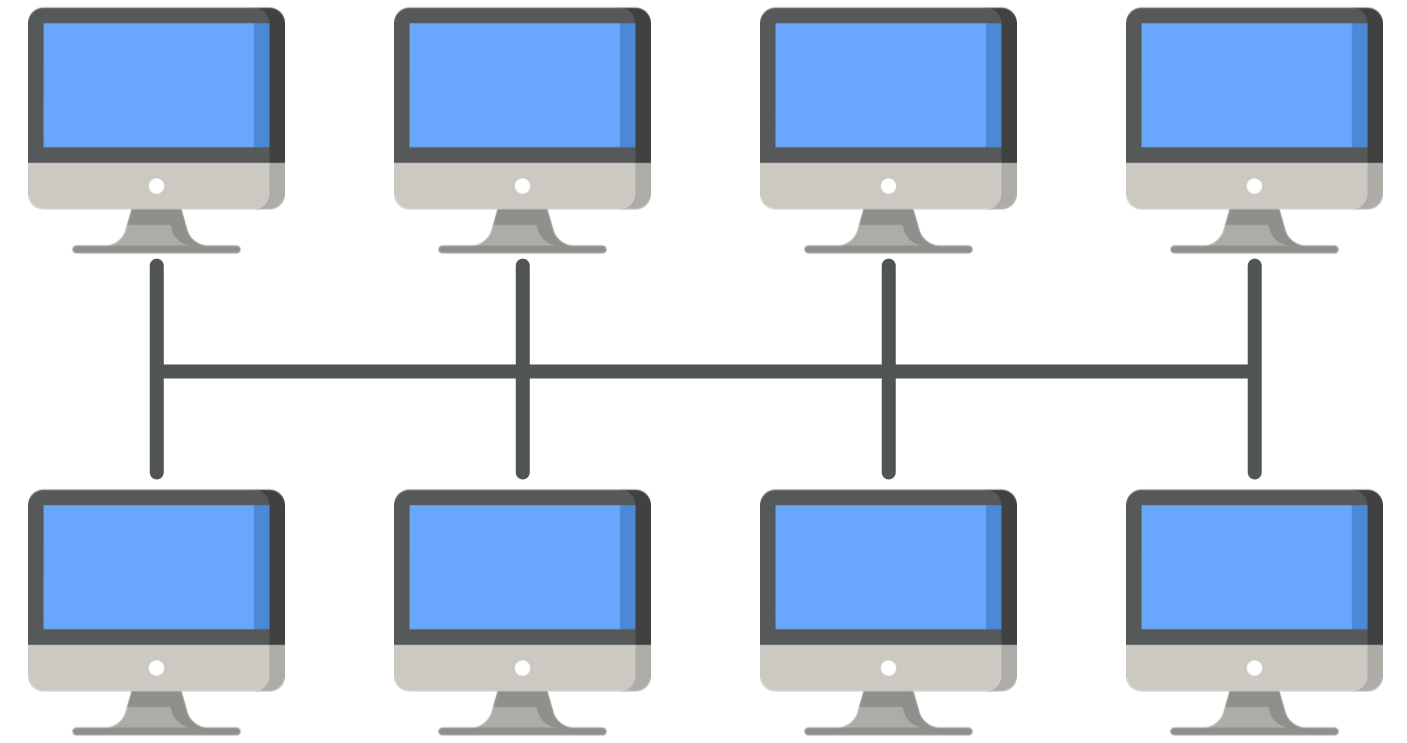
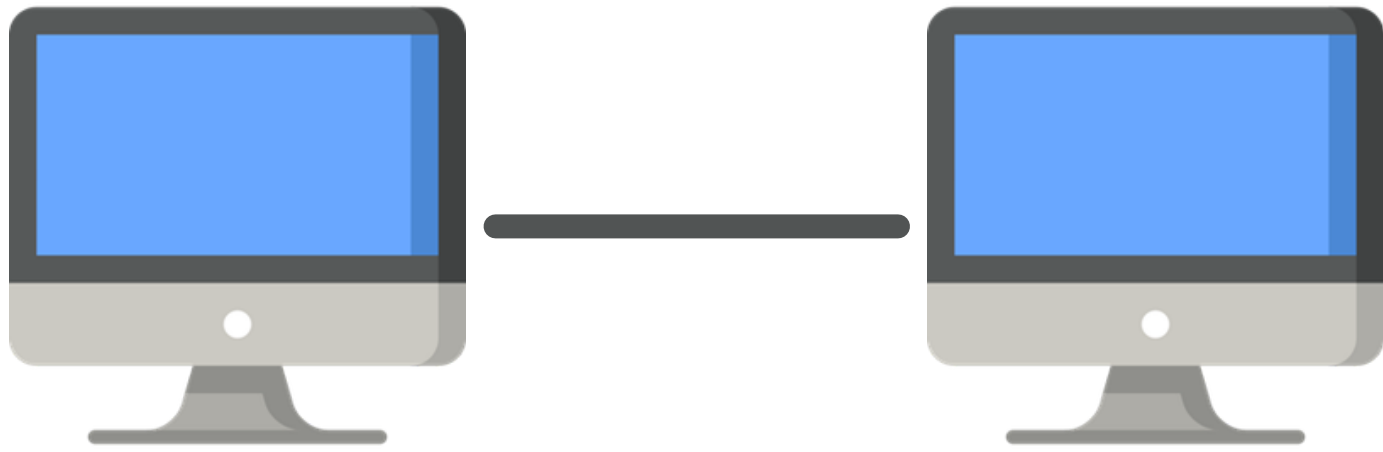
4e:d5:77:5a:c7:c9

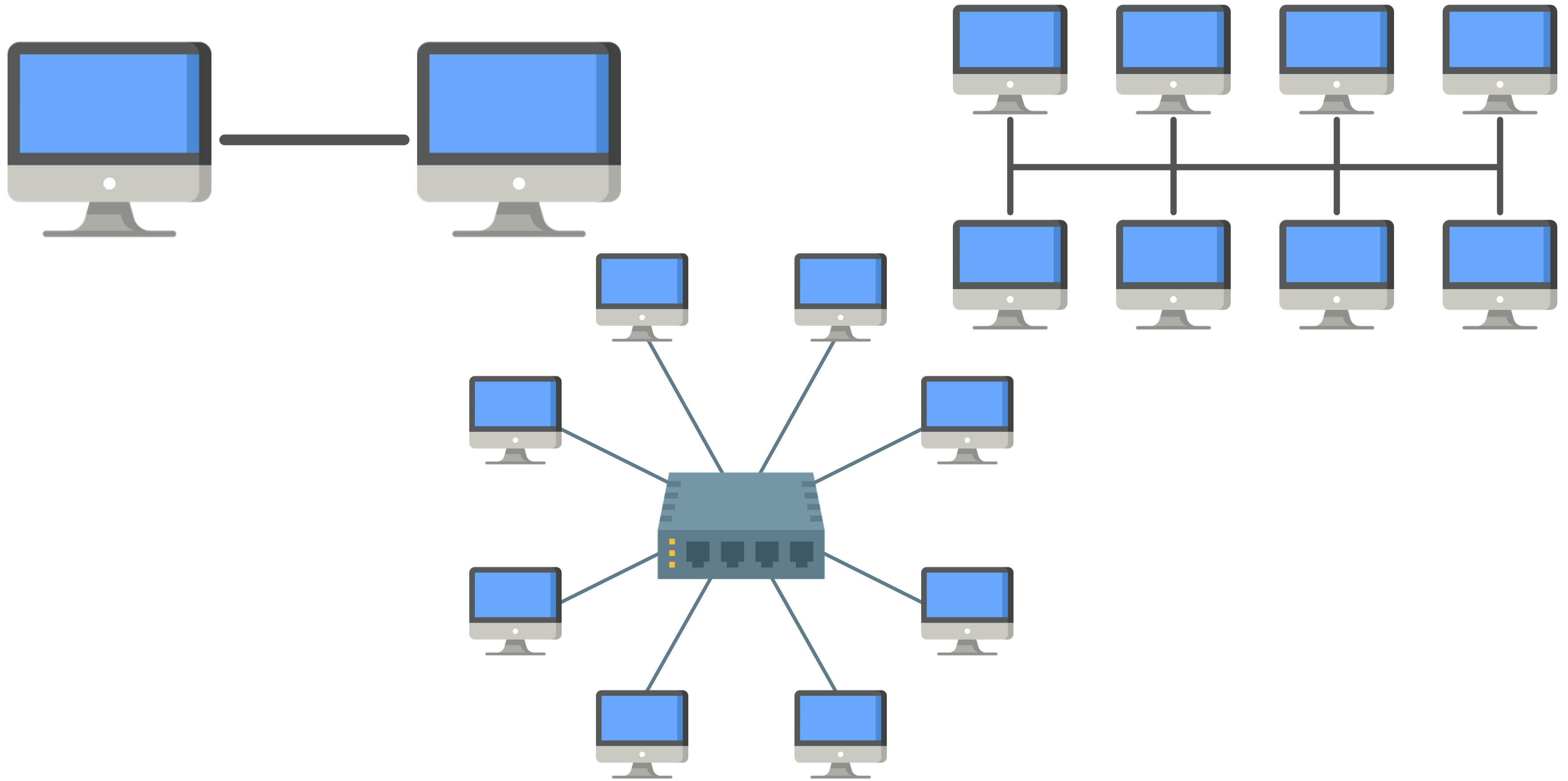
MAC Address

4e:d5:77 5a:c7:c9

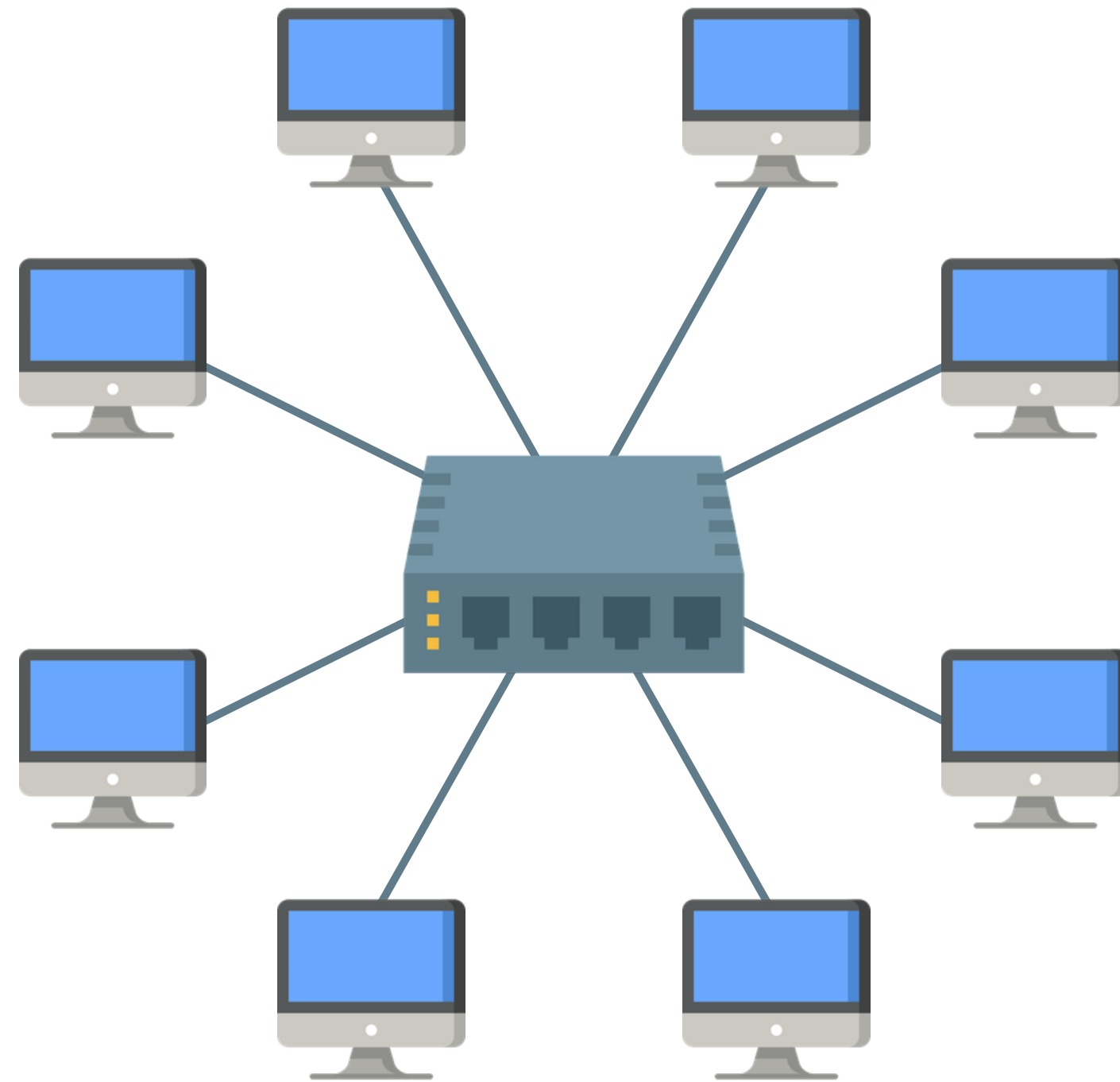
OUI

NIC





LAN





Packet Switch



Packet Switch

A **Packet Switch** is a **Layer 2 device** that receives Ethernet frames and forwards them intelligently based on the destination **MAC address**. It maintains a MAC address table (or **CAM table**) to map ports to devices, enabling efficient data transmission within a **LAN**.



Packet Switch

A Packet?

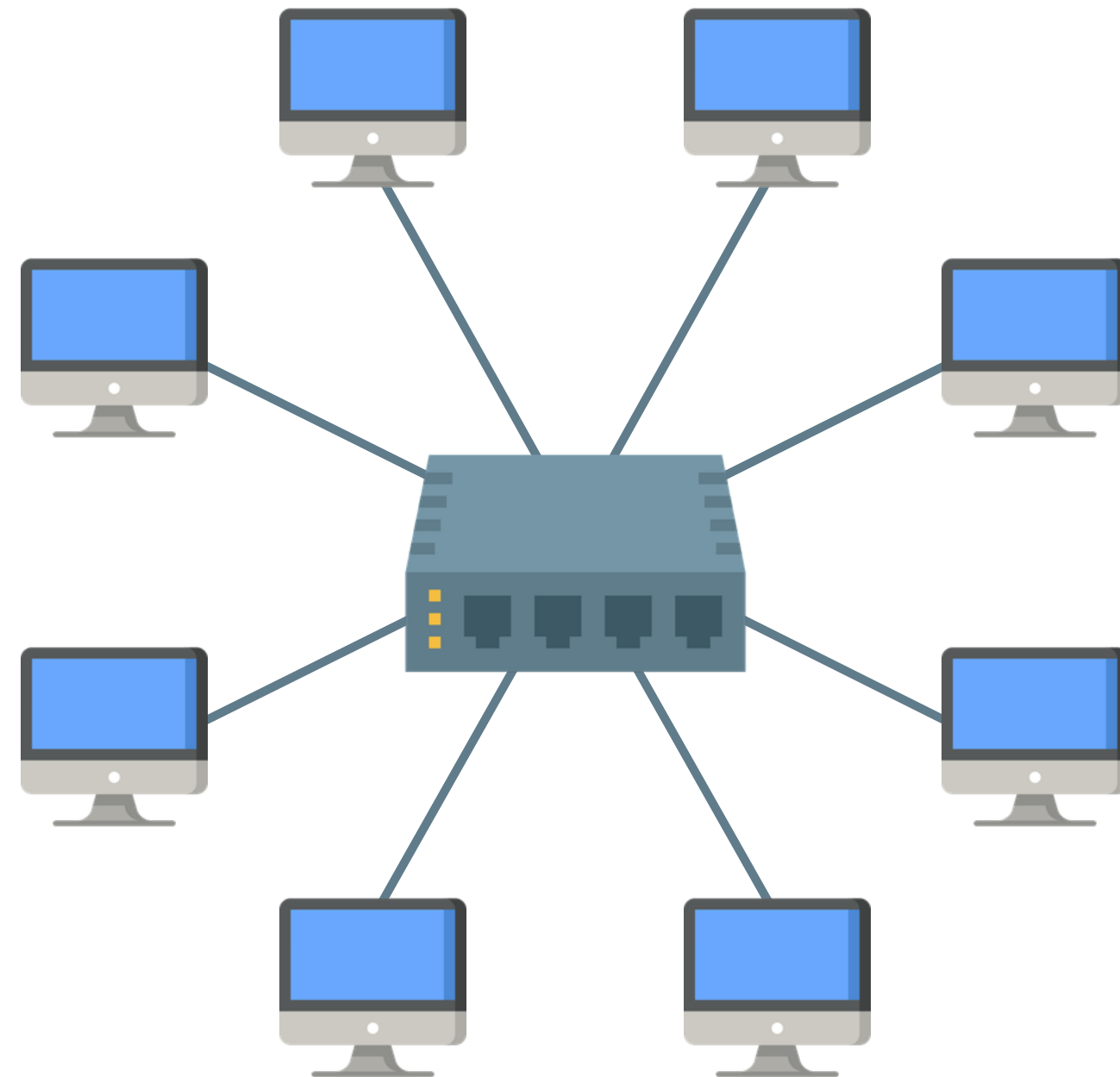
Switching Techniques

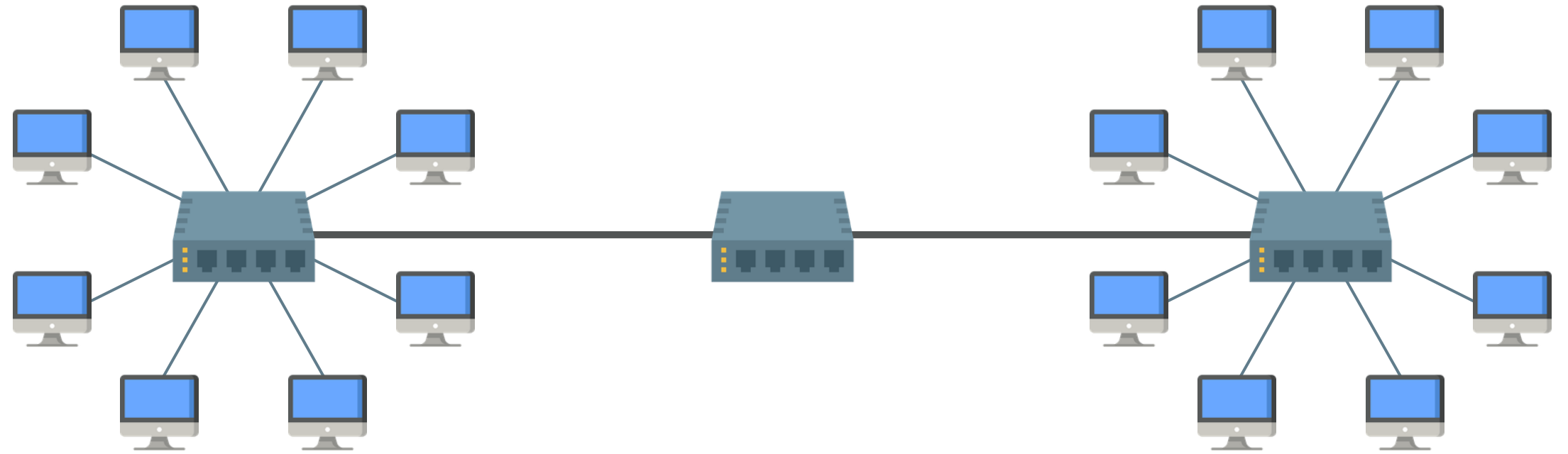
Packet Switching

Circuit Switching

Message Switching

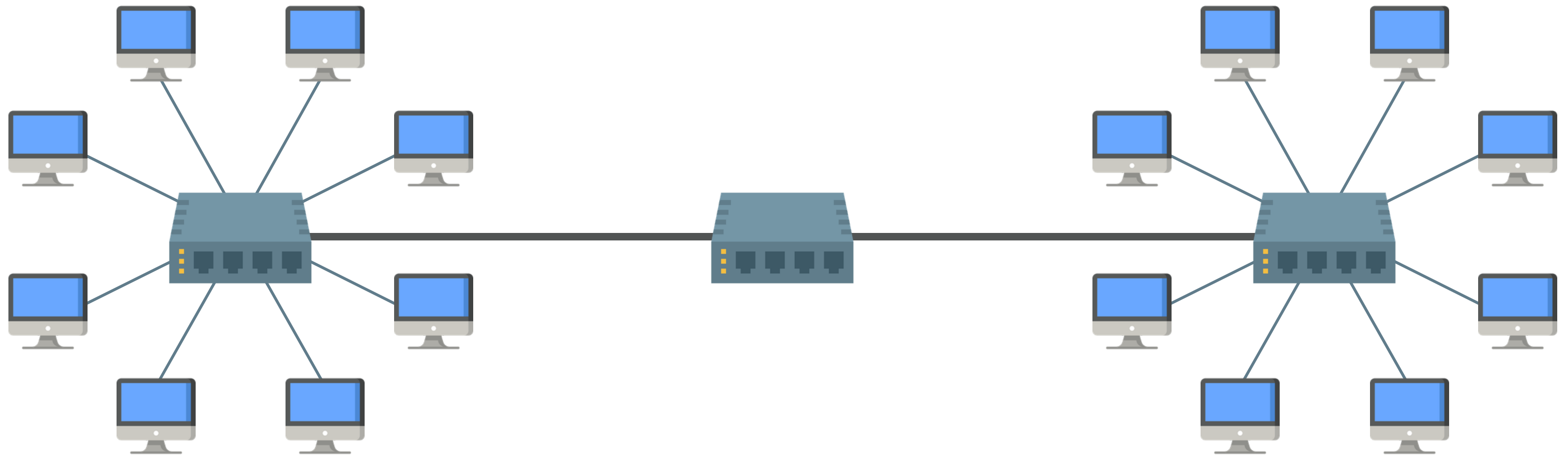
What about the **scaling** **problem?**



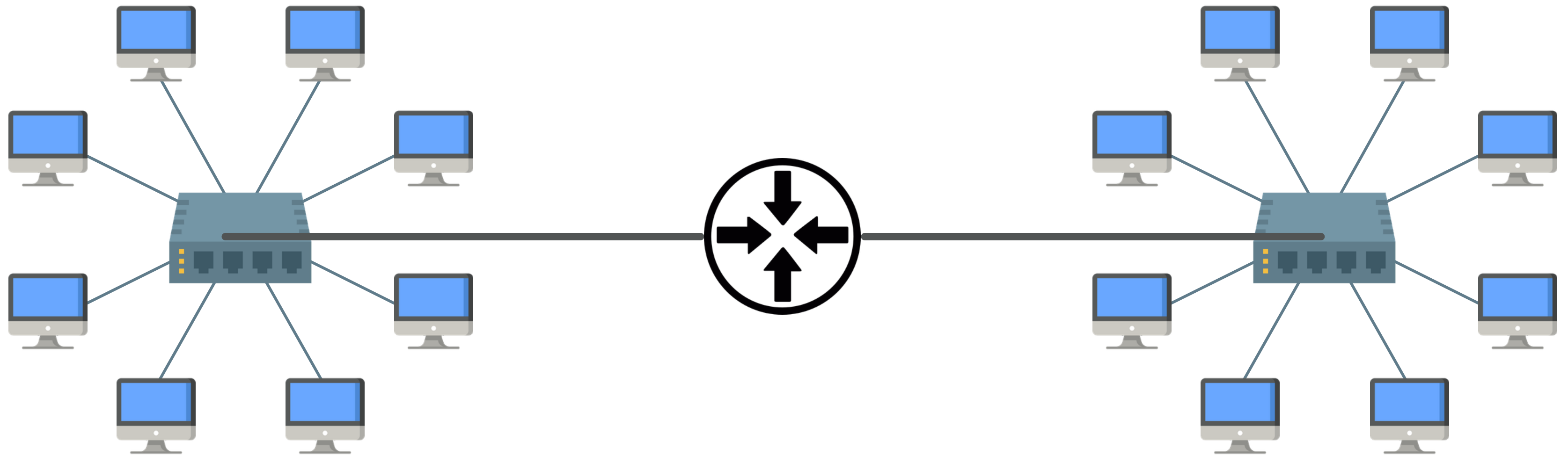


We need to know **how to scale**
our network and **how to get**
from **one network** to **another**?

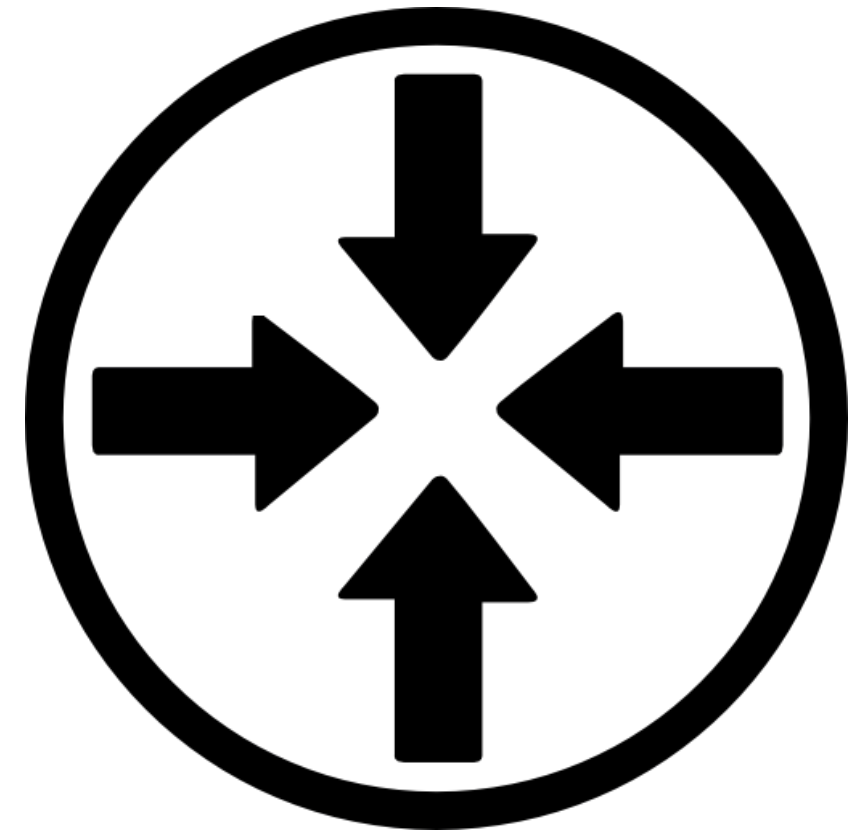
Connecting networks



Connecting networks

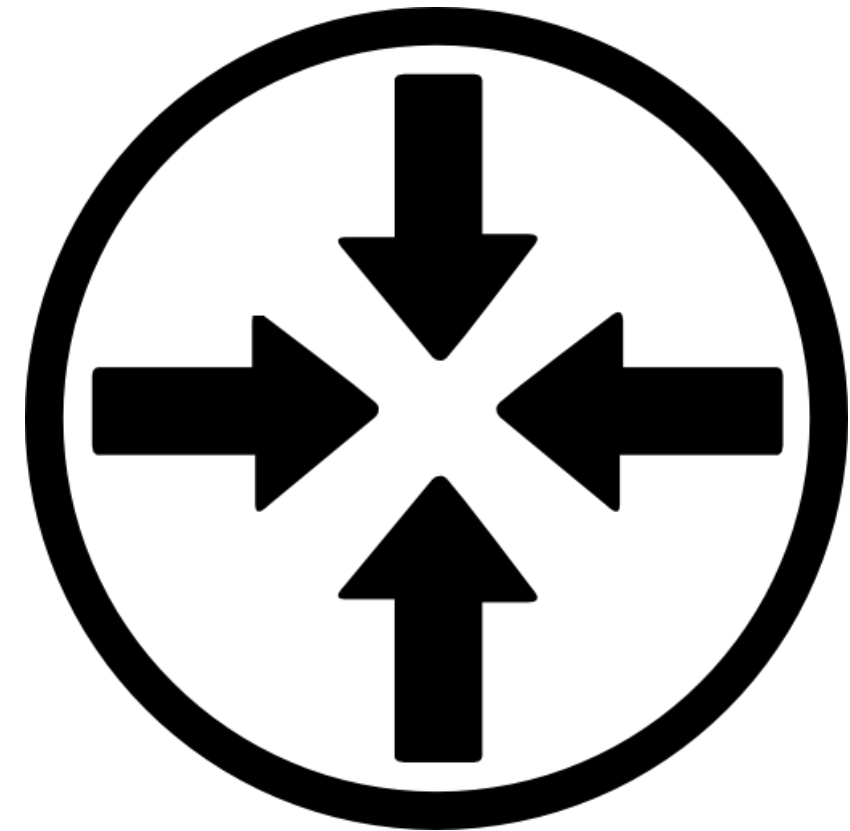


Hi, I'm a Router



A Router is used to connect
networks

Hi, I'm a Router



Hi, I'm a Router



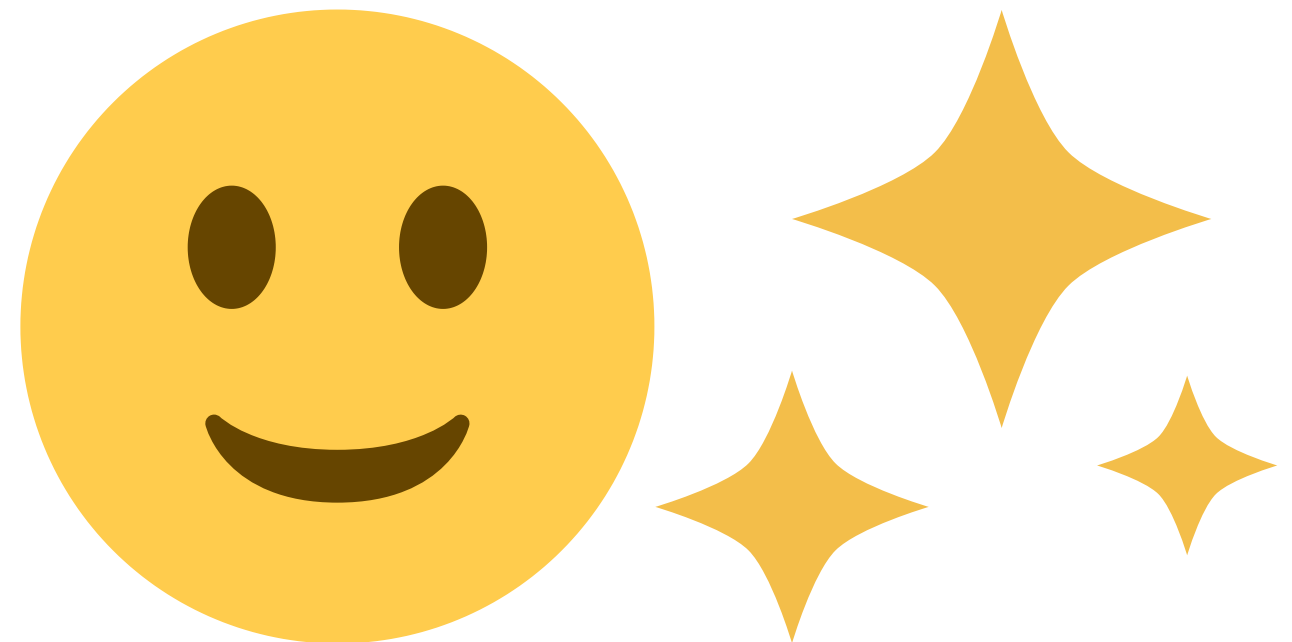
A **Router** is a **Layer 3** device that **connects multiple networks** and directs data packets based on their **IP addresses**. It determines the **optimal path** for packet delivery using **routing tables** and **protocols**, enabling communication between distinct networks such as LANs and the Internet.



OKAAAAYY

FOR NOW

**maybe you
need a break!**



Let's Get Back!

Let's Recap!

General Concepts:

- Network **Topologies**
- Network **Types** (i.e. LAN)
- Some **history**
- **OSI & TCP/IP** Model
- **MAC** Address
- **Switch**
- **Packet Switching**
- **Router**

IPs and Routing

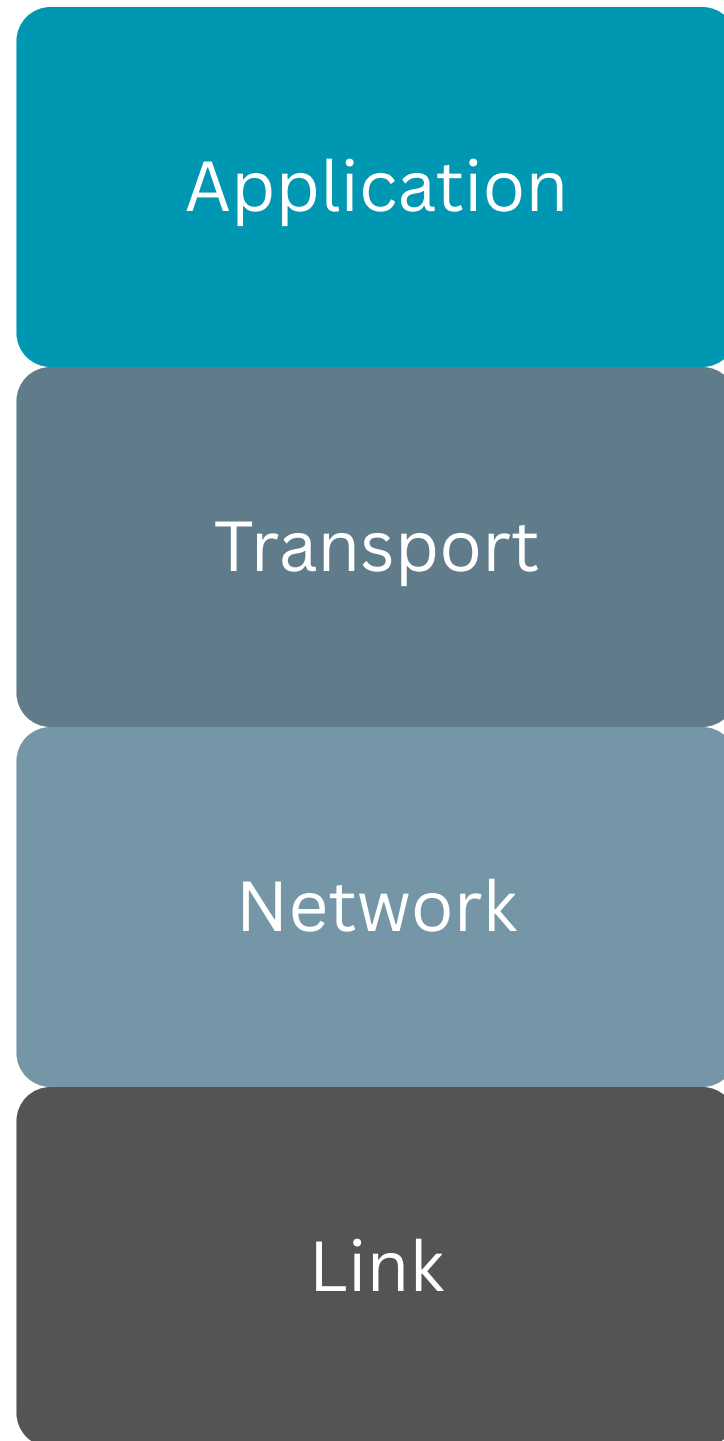
Back to the **TCP/IP Model**

TCP/IP Model Layers	Protocols
Application Layer	HTTP, FTP, SSH, Telnet
Transport Layer	TCP, UDP
Internet Layer	IP
Network Access	Ethernet, Wi-Fi

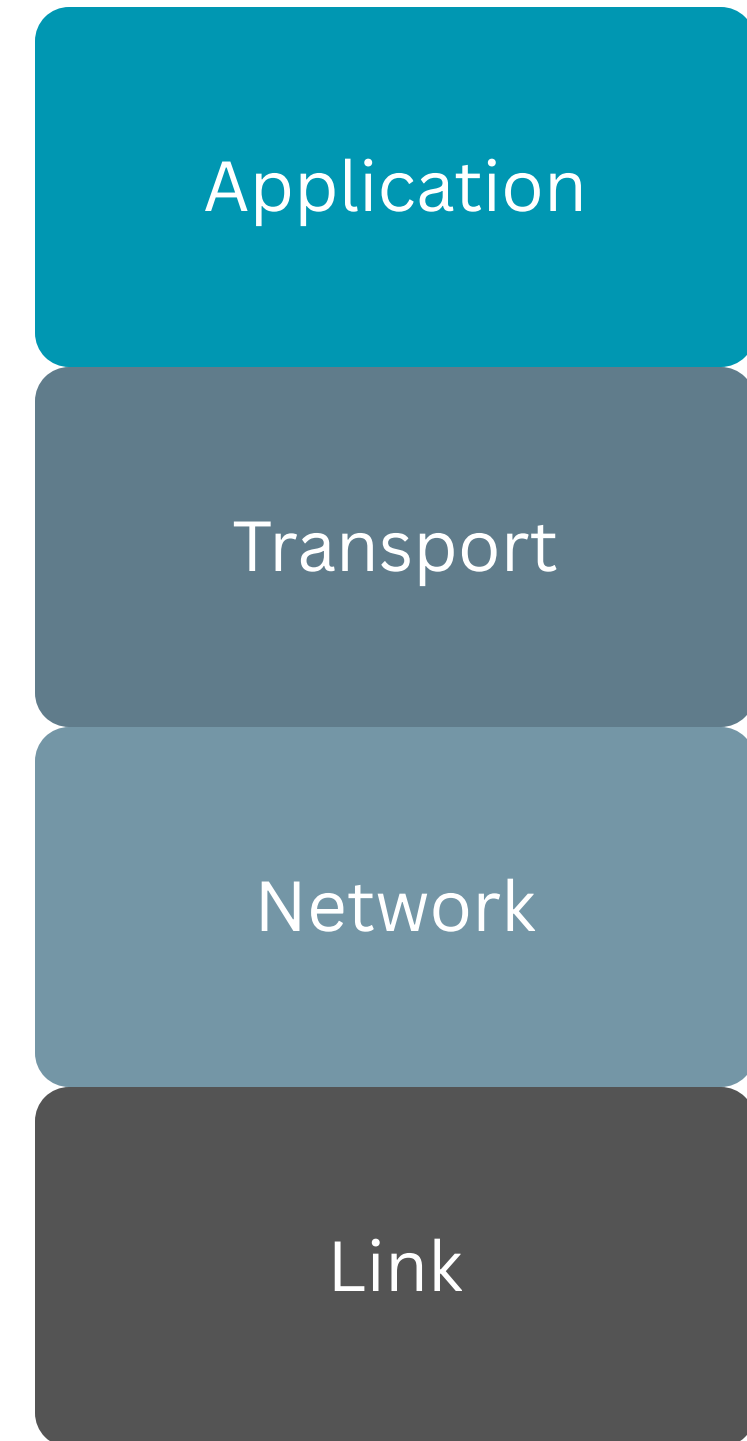
Let's have a **real life scenario**

Let's say PC 1 wants to
communicate with PC 2

PC 1



PC 2



PC 1

Application

Transport

Network

Link

PC 1

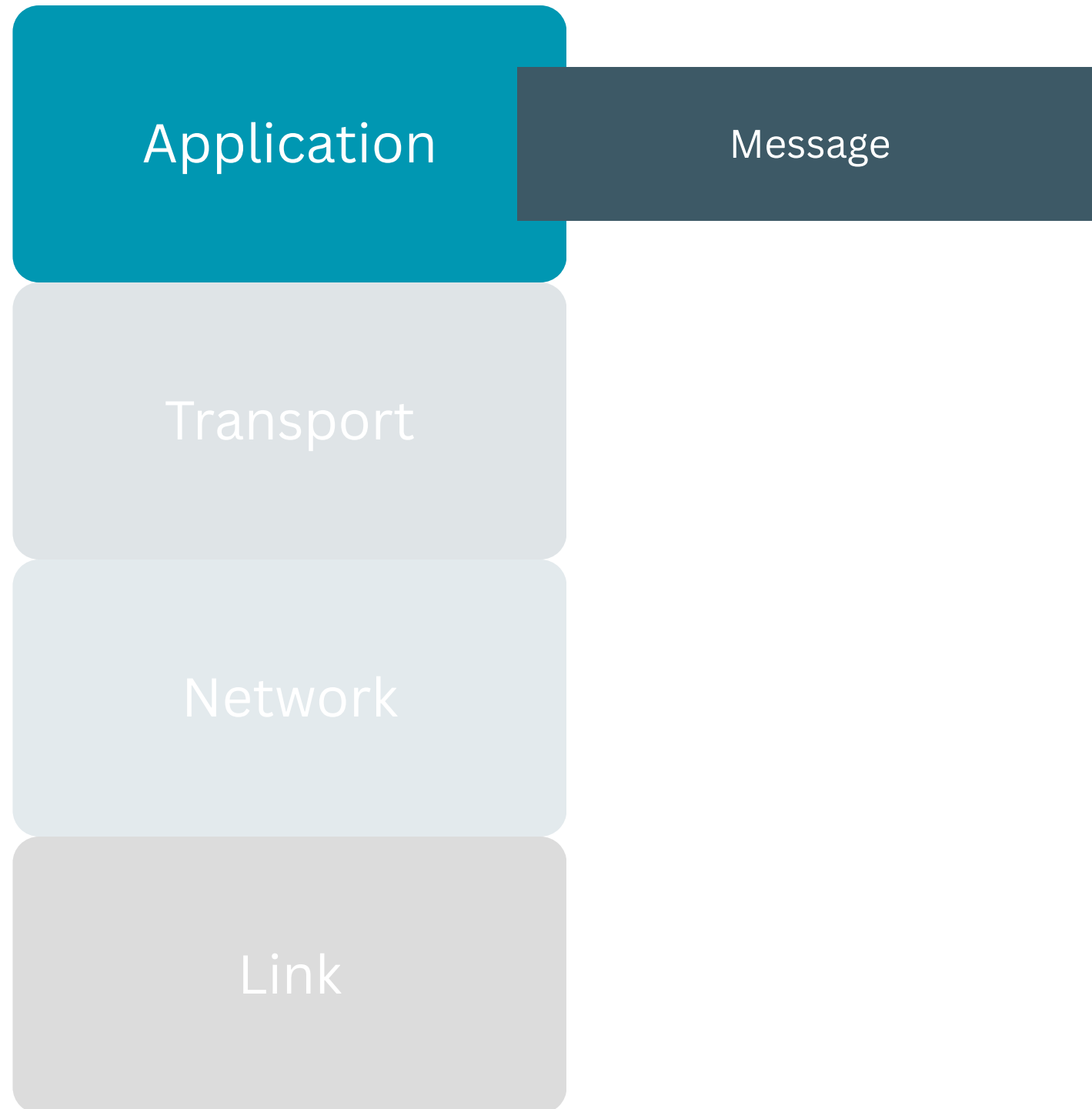
Application

Message

Transport

Network

Link



PC 1

Application

Transport

Network

Link

Message

PC 1

Application

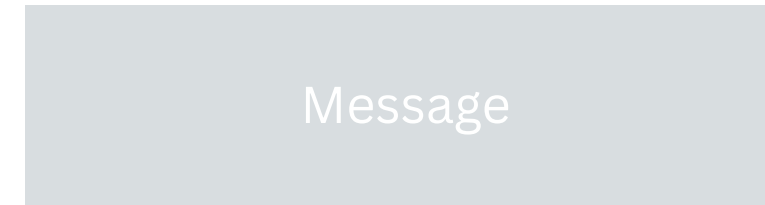
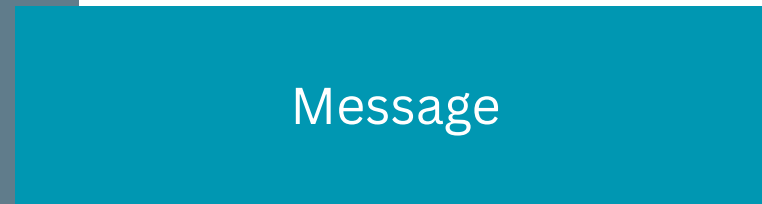
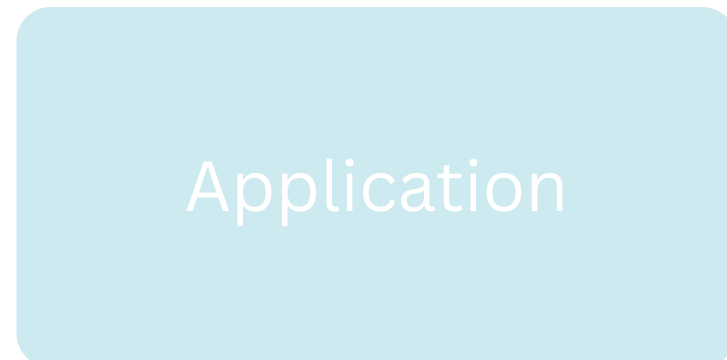
Transport

Network

Link

Message

Message



PC 1

Application

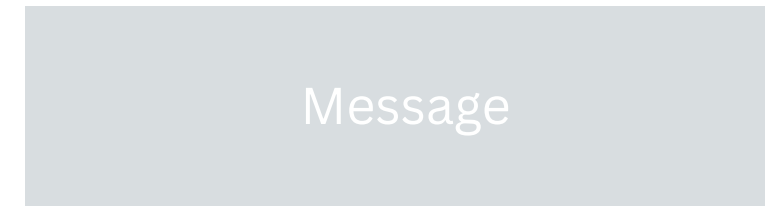
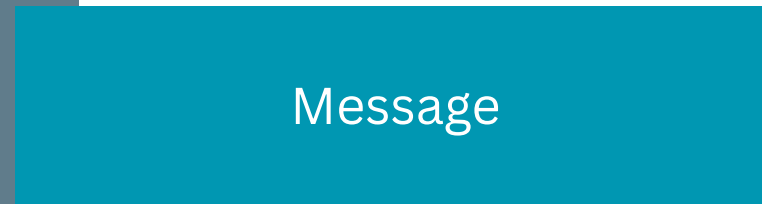
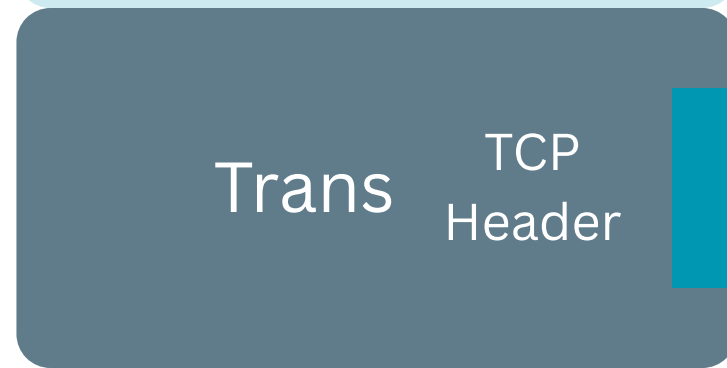
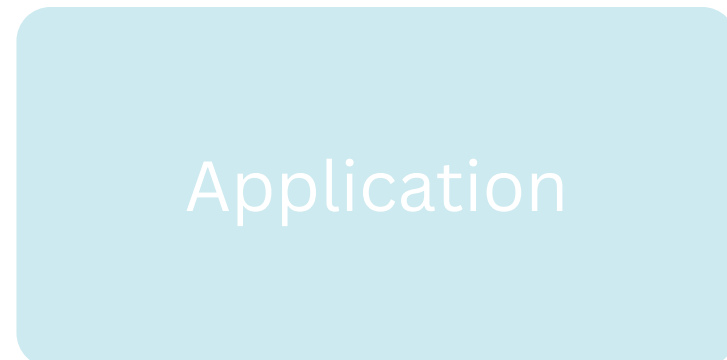
Trans TCP
Header

Network

Link

Message

Message



PC 1

Application

Transport

Network

Link

Message

TCP
Header

Message

PC 1

Application

Transport

Network

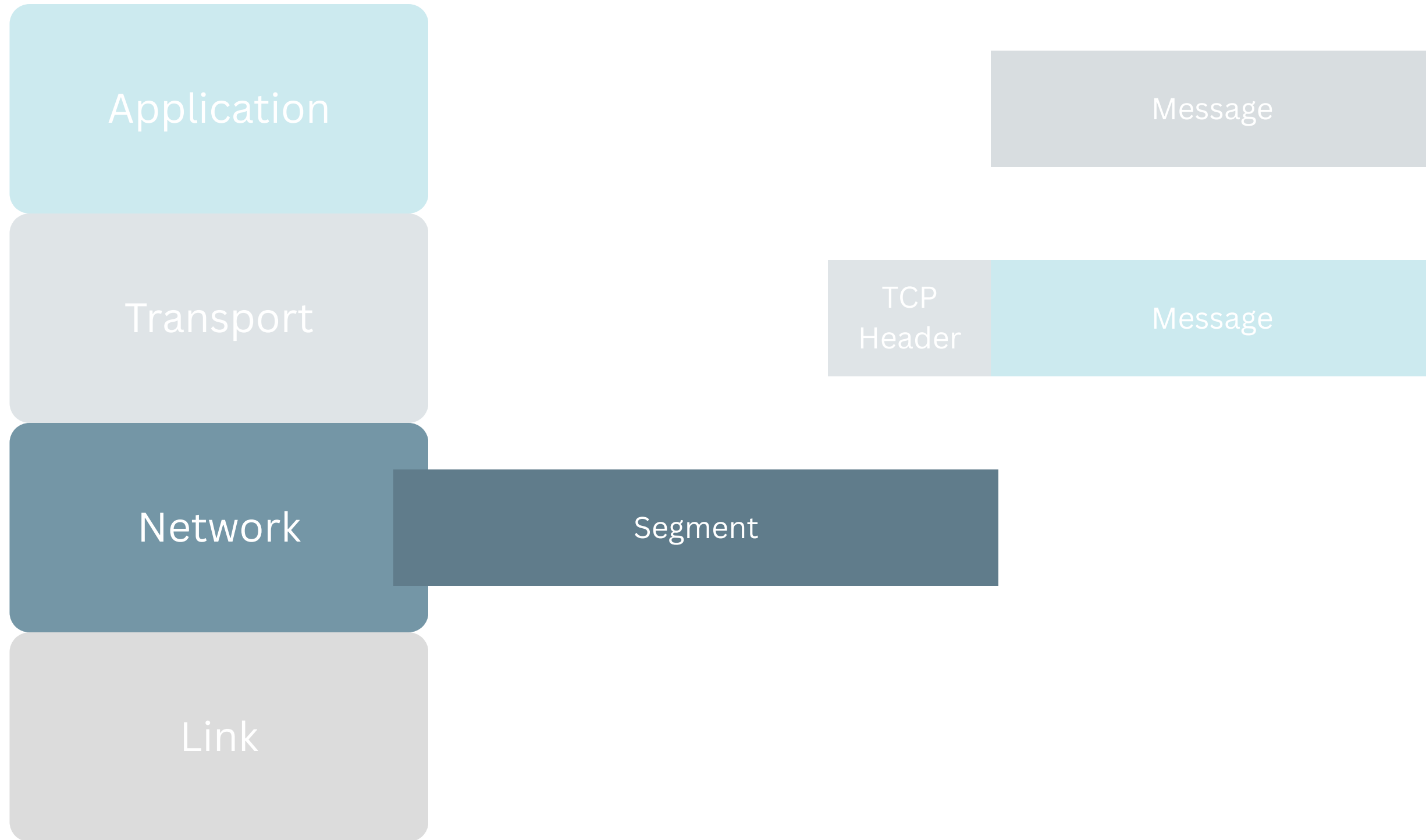
Link

Message

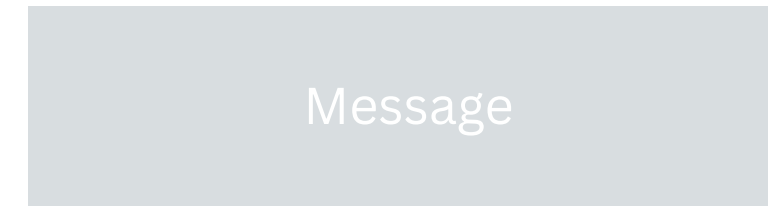
TCP
Header

Message

Segment



PC 1



PC 1

Application

Transport

Network

Link

Message

TCP
Header

Message

IP Header

Segment

PC 1

Application

Message

Transport

TCP
Header

Message

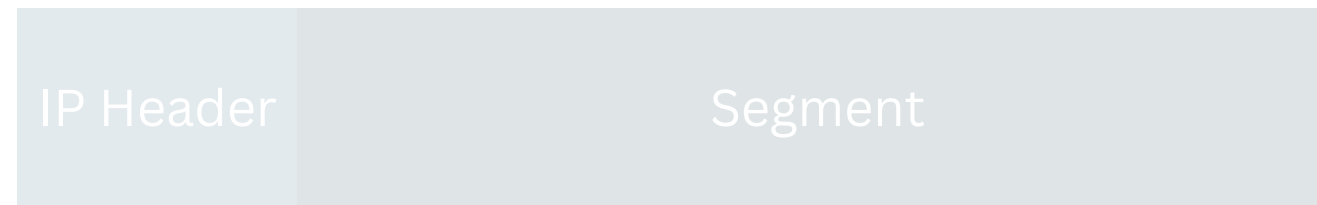
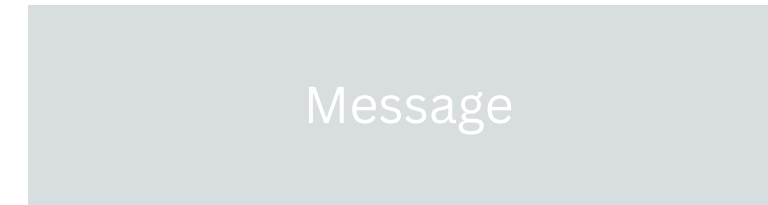
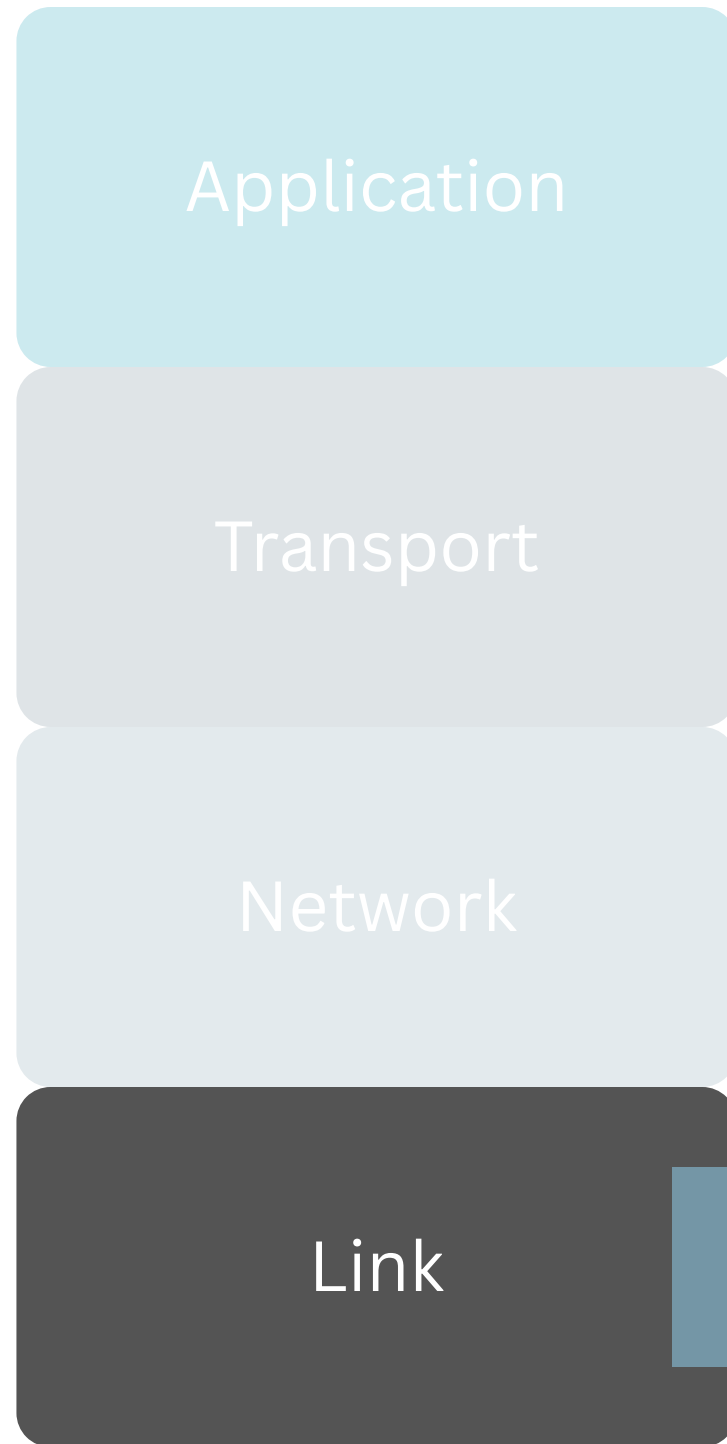
Network

IP Header

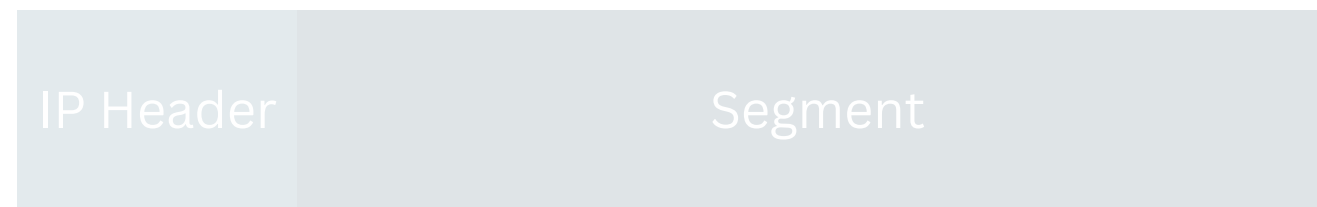
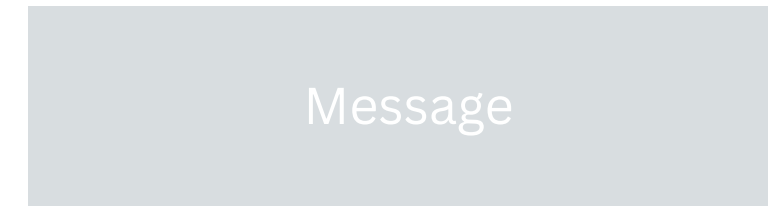
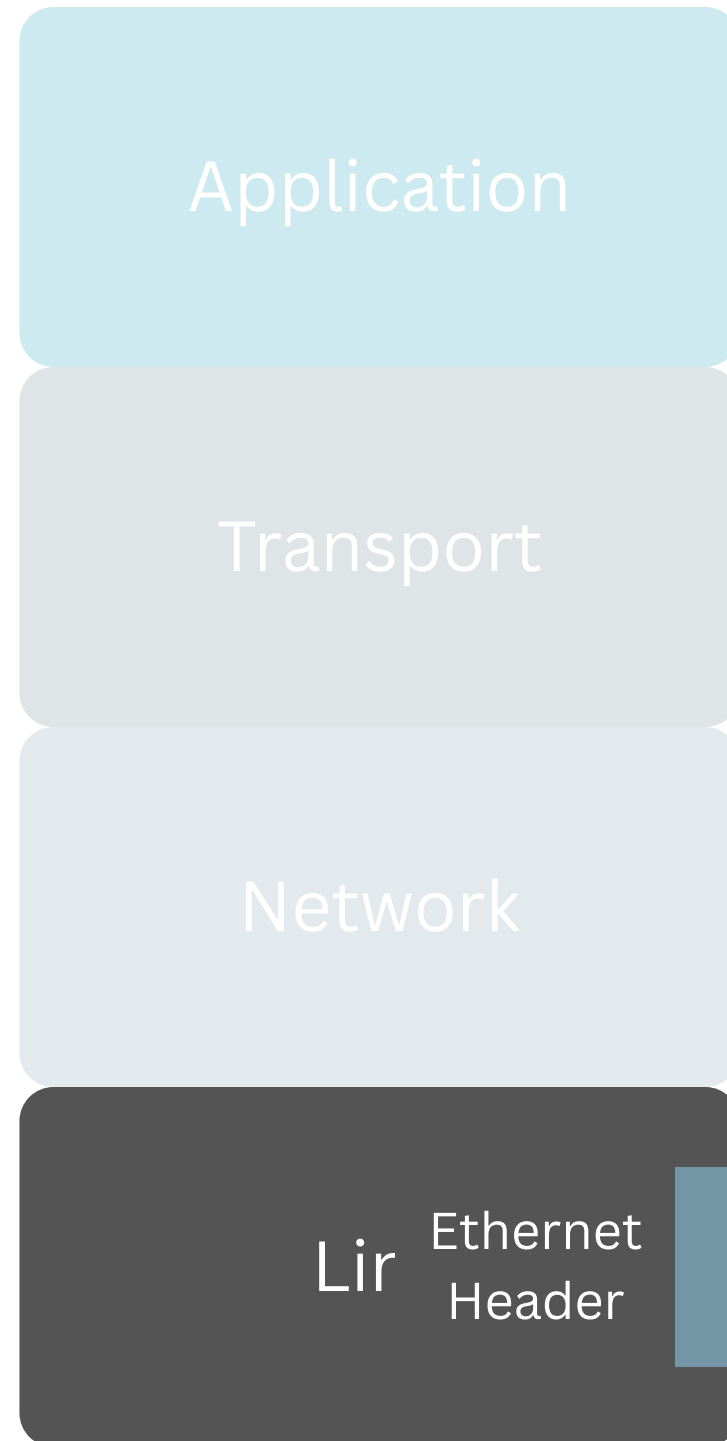
Segment

Link

Packet



PC 1



PC 1

Application

Transport

Network

Link

Message

TCP
Header

Message

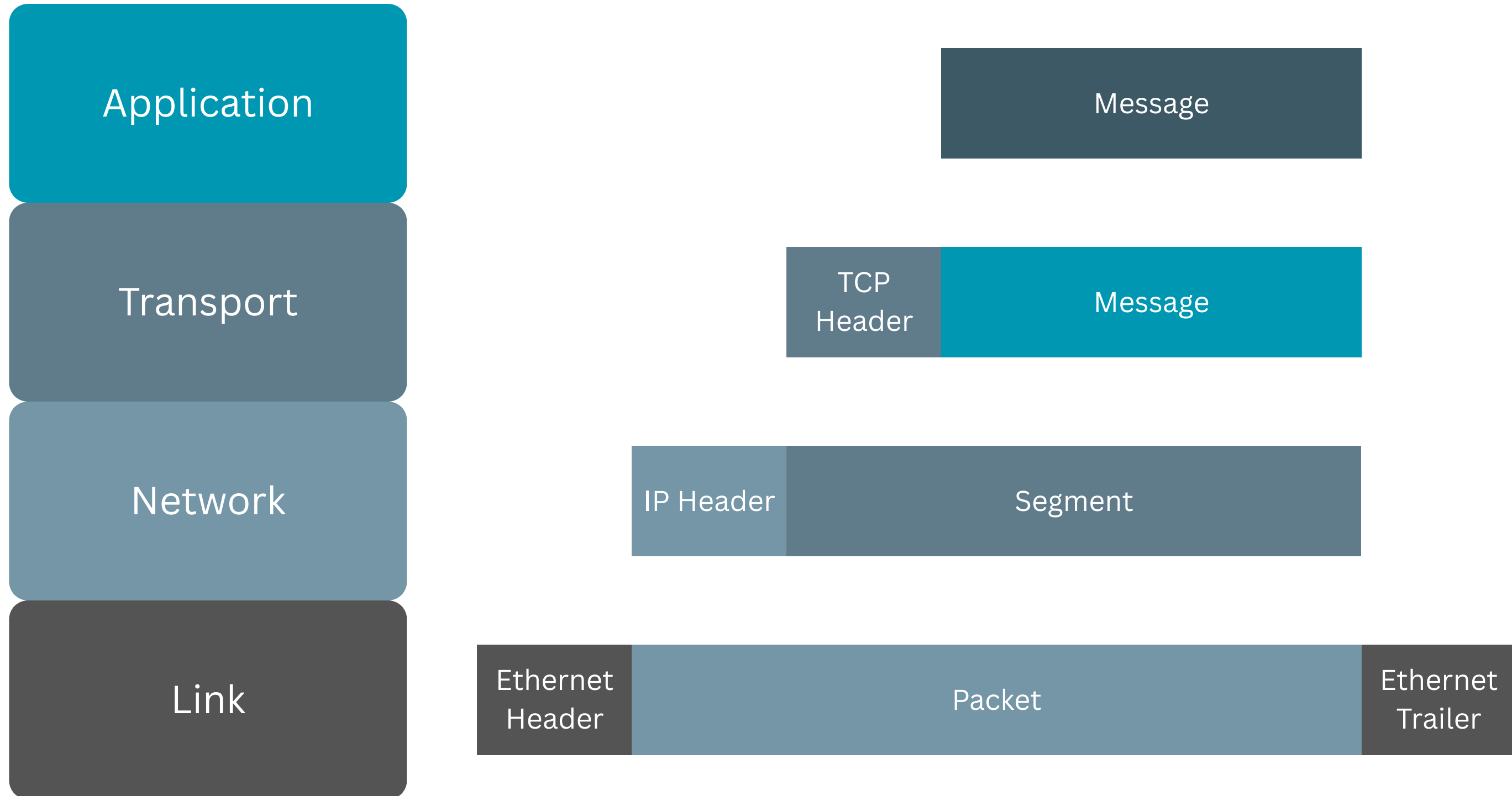
IP Header

Segment

Ethernet
Header

Packet

Ethernet
Trailer



PC 1

Application

Transport

Network

Link

Message

TCP
Header

Message

IP Header

Segment

Ethernet
Header

IP Header

TCP
Header

Message

Ethernet
Trailer

PC 1

Application

Transport

Network

Link

Message

TCP
Header

Message

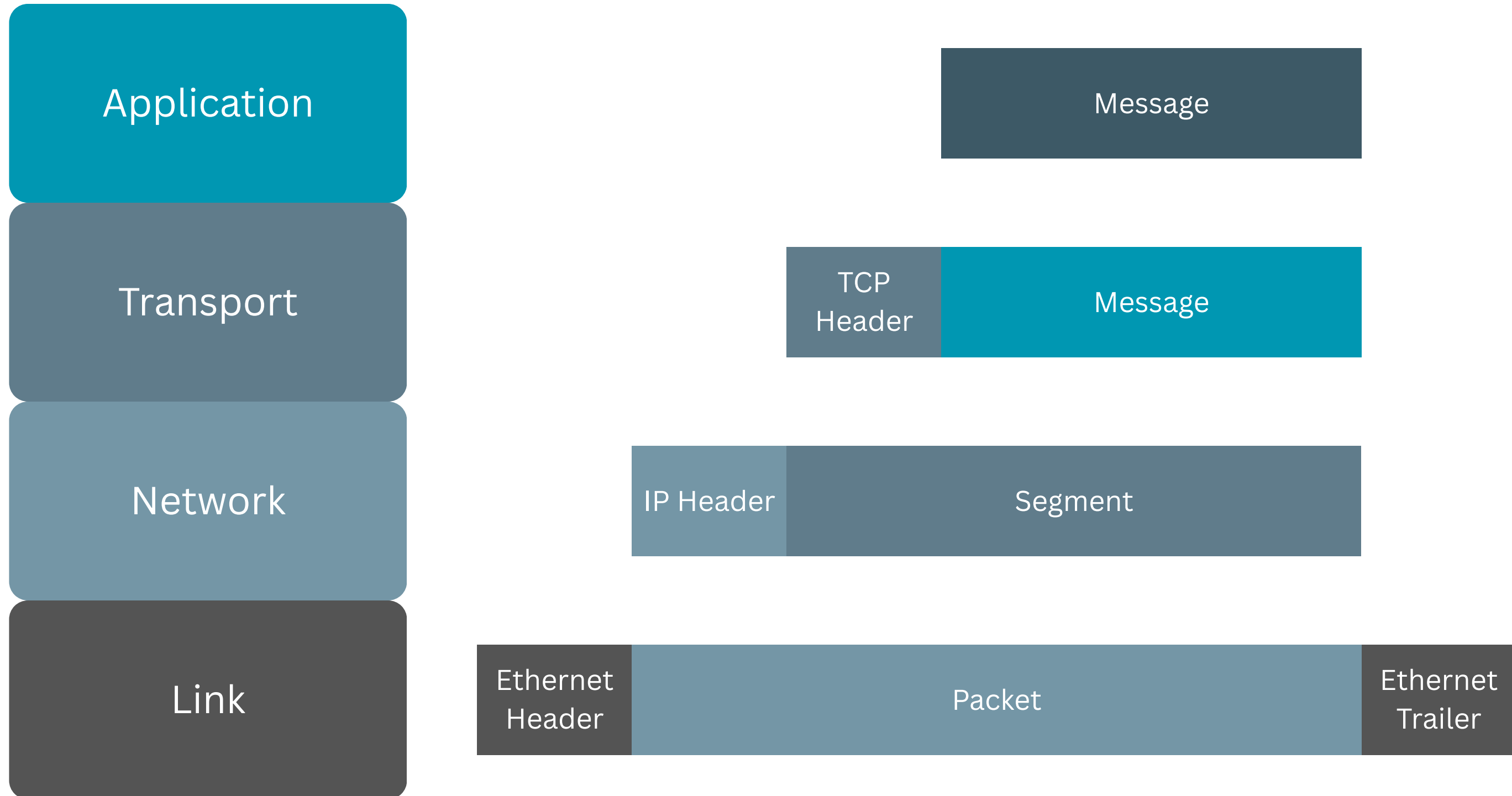
IP Header

Segment

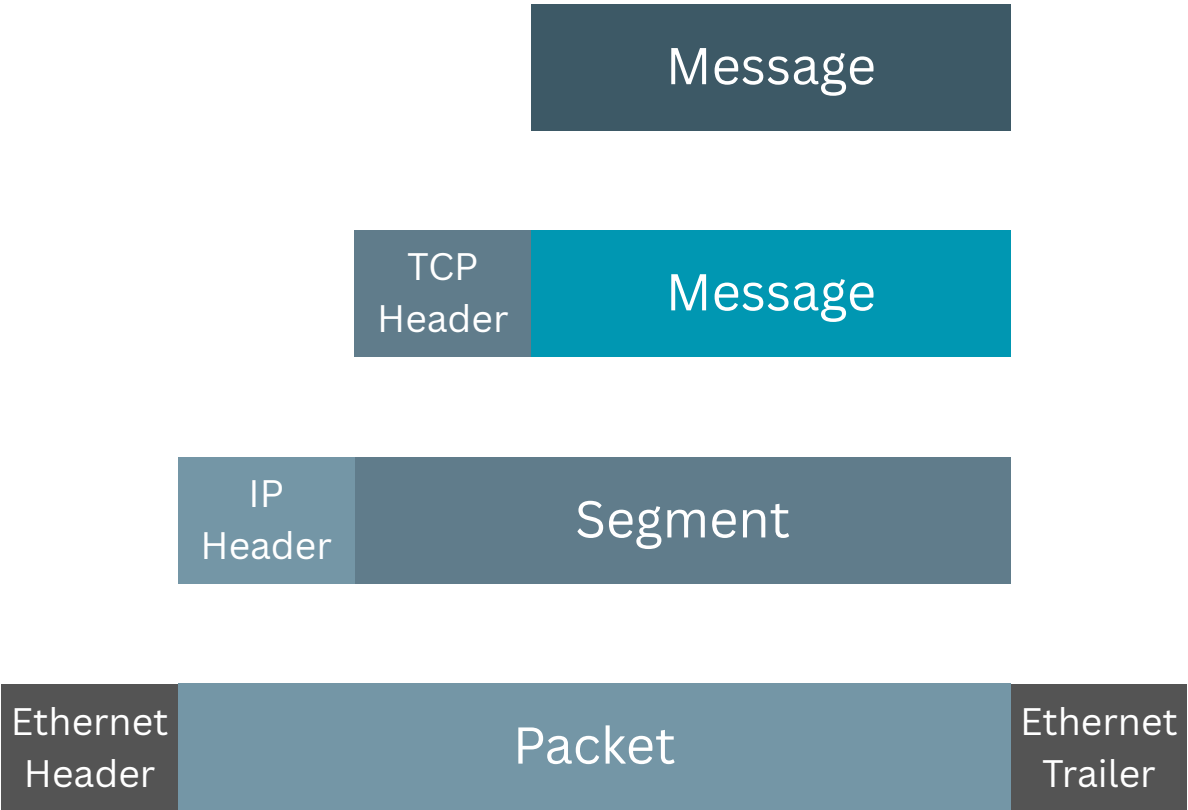
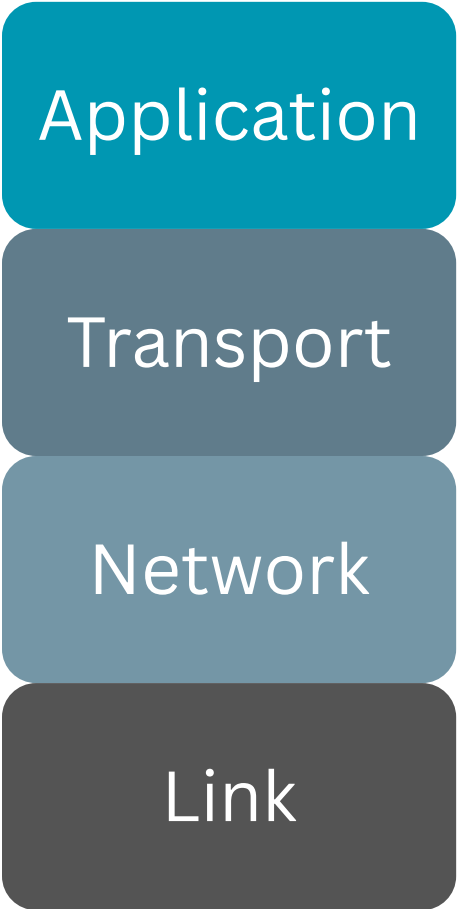
Ethernet
Header

Packet

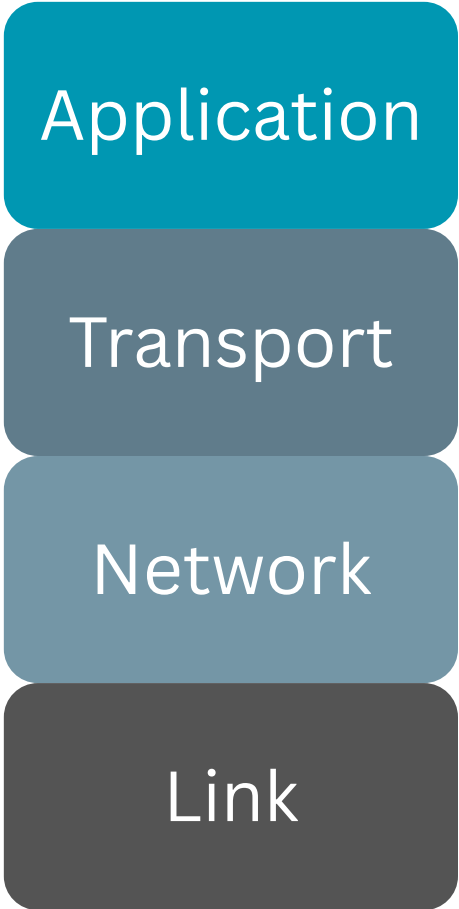
Ethernet
Trailer



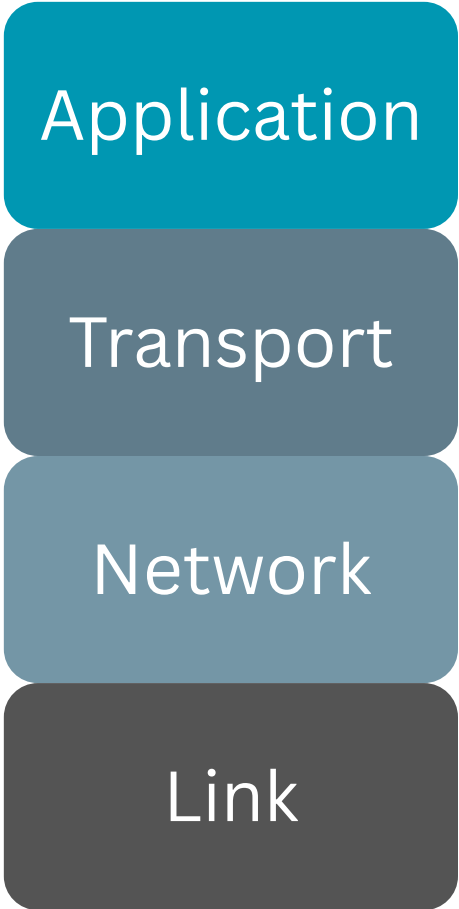
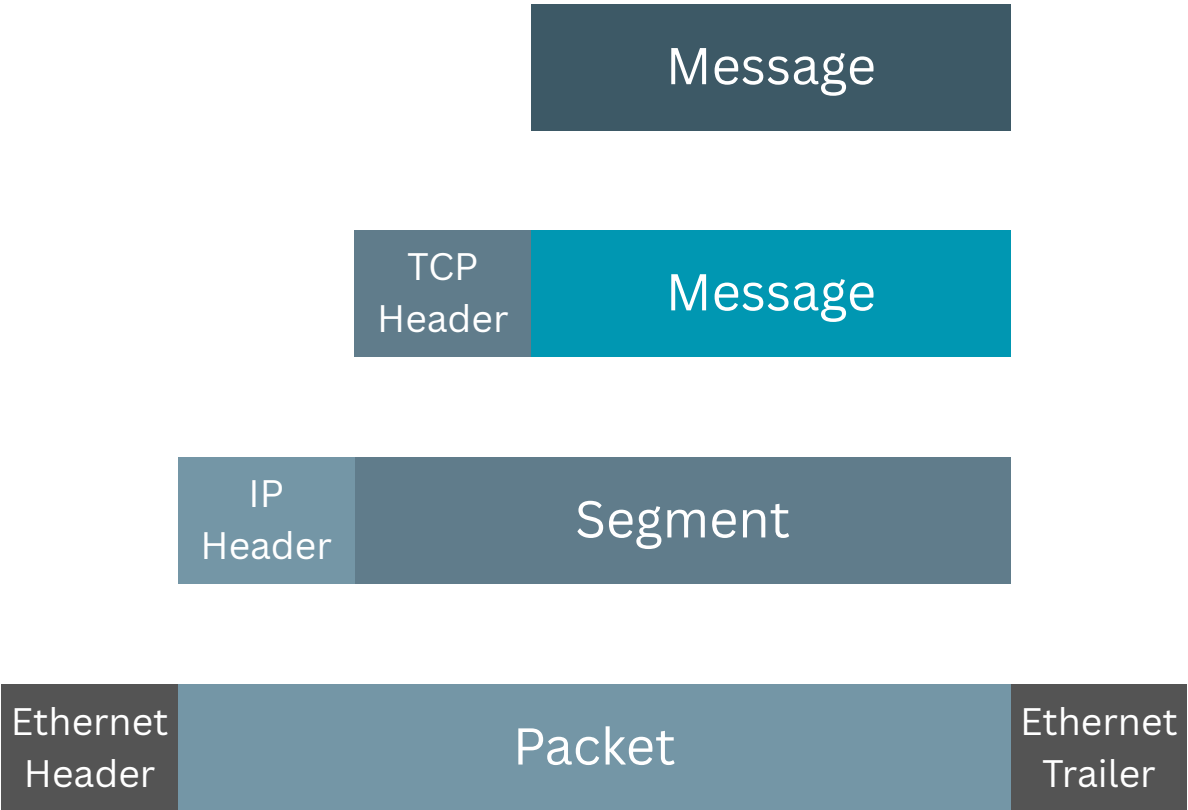
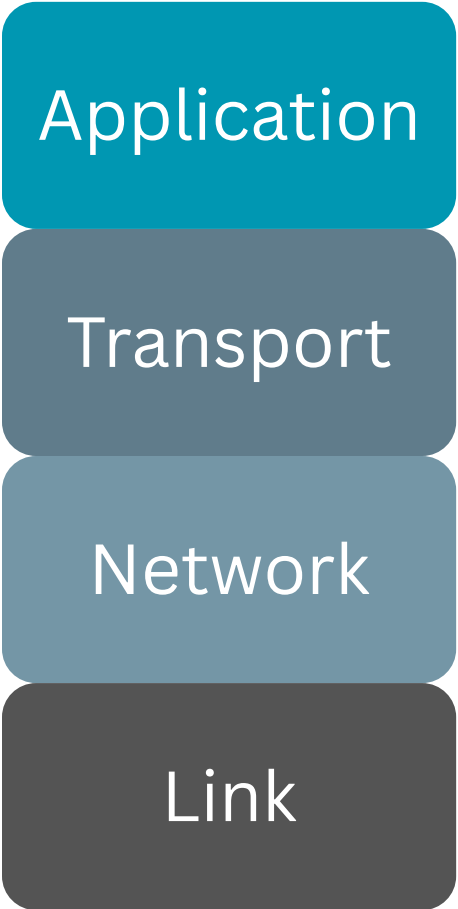
PC 1



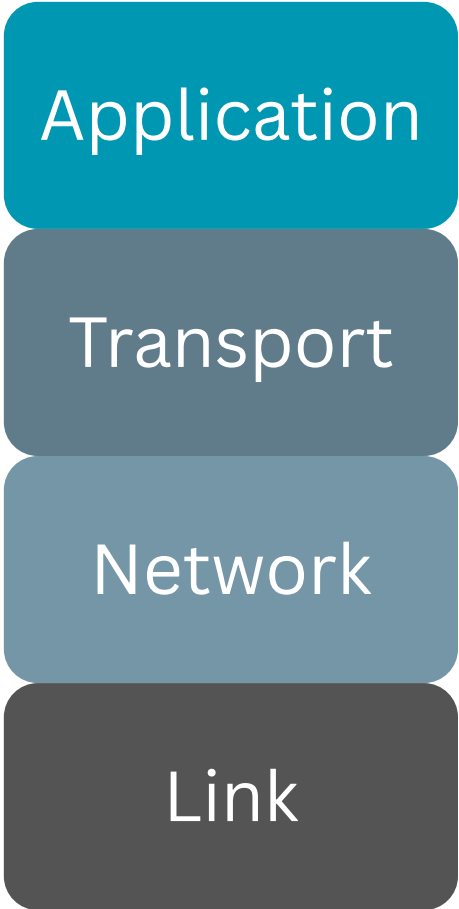
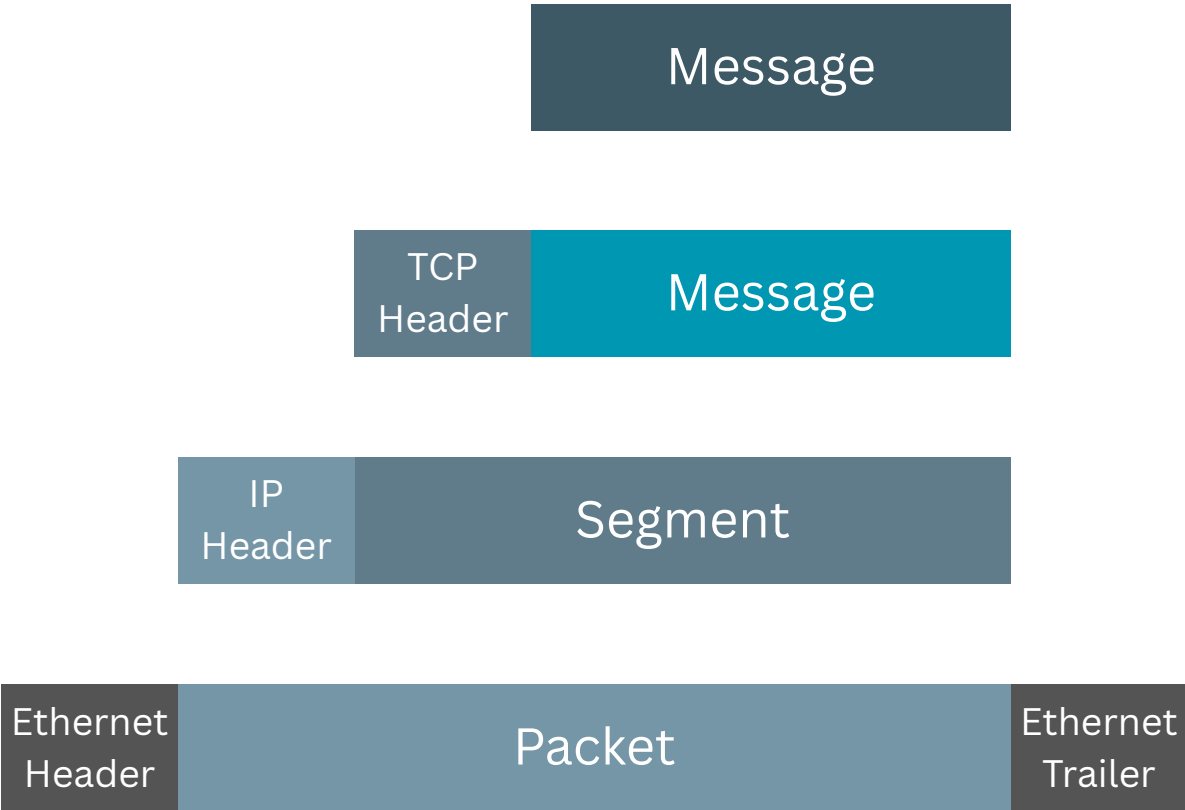
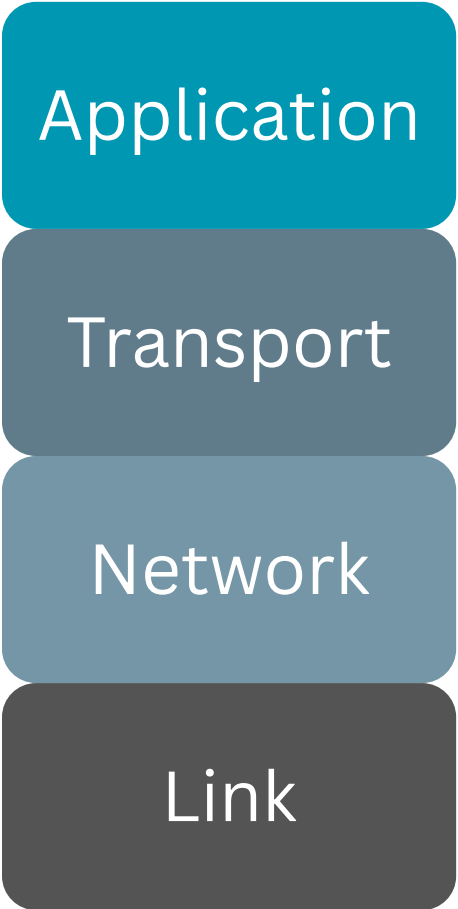
PC 2



PC 1



PC 1



PC 2

Application

Transport

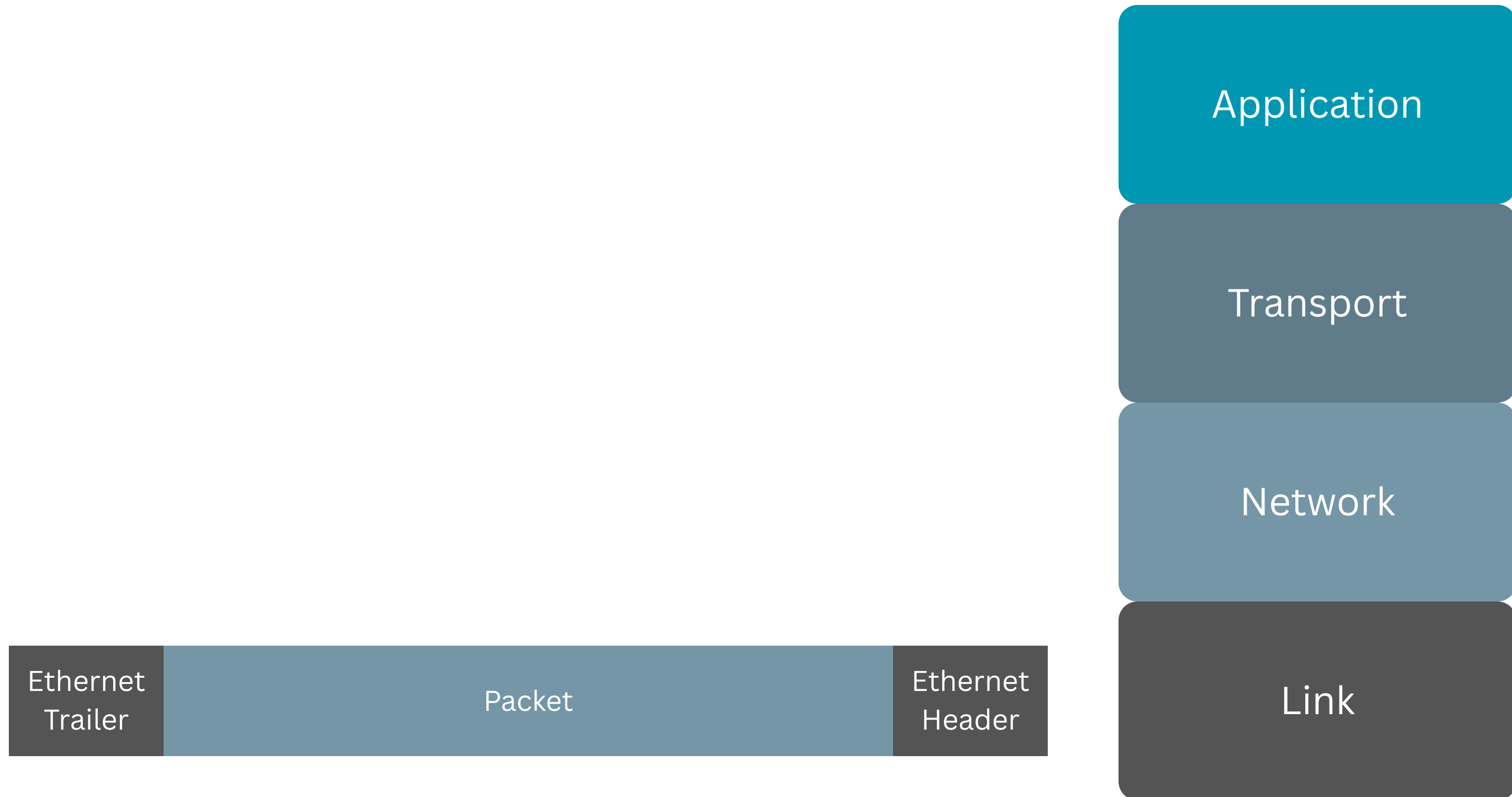
Network

Link

Ethernet
Trailer

Packet

Ethernet
Header



PC 2

Application

Transport

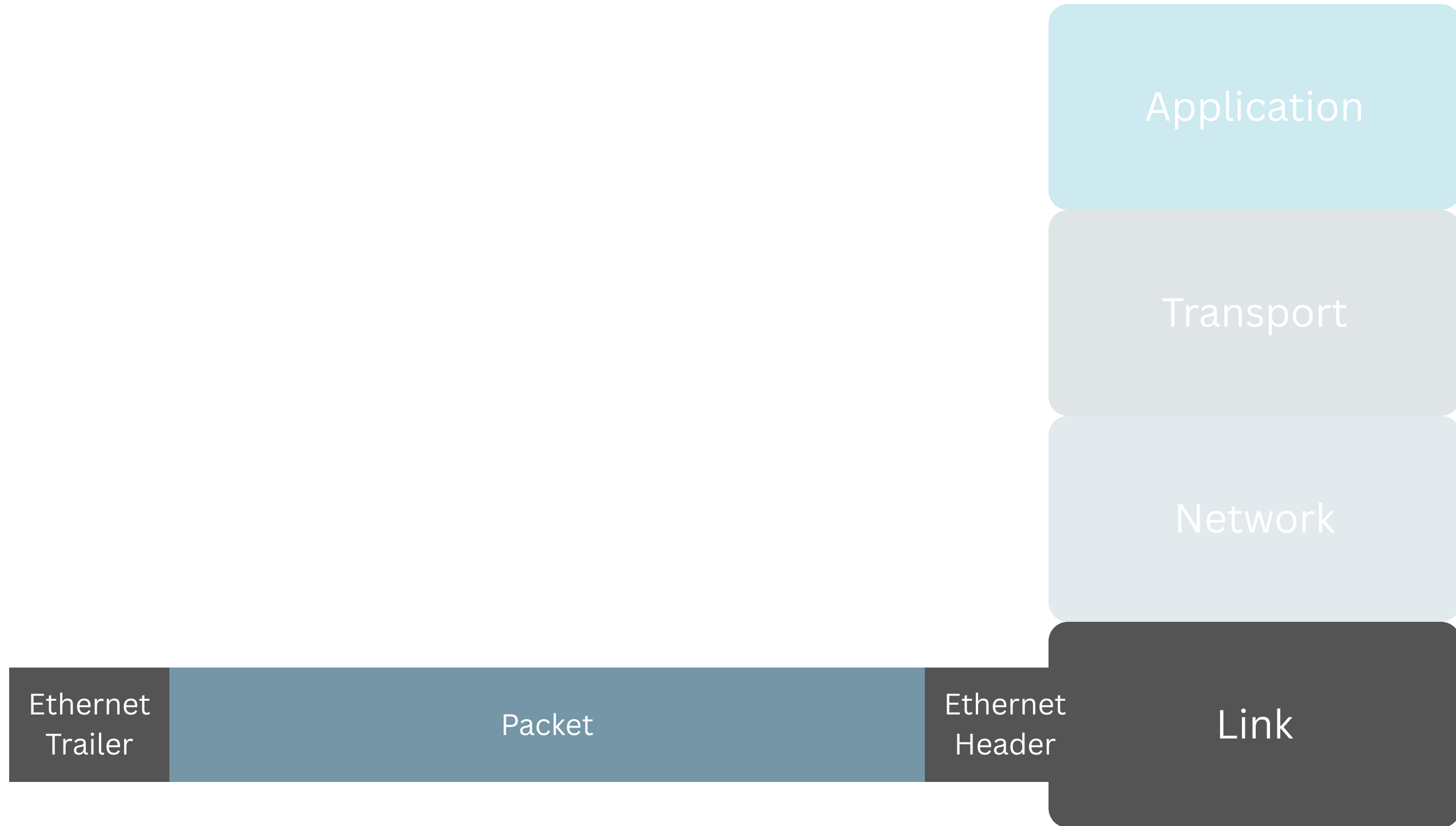
Network

Ethernet
Trailer

Packet

Ethernet
Header

Link



PC 2

Application

Transport

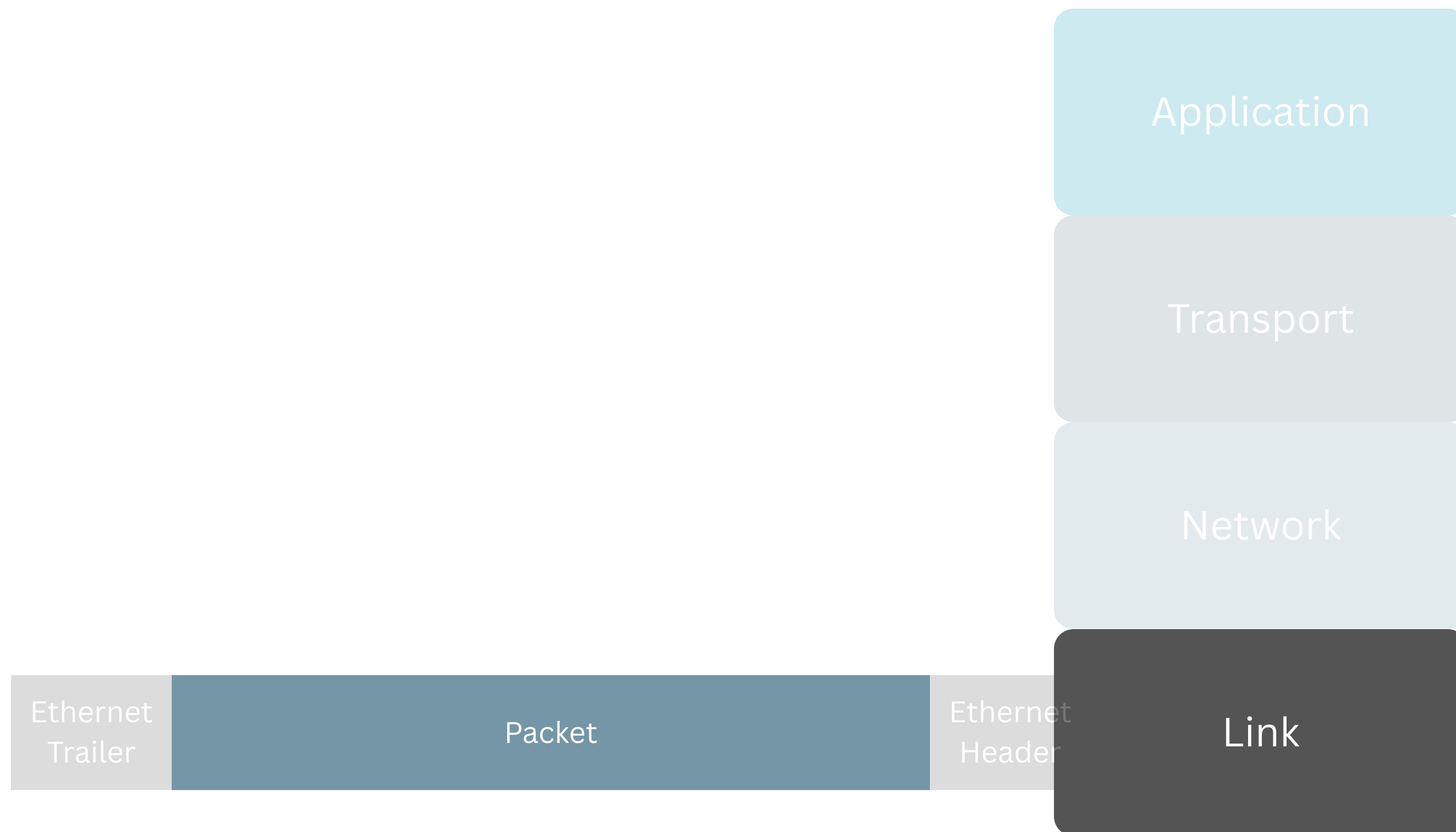
Network

Link

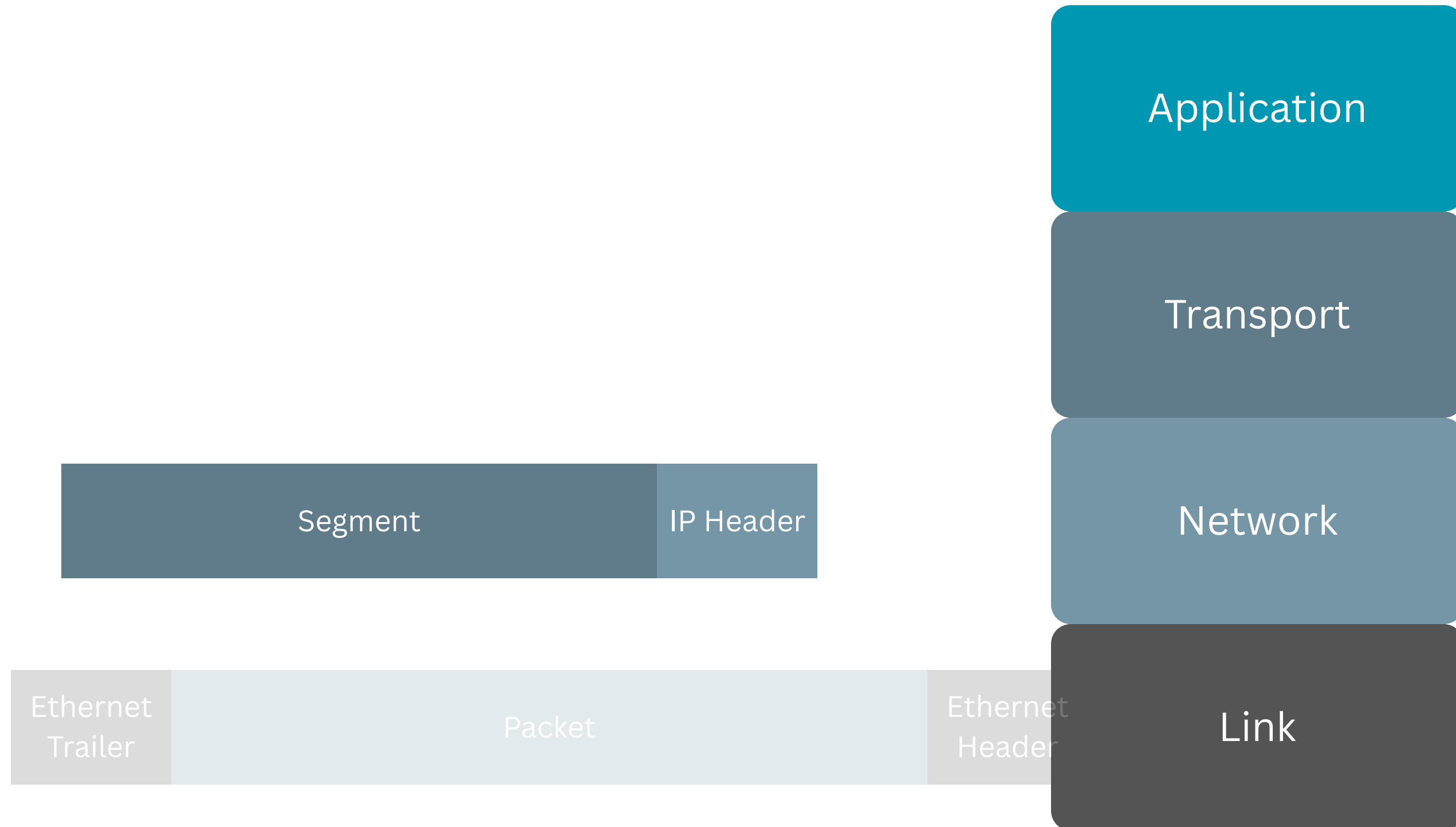
Ethernet
Trailer

Packet

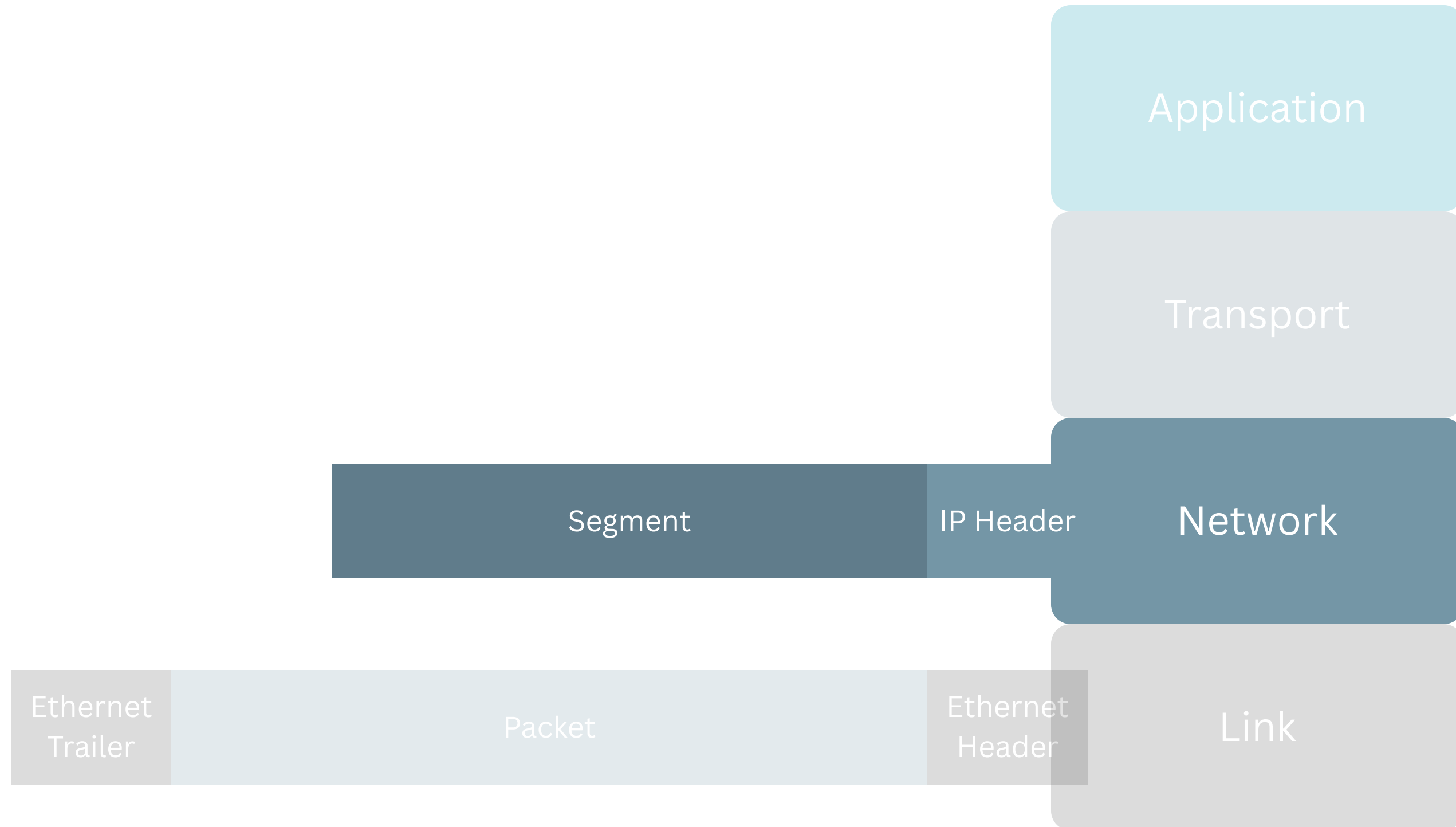
Ethernet
Header



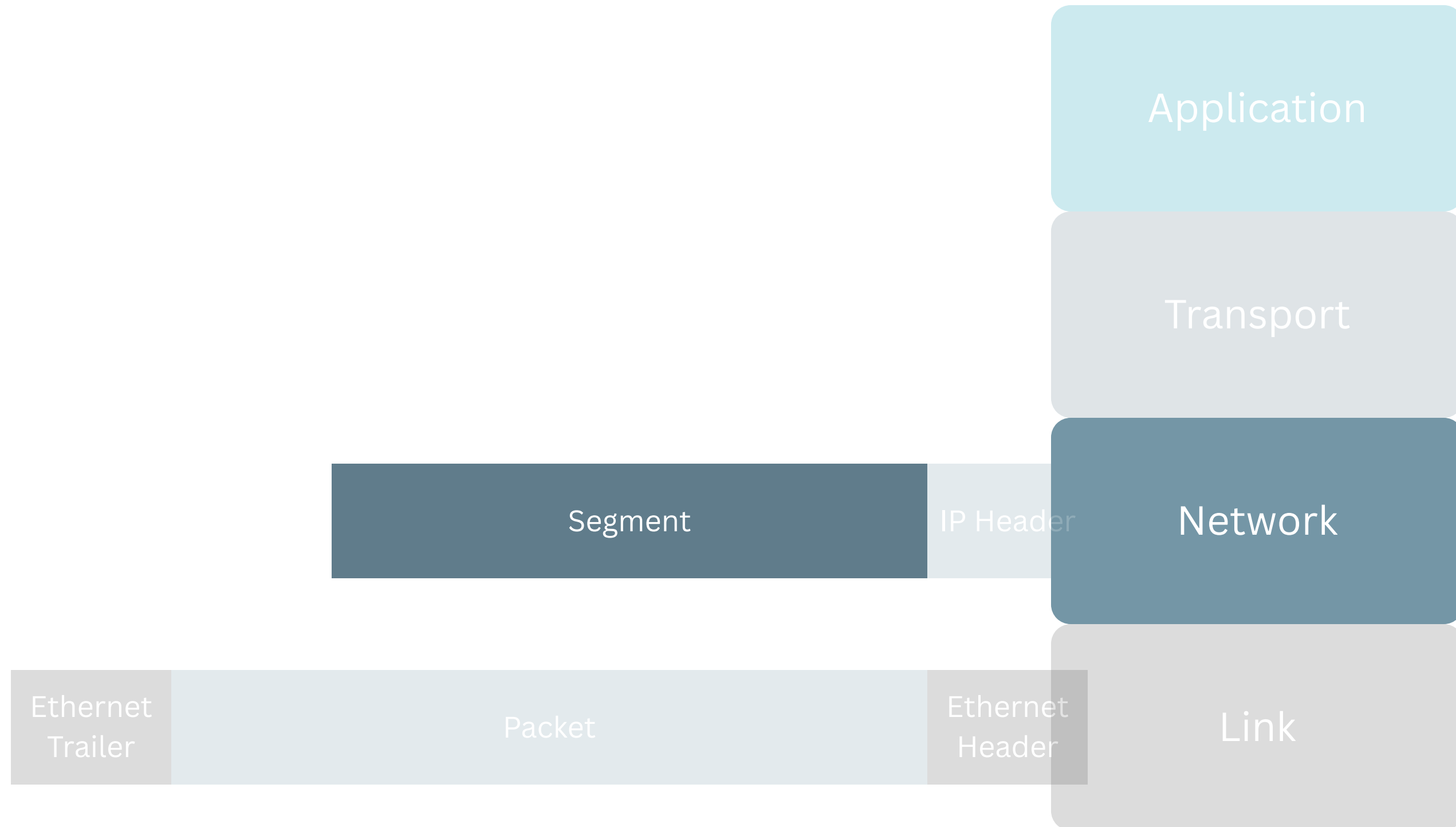
PC 2



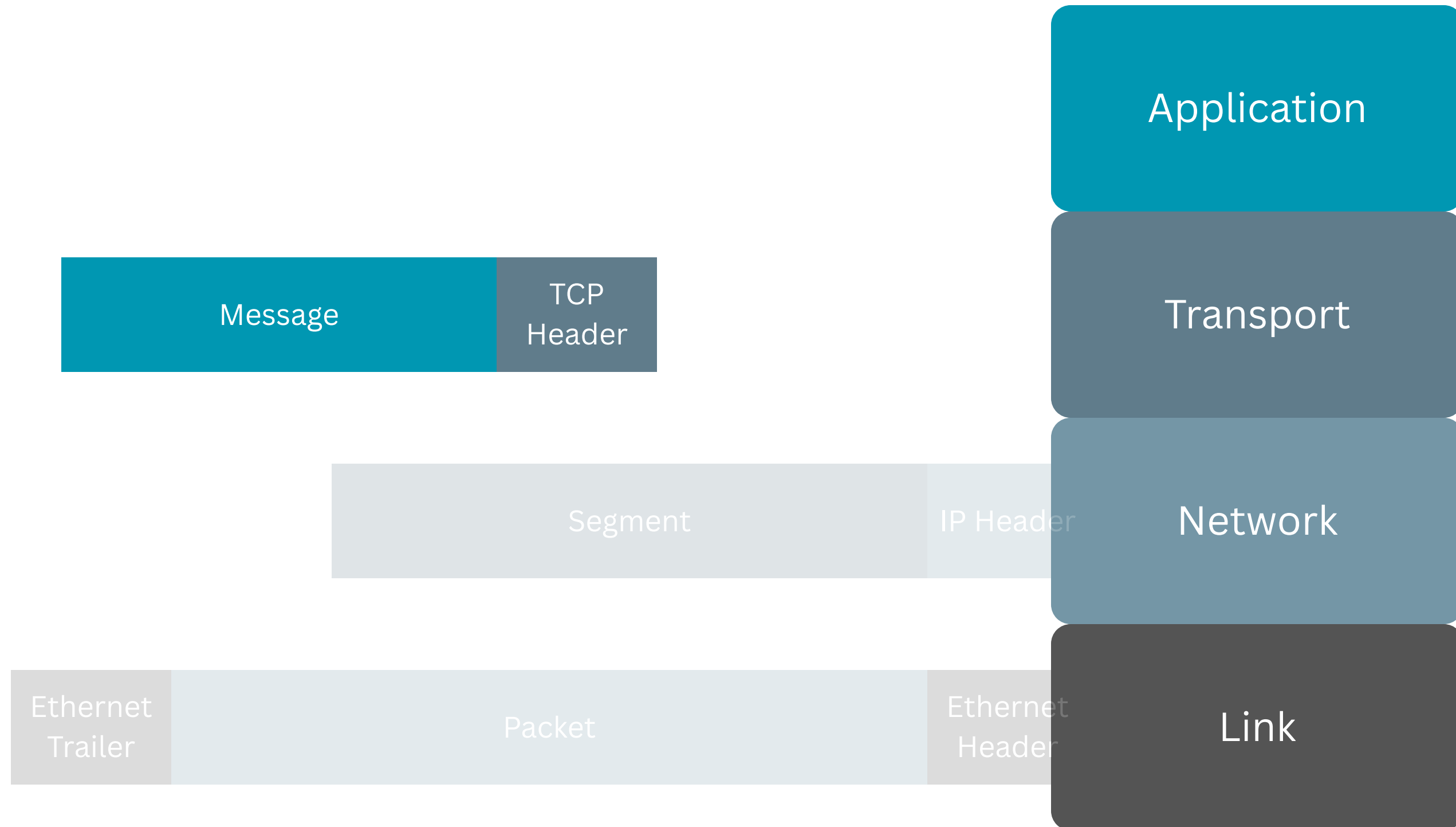
PC 2



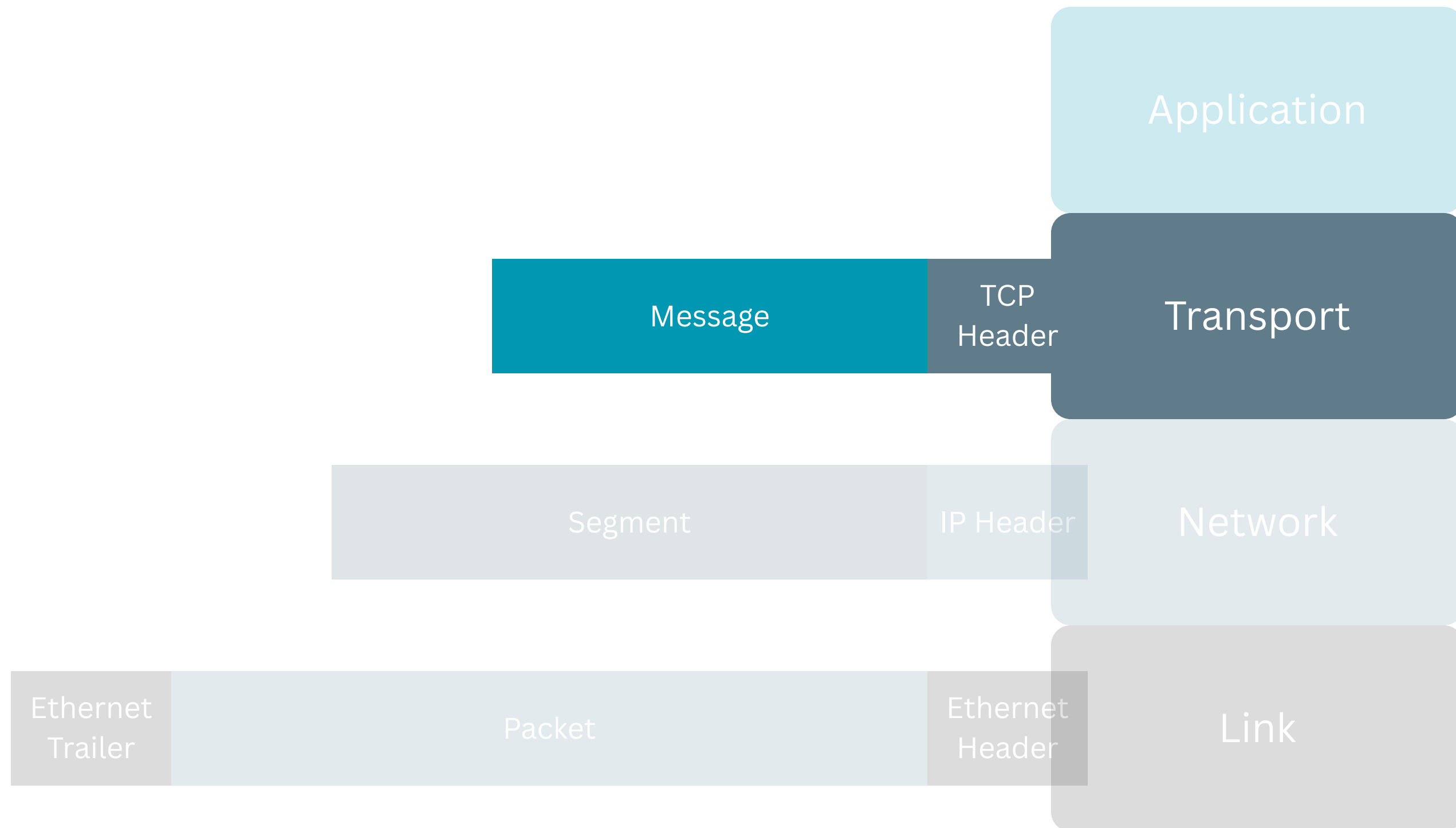
PC 2



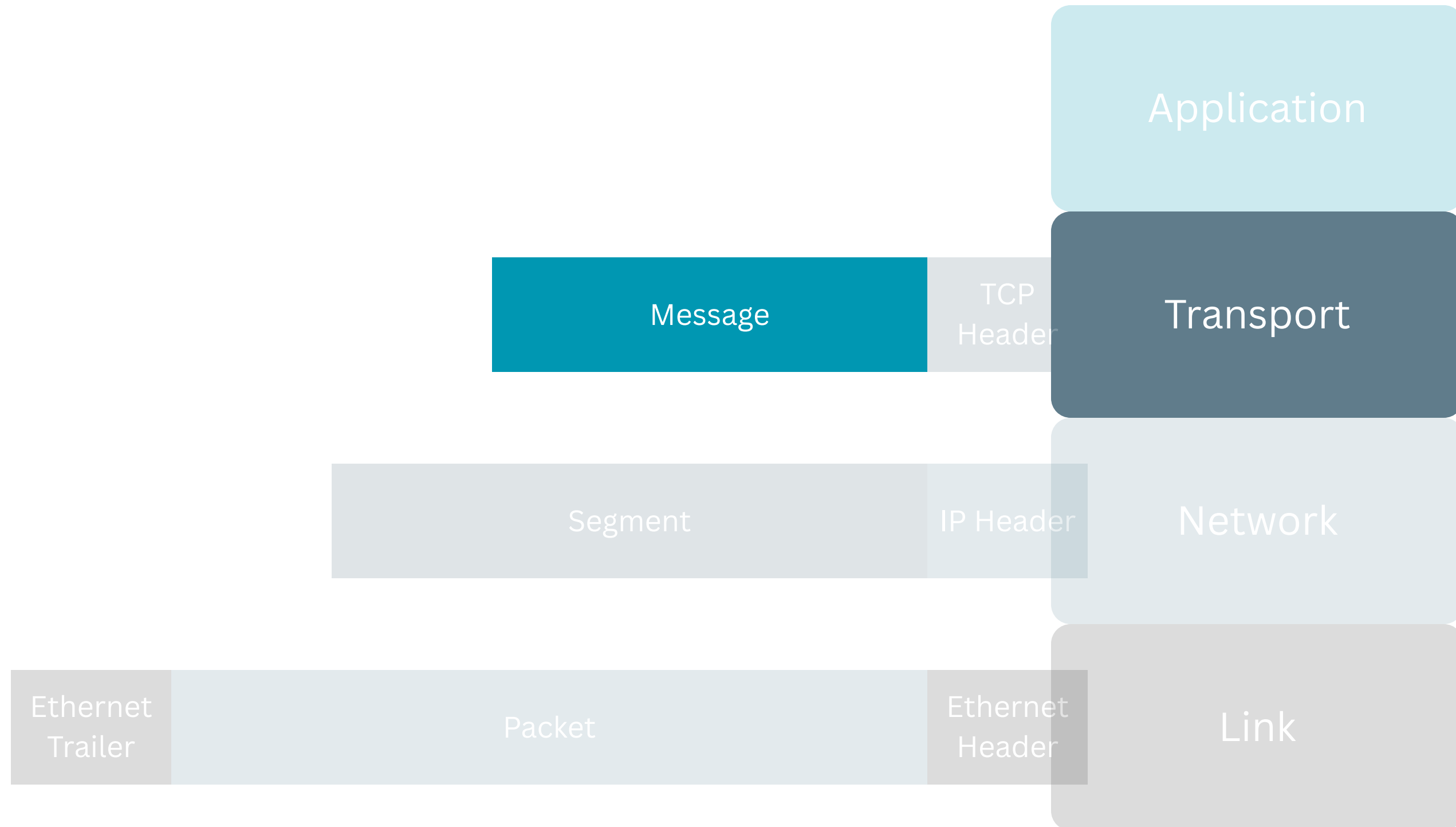
PC 2



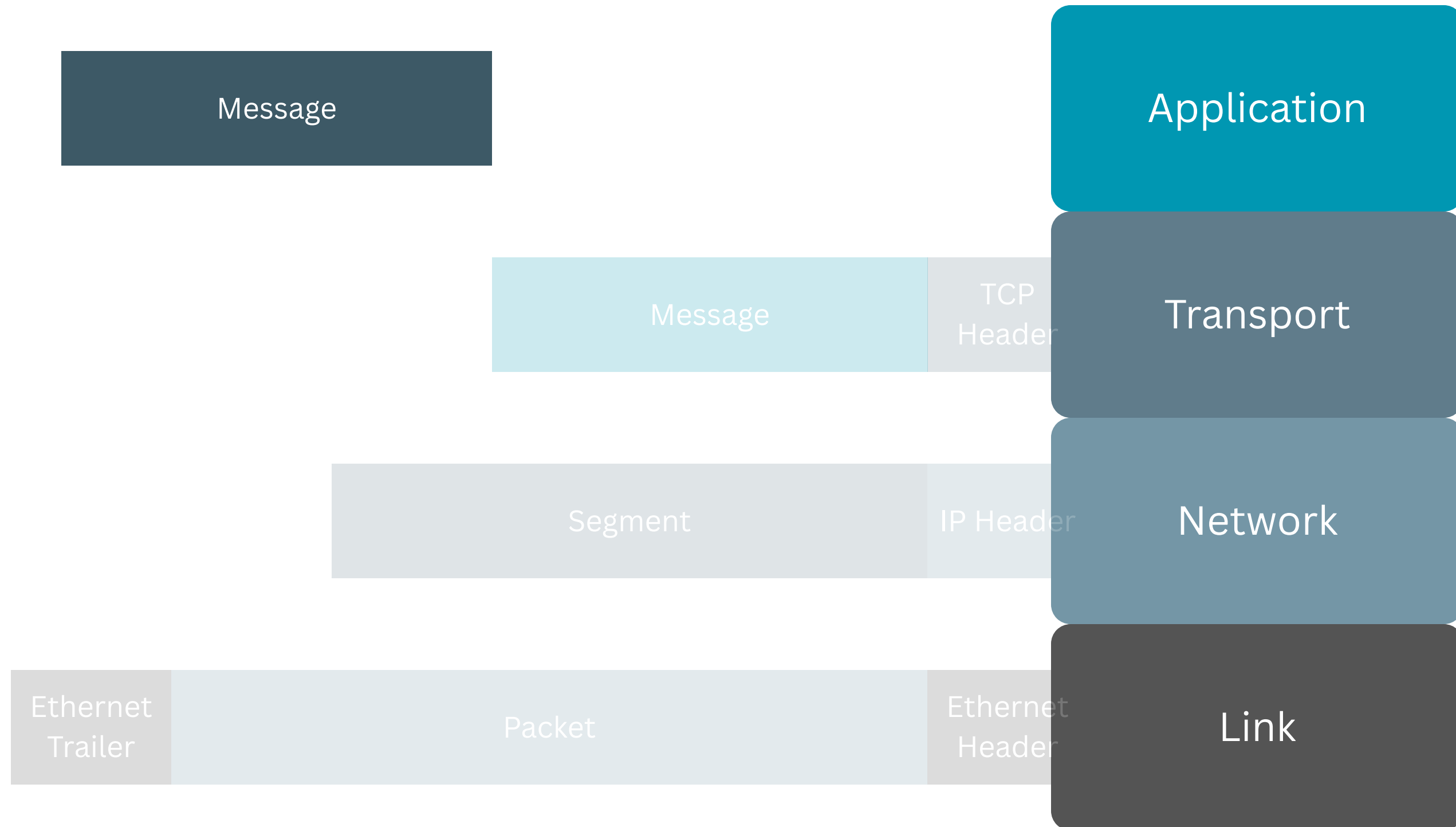
PC 2



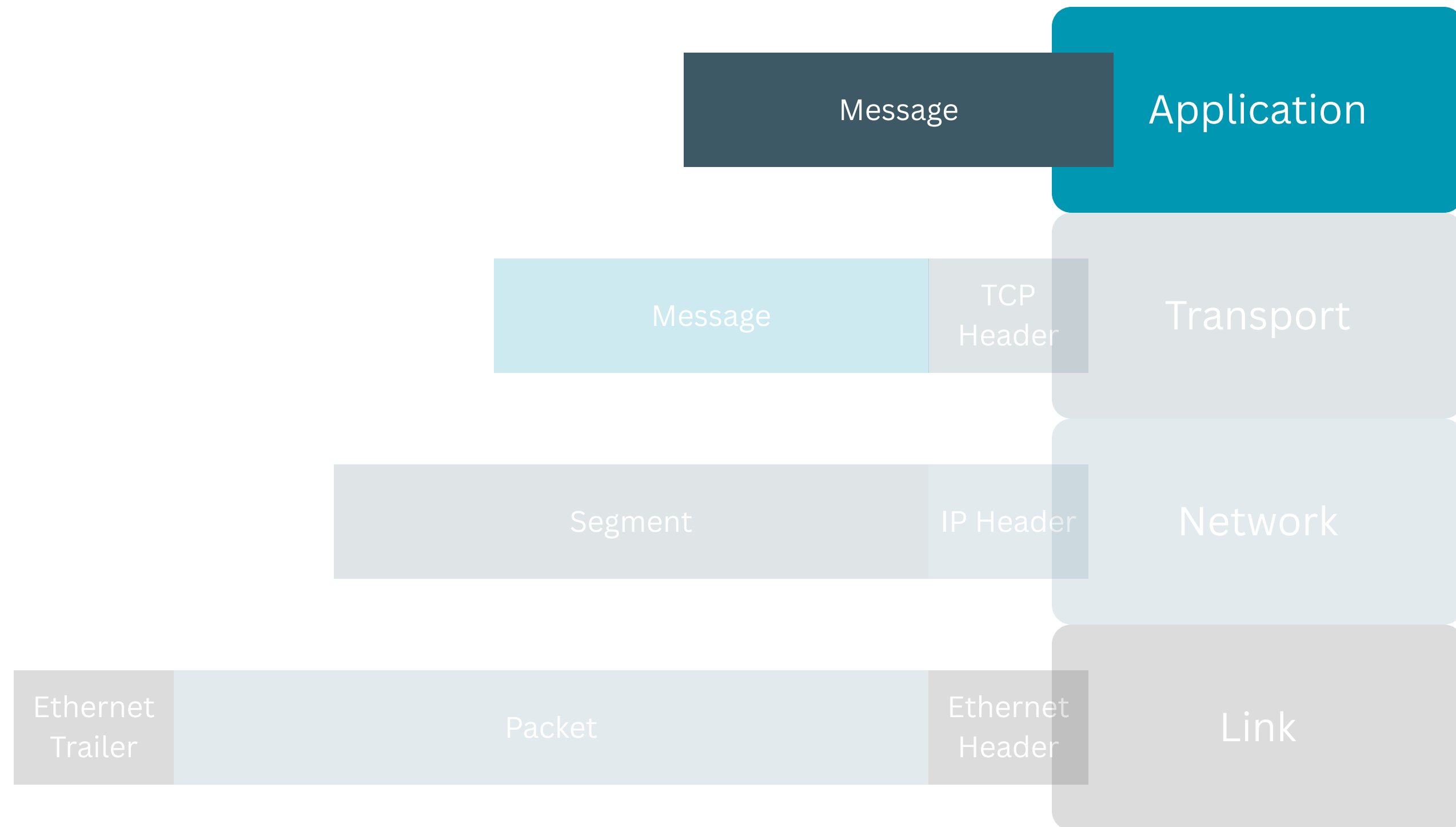
PC 2



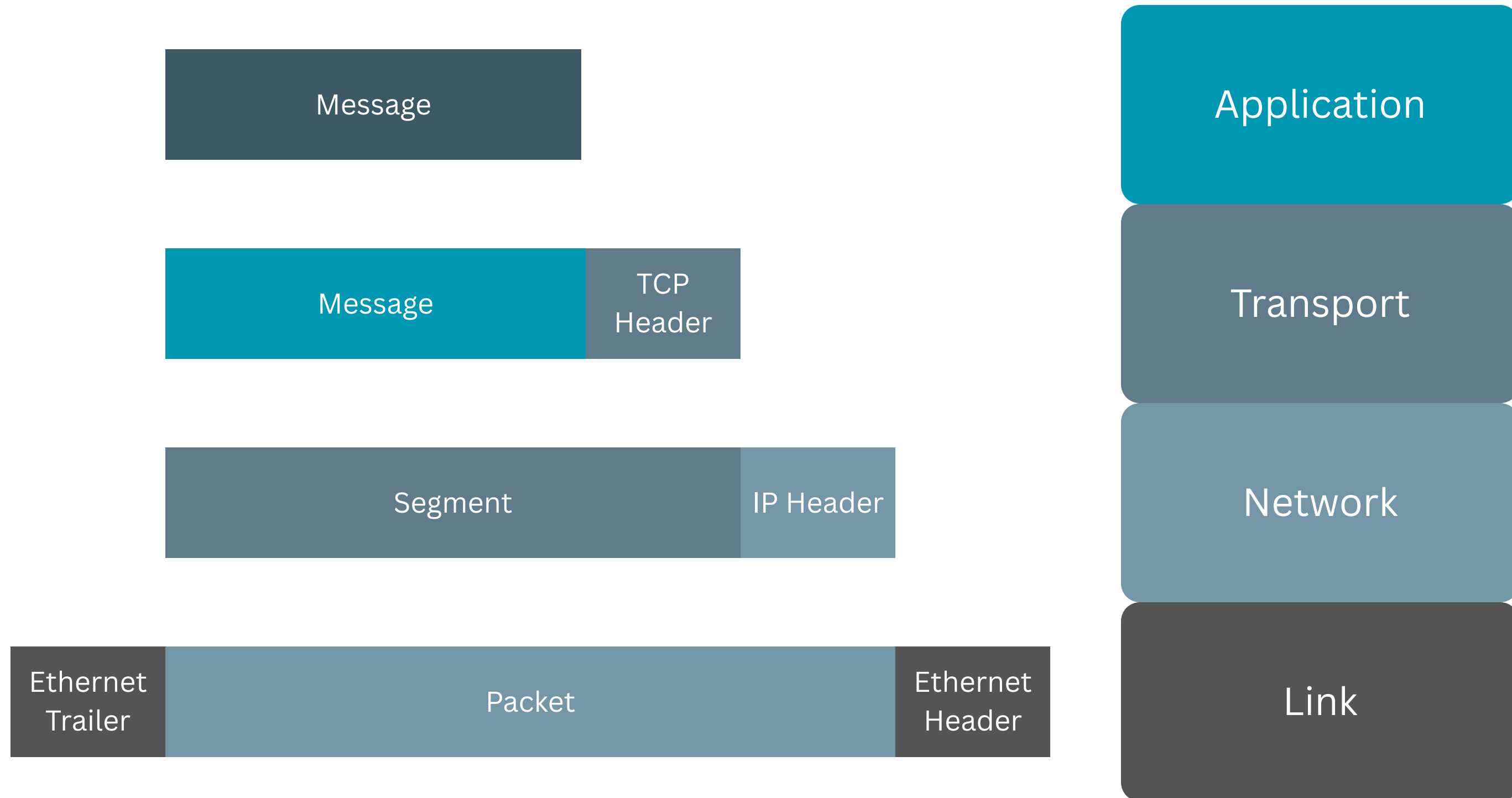
PC 2



PC 2



PC 2



Important Concepts:

- ip **Classes** (temp. solution no. 1)
- ip **CIDR** (classless inter domain routing) (temp solution no. 2)
- **private** and **public ips & NAT**
- **ipv6** (our real long-term solution)
- **TCP vs UDP**
- **DNS**

ip classes

(temp. solution no. 1)

We must have some policy
so, lets make classes

Classes of IP Addresses

Class	From	To
A	0.0.0.0	127.255.255.255
B	128.0.0.0	191.255.255.255
C	192.0.0.0	223.255.255.255
D	224.0.0.0	239.255.255.255
E	240.0.0.0	255.255.255.255

ip CIDR (classless inter
domain routing)

(temp solution no. 2)

classes are not really efficient
so let's ignore them in subnetting

private and public ips & NAT

not all people need to be accessible on the internet.

some may have their private interspace and ips, and they deal with it as they want.

Private IP addresses Ranges:

10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255

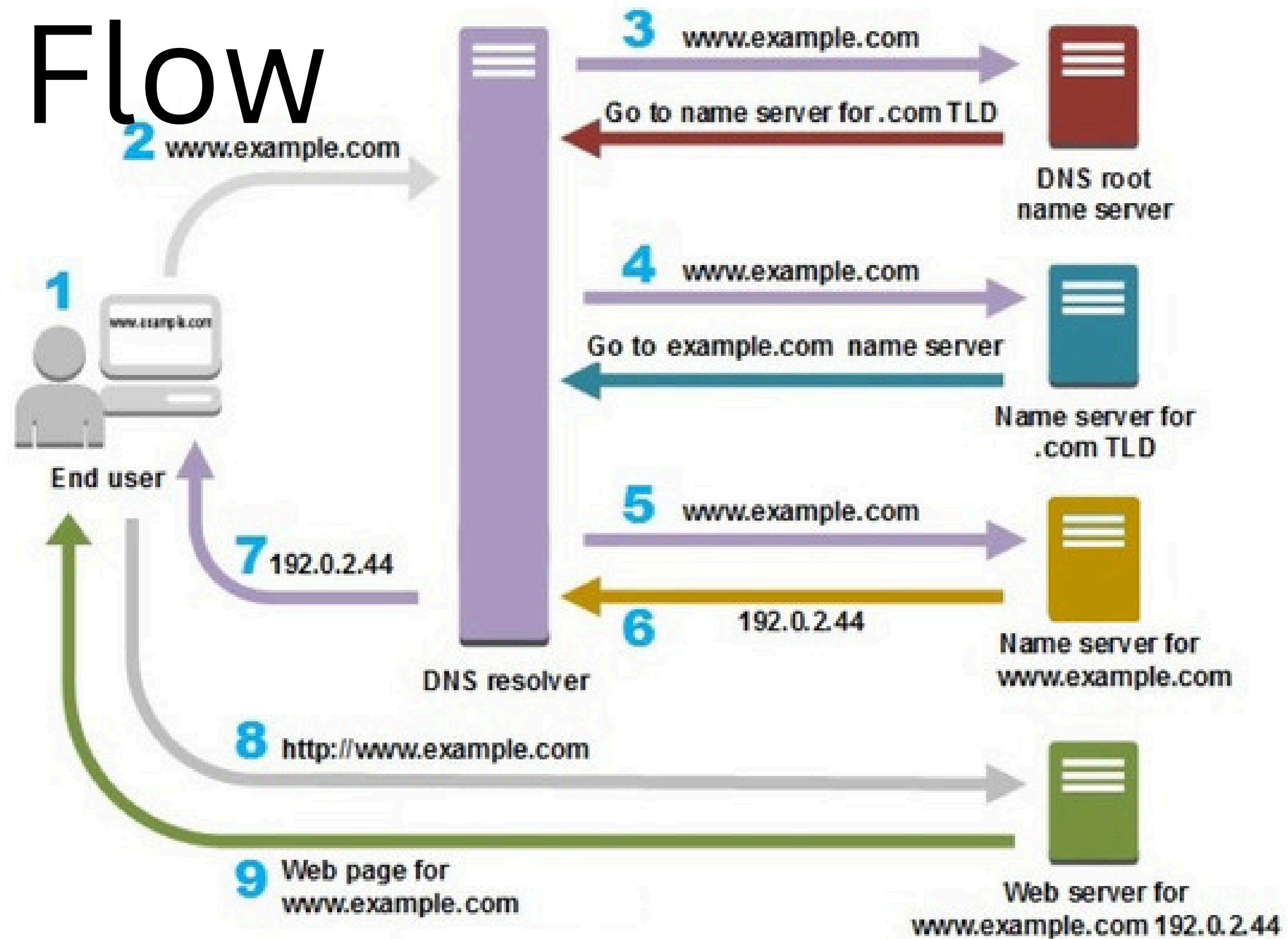
Our real long-term solution

ipv6

TCP vs UDP

DNS

DNS Flow



Some Protocols Headers

HTTP Request example

GET / HTTP/2

Host: example.com

User-Agent: Mozilla/5.0 (Windows NT) Firefox/103.0

Accept: text/html

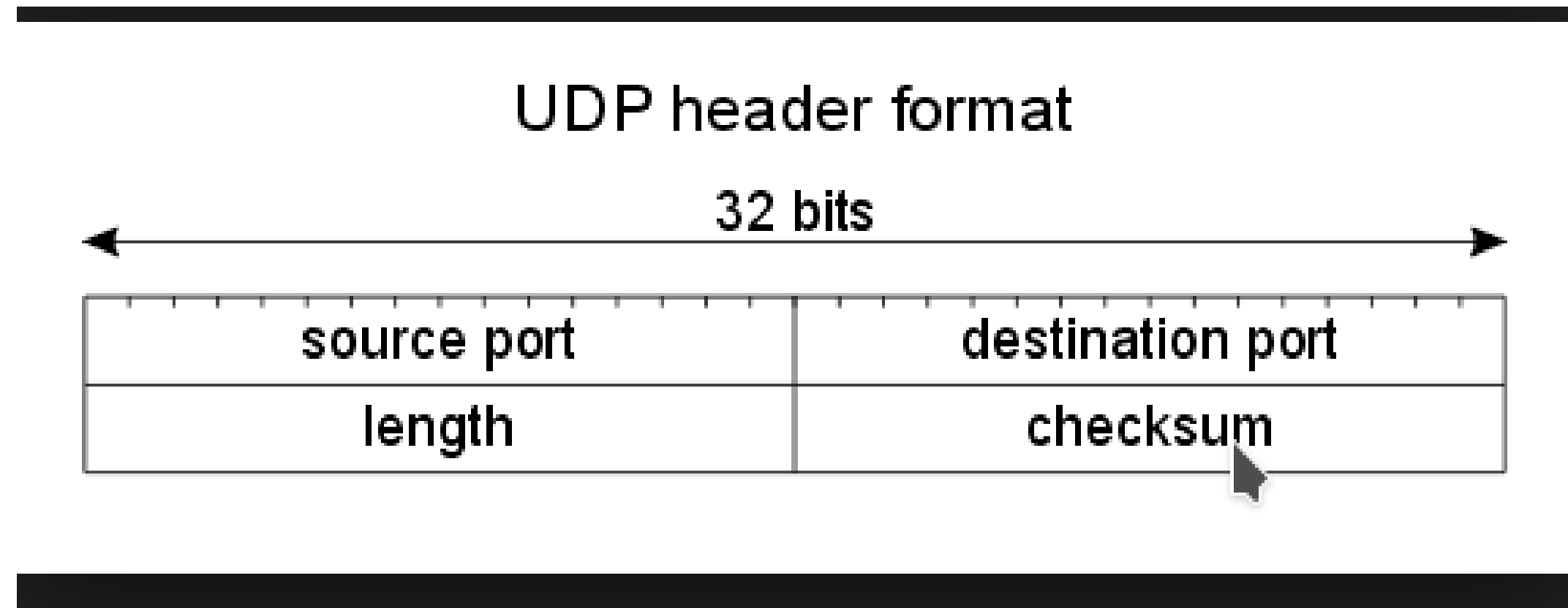
Accept-Language: en-US

Connection: keep-alive

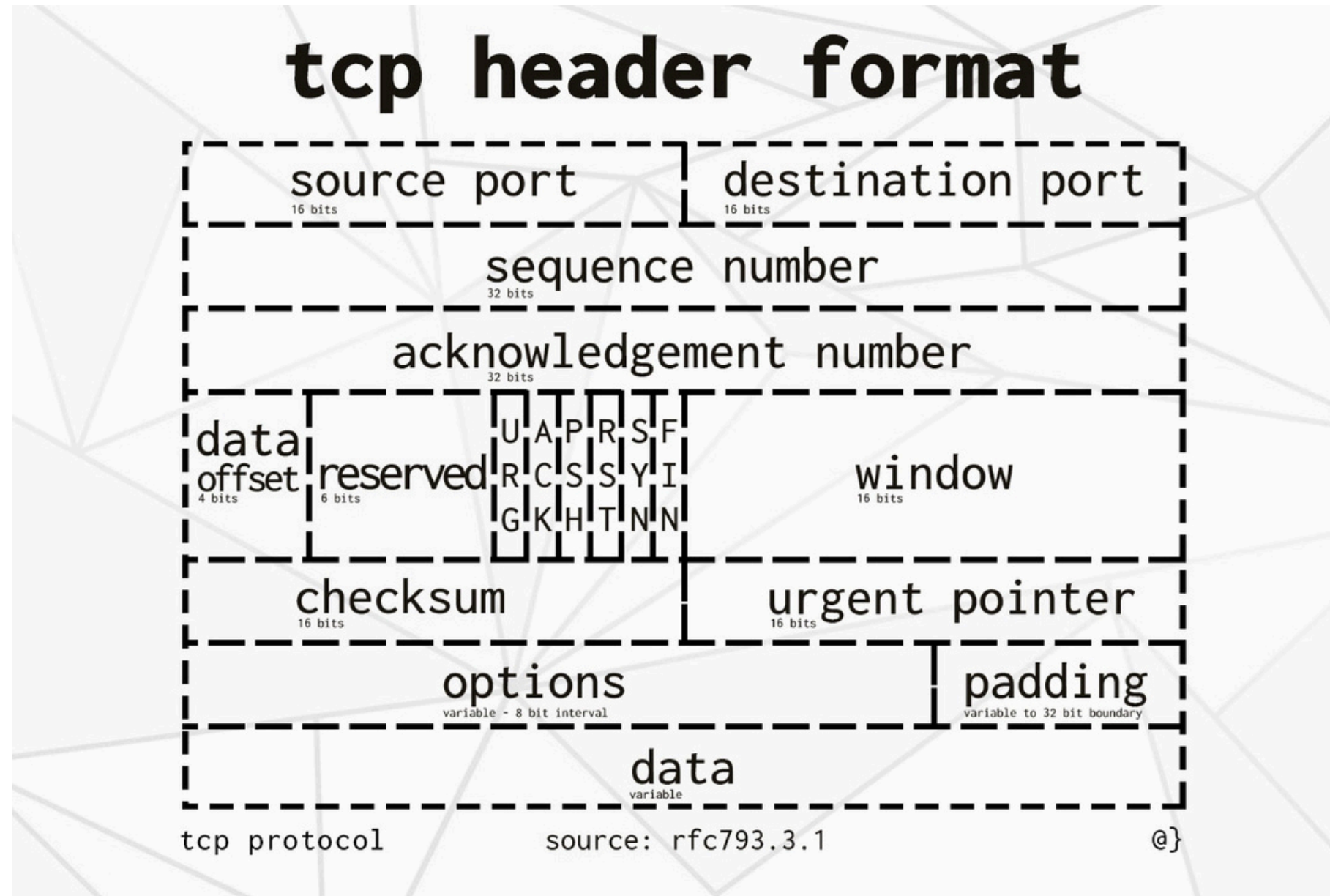
Pragma: no-cache

Cache-Control: no-cache

UDP Header Format



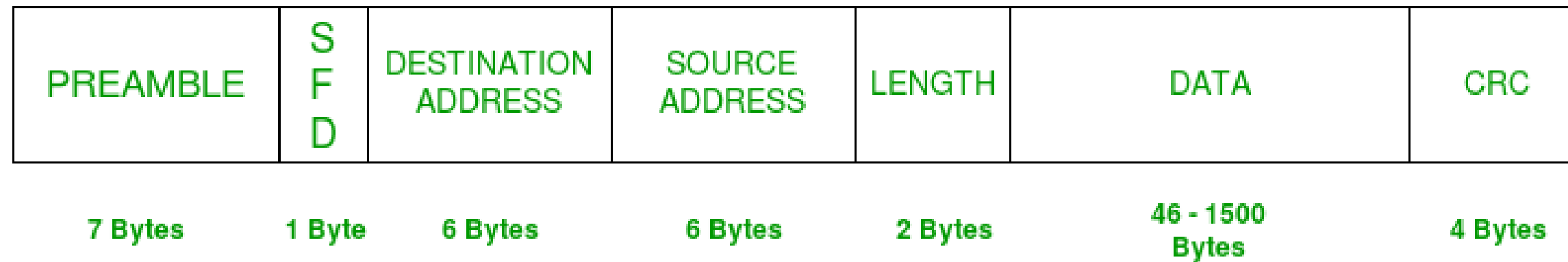
TCP Header Format



IP Header Format

0	4	8	16	19	31
Version	Header Length	Service Type	Total Length		
Identification			Flags	Fragment Offset	
TTL	Protocol		Header Checksum		
Source IP Addr					
Destination IP Addr					
Options				Padding	

Ethernet Frame Format



IEEE 802.3 ETHERNET Frame Format

وېبس کډهو

سلايد ال Thank You!

